

Exercise 19 • Motility Determination

- Avoid delay in using this setup. Water of condensation may develop to decrease clarity and the organisms become less motile with time.
- Review all the characteristics of bacterial motility that are stated on pages 72 and 73 under wet mounts.

Tube Method Inoculate tubes of semisolid or SIM media with each organism according to the following instructions:

- Label the tubes of semisolid (or SIM) media with the names of the organisms. Place your initials on the tubes, also.
- Flame and cool the inoculating needle, and insert it into the culture after flaming the neck of the tube.
- Remove the cap from the tube of medium, flame the neck, and stab it $\frac{2}{3}$ of the way down to the bottom, as shown in figure 19.2. Flame the neck of the tube again before returning the cap to the tube.
- Repeat steps 2 and 3 for the other culture.
- Incubate the tubes at room temperature for 24 to 48 hours.

Plate Method Mark the bottom of a plate of soft agar with two one-half inch circles about one inch apart. Label one circle ML and the other PV. These circles will be targets for your culture stabs. Put your initials on the plate also.

Using proper aseptic techniques, stab the medium in the center of the ML circle with *M. luteus* and the center of the other circle with *P. vulgaris*. Incubate the plate for 24 to 48 hours at room temperature.

SECOND PERIOD

Assemble the following materials that were inoculated during the last period and incubated.

Materials:

- culture tubes of motility medium that have been incubated
- inoculated Petri plate that has been incubated

Compare the two tubes that were inoculated with *M. luteus* and *P. vulgaris*. Look for cloudiness as evidence of motility. *Proteus* should exhibit motility. Does it? Record your results on the Laboratory Report.

Compare the appearance of the two stabs in the soft agar. Describe the differences that exist in the two stabs.

Does the plate method provide any better differentiation of results than the tube method?

LABORATORY REPORT

Complete the Laboratory Report for this exercise.

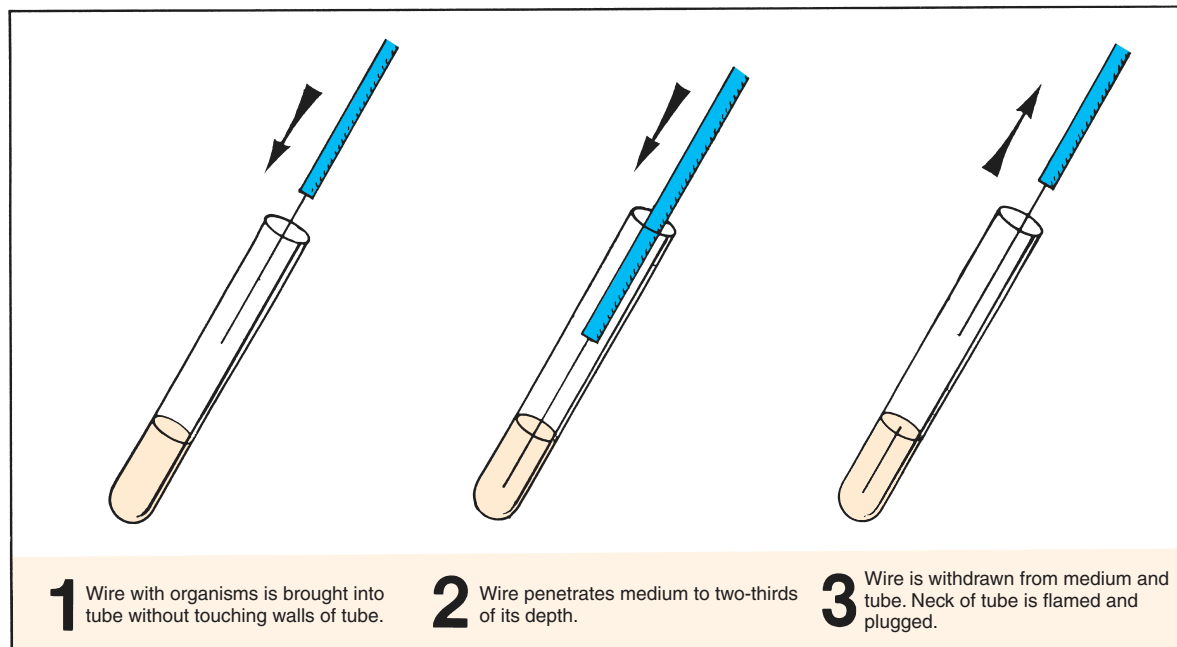


Figure 19.2 Stab technique for motility test

PART

4

Culture Methods

All nutritional types are represented among the protists. This diversity requires a multiplicity of culture methods. An attempt has been made in this unit to present those techniques that have proven most successful for the culture of autotrophic and heterotrophic bacteria, molds, and slime molds.

The first four exercises (20, 21, 22, and 23) pertain to basic techniques applicable to both autotrophs and heterotrophs. The other four exercises are concerned with the culture of specific types. In performing the last four experiments, you should be just as concerned with understanding the growth conditions as with the successful growth of a particular isolate. For example, the use of an enrichment medium, such as in Exercise 27 (Anaerobic Phototrophic Bacteria), has direct application in the culture of many other autotrophic bacteria as well.

This unit culminates the basic techniques phase of this course. A thorough understanding of microscopy, slide techniques, and culture methods provides a substantial foundation for the remainder of the exercises in this manual. If independent study projects are to be pursued as a part of this course, the completion of this unit will round out the background knowledge and skills for such work.

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Culture Media Preparation

From now on, most of the laboratory experiments in this manual will utilize bacteriological media. In most instances it will be provided for you. However, circumstances may arise when you will need some special medium that is not already prepared, and it will be up to you to put it together. It is for situations like this that the information in this exercise will be useful.

The first portion of this exercise pertains to the different types of media and how they relate to the needs of microorganisms. The last part of the exercise pertains to the actual mechanics of making up a batch of medium. Whether you will be provided an opportunity to prepare some media during a designated laboratory period will depend on the availability of time and classroom needs. Your instructor will indicate how this exercise is to be used.

MEDIA CONSISTENCY

A microbiological **medium** (*media*, plural) is the food that we use for culturing bacteria, molds, and other microorganisms. It can exist in three consistencies: liquid, solid, and semisolid. If you have performed all of the exercises in Part 3, you are already familiar with all of them.

Liquid media include nutrient broth, citrate broth, glucose broth, litmus milk, etc. These media are used for the propagation of large numbers of organisms, fermentation studies, and various other tests.

Solid media are made by adding a solidifying agent, such as agar, gelatin, or silica gel, to a liquid medium. A good solidifying agent is one that is not utilized by microorganisms, does not inhibit bacterial growth, and does not liquefy at room temperature. Agar and silica gel do not liquefy at room temperature and are utilized by very few organisms. Gelatin, on the other hand, is hydrolyzed by quite a few organisms and liquefies at room temperature.

Nutrient agar, blood agar, and Sabouraud's agar are examples of solid media that are used for developing surface colony growth of bacteria and molds. As we will see in the next exercise, the development of colonies on the surface of a medium is essential when trying to isolate organisms from mixed cultures.

Semisolid media fall in between liquid and solid media. Although they are similar to solid media in that they contain solidifying agents such as agar and gelatin, they are more jellylike due to lower percentages of these solidifiers. These media are particularly useful in determining whether certain bacteria are motile (Exercise 19).

NUTRITIONAL NEEDS OF BACTERIA

Before one can construct a medium that will achieve a desired result in the growth of organisms, one must understand their basic needs. Any medium that is to be suitable for a specific group of organisms must take into account the following seven factors: water, carbon, energy, nitrogen, minerals, growth factors, and pH. The role of each one of these factors follows.

Water Protoplasm consists of 70% to 85% water. The water in a single-celled organism is continuous with the water of its environment, and the molecules pass freely in and out of the cell, providing a vehicle for nutrients inward and secretions or excretions outward. All the enzymatically controlled chemical reactions that occur within the cell occur only in the presence of an adequate amount of water.

The quality of water used in preparing media is important. Hard tap water, high in calcium and magnesium ions, should not be used. Insoluble phosphates of calcium and magnesium may precipitate in the presence of peptones and beef extract. The best policy is to **always use distilled water**.

Carbon Organisms are divided into two groups with respect to their sources of carbon. Those that can utilize the carbon in carbon dioxide for synthesis of all cell materials are called **autotrophs**. If they must have one or more organic compounds for their carbon source, they are called **heterotrophs**. In addition to organic sources of carbon, the heterotrophs are also dependent on carbon dioxide. If this gas is completely excluded from their environment, their growth is greatly retarded, particularly in the early stages of starting a culture.

Specific organic carbon needs are as diverse as the organisms themselves. Where one organism may

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require only a single simple compound such as acetic acid, another may require a dozen or more organic nutrients of various degrees of complexity.

Energy Organisms that have pigments that enable them to utilize solar energy are called *photoautotrophs* (photosynthetic autotrophs). Media for such organisms will not include components to provide energy.

Autotrophs that cannot utilize solar energy but are able to oxidize simple inorganic substances for energy are called *chemoautotrophs* (chemosynthetic autotrophs). The essential energy-yielding substance for these organisms may be as elemental as nitrite, nitrate, or sulfide.

Most bacteria fall into the category of *chemoheterotrophs* (chemosynthetic heterotrophs) that require an organic source of energy, such as glucose or amino acids. The amounts of energy-yielding ingredients in media for both chemosynthetic types is on the order of 0.5%.

A small number of bacteria are classified as *photoheterotrophs* (photosynthetic heterotrophs). These organisms have photosynthetic pigments that enable them to utilize sunlight for energy. Their carbon source must be an organic compound such as alcohol.

Nitrogen Although autotrophic organisms can utilize inorganic sources of nitrogen, the heterotrophs get their nitrogen from amino acids and intermediate protein compounds such as peptides, proteoses, and peptones. Beef extract and peptone, as used in nutrient broth, provide the nitrogen needs for the heterotrophs grown in this medium.

Minerals All organisms require several metallic elements such as sodium, potassium, calcium, magnesium, manganese, iron, zinc, copper, phosphorus, and

cobalt for normal growth. Bacteria are no exception. The amounts required are very small.

Growth Factors Any essential component of cell material that an organism is unable to synthesize from its basic carbon and nitrogen sources is classified as being a *growth factor*. This may include certain amino acids or vitamins. Many heterotrophs are satisfied by the growth factors present in beef extract of nutrient broth. Most fastidious pathogens require enriched media such as blood agar for ample growth factors.

Hydrogen Ion Concentration The growth of organisms in a particular medium may be completely inhibited if the pH of the medium is not within certain limits. The enzymes of microorganisms are greatly affected by this factor. Since most bacteria grow best around pH 7 or slightly lower, the pH of nutrient broth should be adjusted to pH 6.8. Pathogens, on the other hand, usually prefer a more alkaline pH. Trypticase soy broth, a suitable medium for the more fastidious pathogens, should be adjusted to pH 7.3.

EXACT COMPOSITION MEDIA

Media can be prepared to exact specifications so that the exact composition is known. These media are generally made from chemical compounds that are highly purified and precisely defined. Such media are readily reproducible. They are known as **synthetic media**. Media such as nutrient broth that contain ingredients of imprecise composition are called **nonsynthetic media**. Both the beef extract and peptone in nutrient broth are inexact in composition.



Figure 20.1 Basic supplies and equipment needed for making up a batch of medium.



Figure 20.2 Correct amount of dehydrated medium is weighed on balance.

Exercise 20 • Culture Media Preparation

SPECIAL MEDIA

Two kinds of special media that will be widely used in this manual are selective and differential media.

Selective media are media that allow only certain types of organisms to grow in or on them because of (1) the absence of certain critical nutrients that make it unfavorable for most, but not all, organisms, or (2) the presence of inhibitory substances that prevent certain types of organisms to grow on them. The inhibitory substance may be salt (NaCl), acid, a toxic chemical (crystal violet), an antibiotic (streptomycin), or some other substance.

Differential media are media that contain substances that cause some bacteria to take on a different appearance from other species, allowing one to differentiate one species from another.

In some cases media have been formulated that are both selective and differential. A good example is Levine EMB agar, which is used to determine the presence of coliforms in water analysis (Exercise 63).

DEHYDRATED MEDIA

Until around 1930, the laboratory worker had to spend a good deal of time preparing laboratory media from various raw materials. If a medium contained five or six ingredients, it was not only necessary to measure the various materials, but, also, in many instances, to fabricate some of the components such as beef extract or veal infusion by long tedious cooking methods. Today, dehydrated media have revolutionized media preparation techniques in much the same way that commercial cake mixes have taken over in the kitchen. For most routine bacteriological work, media preparation has been simplified to the extent that all that is necessary is to dissolve a measured amount of

dehydrated medium in water, adjust the pH, dispense into tubes, and sterilize. In many cases pH adjustment is not even necessary.

MEDIA PREPARATION ASSIGNMENT

In this laboratory period you will work with your laboratory partner to prepare tubes of media that will be used in future laboratory experiments. Your instructor will indicate which media you are to prepare. Record in the space below the number of tubes of specific media that have been assigned to you and your partner.

nutrient broth	_____
nutrient agar pours	_____
nutrient agar slants	_____
other	_____

Several different sizes of test tubes are used for media, but the two sizes most generally used are either 16 mm or 20 mm diameter by 15 cm long. Select the correct size tubes first, according to these guidelines:

Large tubes (20 mm dia): Use these test tubes for *all pours*: i.e., nutrient agar, Sabouraud's agar, EMB agar, etc. Pours are used for filling Petri plates.

Small tubes (16 mm dia): Use these tubes for all *broths, deeps, and slants*.

If the tubes are clean and have been protected from dust or other contamination, they can be used without cleaning. If they need cleaning, scrub out the insides with warm water and detergent, using a test-tube brush. Rinse twice, first with tap water, and finally with distilled water to rid them of all traces of detergent. Place them in a wire basket or rack, inverted, so that they can drain. Do not dry with a towel.



Figure 20.3 Dehydrated medium is dissolved in a measured amount of distilled water.



Figure 20.4 If medium contains agar, it must be brought to a boil to bring agar into solution.

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Measurement and Mixing

The amount of medium you make for a batch should be determined as precisely as possible to avoid shortage or excess.

Materials:

graduate, beaker, glass stirring rod
bottles of dehydrated media
Bunsen burner and tripod, or hot plate

1. Measure the correct amount of water needed to make up your batch. The following volumes required per tube must be taken into consideration:

pours 12 ml
deeps6 ml
slants4 ml
broths5 ml
broths with fermentation tubes5–7 ml

2. Consult the label on the bottle to determine how much powder is needed for 1,000 ml and then determine by proportionate methods how much you need for the amount of water you are using. Weigh this amount on a balance and add it to the beaker of water. If the medium does not contain agar, the mixture usually goes into solution without heating.
3. **If the medium contains agar**, heat the mixture over a Bunsen burner (figure 20.4) or on an electric hot plate until it comes to a boil. To safeguard against water loss, *before heating, mark the level of the top of the medium on the side of the beaker with a china marking pencil*. As soon as it “froths up,” turn off the heat. If an electric hot plate is used, the medium must be removed from the hot plate or it will boil over the sides of the container.

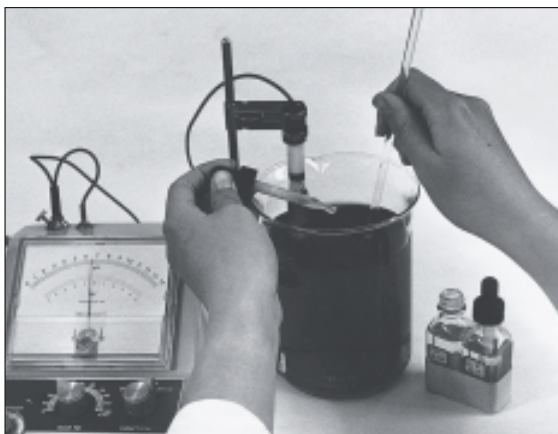


Figure 20.5 The hydrogen ion concentration of a medium must be adjusted to its recommended pH.

Caution: Be sure to keep stirring from the bottom with a glass stirring rod so that the medium does not char on the bottom of the beaker.

4. Check the level of the medium with the mark on the beaker to note if any water has been lost. Add sufficient distilled water as indicated. Keep the temperature of the medium at about 60° C to avoid solidification. The medium will solidify at around 40° C.

Adjusting the pH

Although dehydrated media contain buffering agents to keep the pH of the medium in a desired range, the pH of a batch of medium may differ from that stated on the label of the bottle. Before the medium is tubed, therefore, one should check the pH and make any necessary adjustments.

If a pH meter (figure 20.5) is available and already standardized, use it to check the pH of your medium. If the medium needs adjustment use the bottles of HCl and NaOH to correct the pH. If no meter is available pH papers will work about as well. Make pH adjustment as follows:

Materials:

beaker of medium
acid and base kits (dropping bottles of 1N and 0.1N HCl and NaOH)
glass stirring rod
pH papers
pH meter (optional)

1. Dip a piece of pH test paper into the medium to determine the pH of the medium.
2. **If the pH is too high**, add a drop or two of HCl to lower the pH. For large batches use 1N HCl. If the pH difference is slight, use the 0.1N HCl. Use a glass stirring rod to mix the solution as the drops are added.

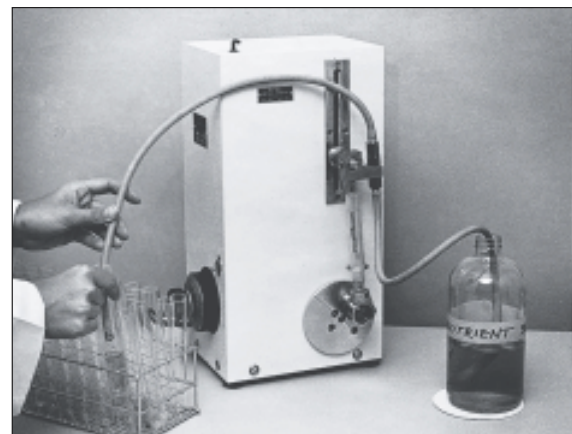


Figure 20.6 An automatic pipetting machine will deliver precise amounts of media at a controlled rate.

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3. **If the pH is too low**, add NaOH, one drop at a time, to raise the pH. For slight pH differences, use 0.1N NaOH; for large differences use 1N NaOH. Use a glass stirring rod to mix the solution as the drops are added.

FILLING THE TEST TUBES

Once the pH of the medium is adjusted it must be dispensed into test tubes. If an automatic pipetting machine is to be used, as shown in figure 20.6, it will have to be set up for you by your instructor. These machines can be adjusted to deliver any amount of medium at any desired speed. When large numbers of tubes are to be filled, the automatic pipetting machine should be used. For smaller batches, the funnel method shown in figure 20.7 is adequate. Use the following procedure when filling tubes with a funnel assembly.

Materials:

ring stand assembly
funnel assembly (glass funnel, rubber tubing,
hose clamp, and glass tip)
graduate (small size)

1. Fill one test tube with a measured amount of medium. This tube will be your guide for filling the other tubes.
2. Fill the funnel and proceed to fill the test tubes to the proper level, holding the guide tube alongside of each empty tube to help you to determine the amount to allow into each tube.
3. Keep the beaker of medium over heat if it contains agar.
4. If fermentation tubes are to be used, add one to each tube at this time *with the open end down*.

Capping the Tubes

The last step before sterilization is to provide a closure for each tube. Plastic (polypropylene) caps are suitable in most cases. All caps that slip over the tube end have inside ridges that grip the side of the tube and provide an air gap to allow steam to escape during sterilization. If you are using tubes with plastic screw-caps, *the caps should not be screwed tightly before sterilization; instead, each one must be left partly unscrewed*.

If no slip-on caps of the correct size are available, it may be necessary to make up some cotton plugs. A properly made cotton plug should hold firmly in the tube so that it is not easily dislodged.

Sterilization

As soon as the tubes of media have been stoppered they must be sterilized. Organisms on the walls of the



Figure 20.7 A glass funnel assembly and hose clamp are adequate for filling small batches of tubes.

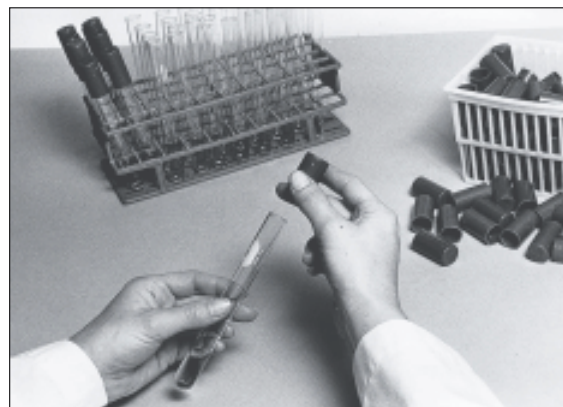


Figure 20.8 Once the medium has been dispensed to all the tubes, they are capped prior to sterilization.



Figure 20.9 Tubes of media are sterilized in an autoclave for 20 to 30 minutes at 15 psi steam pressure.

tubes, in the distilled water, and in the dehydrated medium will begin to grow within a short period of time at room temperature, destroying the medium.

Prior to sterilization, the tubes of media should be placed in a wire basket with a label taped on the outside of the basket. The label should indicate the type of medium, the date, and your name.

Sterilization must be done in an autoclave. The following considerations are important in using an autoclave:

- Check with your instructor on the procedure to be used with your particular type of autoclave. Complete sterilization occurs at 250° F (121.6° C). To achieve this temperature the autoclave has to develop 15 pounds per square inch (psi) of steam pressure. To reach the correct temperature there must be some provision in the chamber for the escape of air. On some of the older units it is necessary to allow the steam to force air out through the door before closing it.
- *Don't overload the chamber.* One should not attempt to see how much media can be packed into it. Provide ample space between baskets of media to allow for circulation of steam.
- *Adjust the time of sterilization to the size of load.* Small loads may take only 10 to 15 minutes. An

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autoclave full of media may require 30 minutes for complete sterilization.

After Sterilization

Slants If you have a basket of tubes that are to be converted to slants, it is necessary to lay the tubes down in a near-horizontal manner as soon as they are removed from the autoclave. The easiest way to do this is to use a piece of rubber tubing (1/2" dia) to support the capped end of the tube as it rests on the countertop. Solidification should occur in about 30–60 minutes.

Other Media Tubes of broth, agar deeps, nutrient gelatin, etc., should be allowed to cool to room temperature after removal from the autoclave. Once they have cooled down, place them in a refrigerator or cold-storage room.

Storage If tubes of media are not to be used immediately, they should be stored in a cool place. When stored for long periods of time at room temperature media tend to lose moisture. At refrigerated temperatures media will keep for months.

LABORATORY REPORT

Complete the Laboratory Report for this exercise.

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Pure Culture Techniques

When we try to study the bacterial flora of the body, soil, water, food, or any other part of our environment, we soon discover that bacteria exist in mixed populations. It is only in very rare situations that they occur as a single species. To be able to study the cultural, morphological, and physiological characteristics of an individual species, it is essential, first of all, that the organism be separated from the other species that are normally found in its habitat; in other words, we must have a **pure culture** of the microorganism.

Several different methods of getting a pure culture from a mixed culture are available to us. The two most frequently used methods involve making a streak plate or a pour plate. Both plate techniques involve thinning the organisms so that the individual species can be selected from the others.

In this exercise you will have an opportunity to use both methods in an attempt to separate three distinct species from a tube that contains a mixture. The principal difference between the three organisms will be their colors: *Serratia marcescens* is red, *Micrococcus luteus* is yellow, and *Escherichia coli* is white. If *Chromobacterium violaceum* is used in place of *M. luteus*, the three colors will be red, white, and purple.

STREAK PLATE METHOD

For economy of materials and time, this method is best. It requires a certain amount of skill, however, which is forthcoming with experience. A properly executed streak plate will give as good an isolation as is desired for most work. Figure 21.1 illustrates how colonies of a mixed culture should be spread out on a properly made streak plate. The important thing is to produce good spacing between colonies.

Materials:

- electric hot plate (or tripod and wire gauze)
- Bunsen burner and beaker of water
- wire loop, thermometer, and china marking pencil
- 1 nutrient agar pour and 1 sterile Petri plate
- 1 mixed culture of *Serratia marcescens*, *Escherichia coli*, and *Micrococcus luteus* (or *Chromobacterium violaceum*)



Figure 21.1 If your streak plate reveals well-isolated colonies of three colors (red, white, and yellow), you will have a plate suitable for subculturing.

1. Prepare your tabletop by disinfecting its surface with the disinfectant that is available in the laboratory (Roccal, Zephiran, Betadine, etc.). Use a sponge to scrub it clean.
2. Label the bottom surface of a sterile Petri plate with your name and date. Use a china marking pencil.
3. Liquefy a tube of nutrient agar, cool to 50° C, and pour the medium into the bottom of the plate, following the procedure illustrated in figure 21.2. Be sure to flame the neck of the tube prior to pouring to destroy any bacteria around the end of the tube.

After pouring the medium into the plate, gently rotate the plate so that it becomes evenly distributed, but do not splash any medium up over the sides.

Agar-agar, the solidifying agent in this medium becomes liquid when boiled and resolidifies at around 42° C. Failure to cool it prior to pouring into the plate will result in condensation of moisture on the cover. Any moisture on the cover is undesirable because if it drops down on the colonies, the organisms of one colony can spread to other colonies, defeating the entire isolation technique.

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4. Streak the plate by one of the methods shown in figure 21.4. Your instructor will indicate which technique you should use.

Caution: Be sure to follow the routine in figure 21.3

for getting the organism out of culture.

5. Incubate the plate in an *inverted position* at 25° C for 24–48 hours. By incubating plates upside down, the problem of moisture on the cover is minimized.

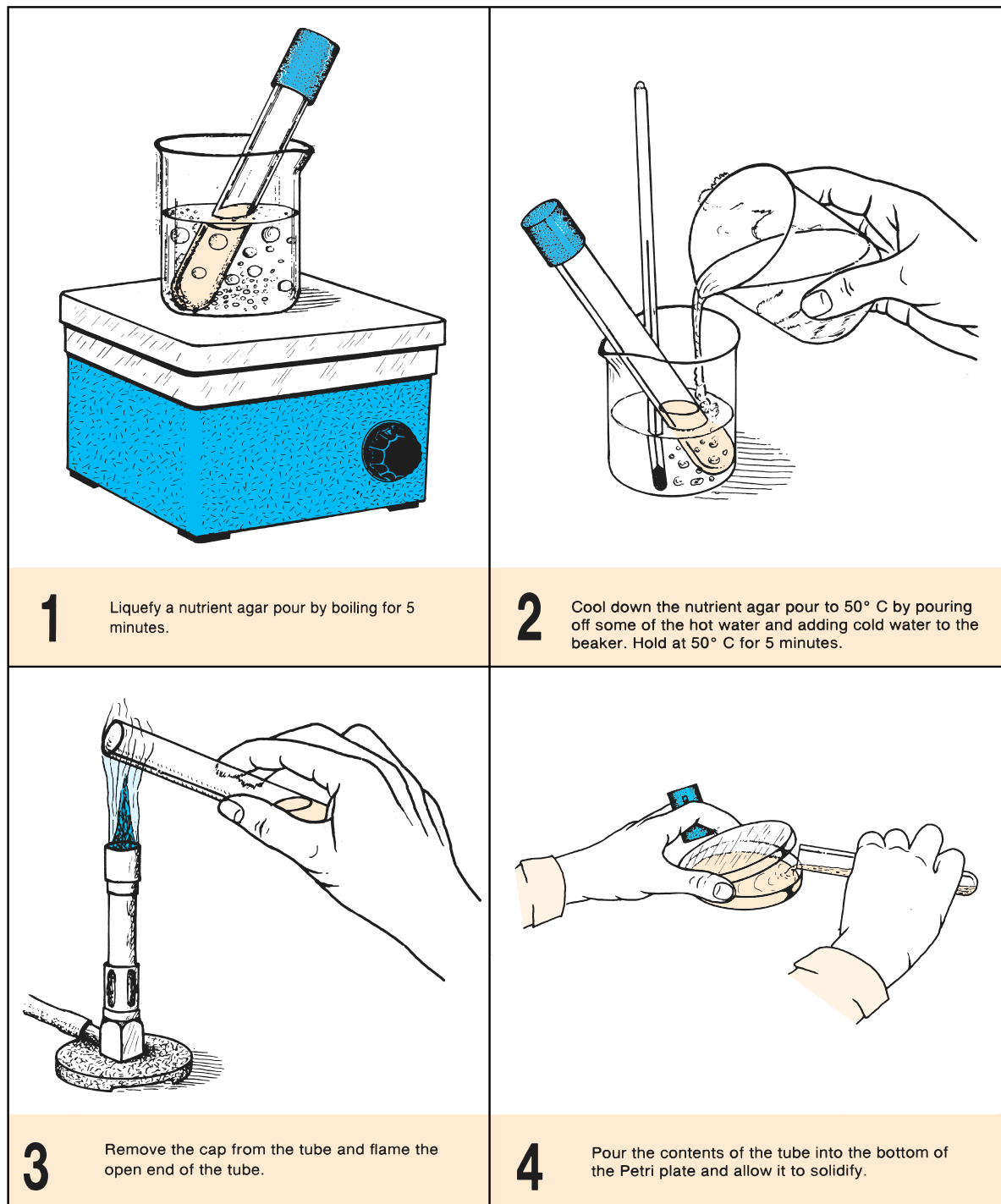


Figure 21.2 Procedure for pouring an agar plate for streaking

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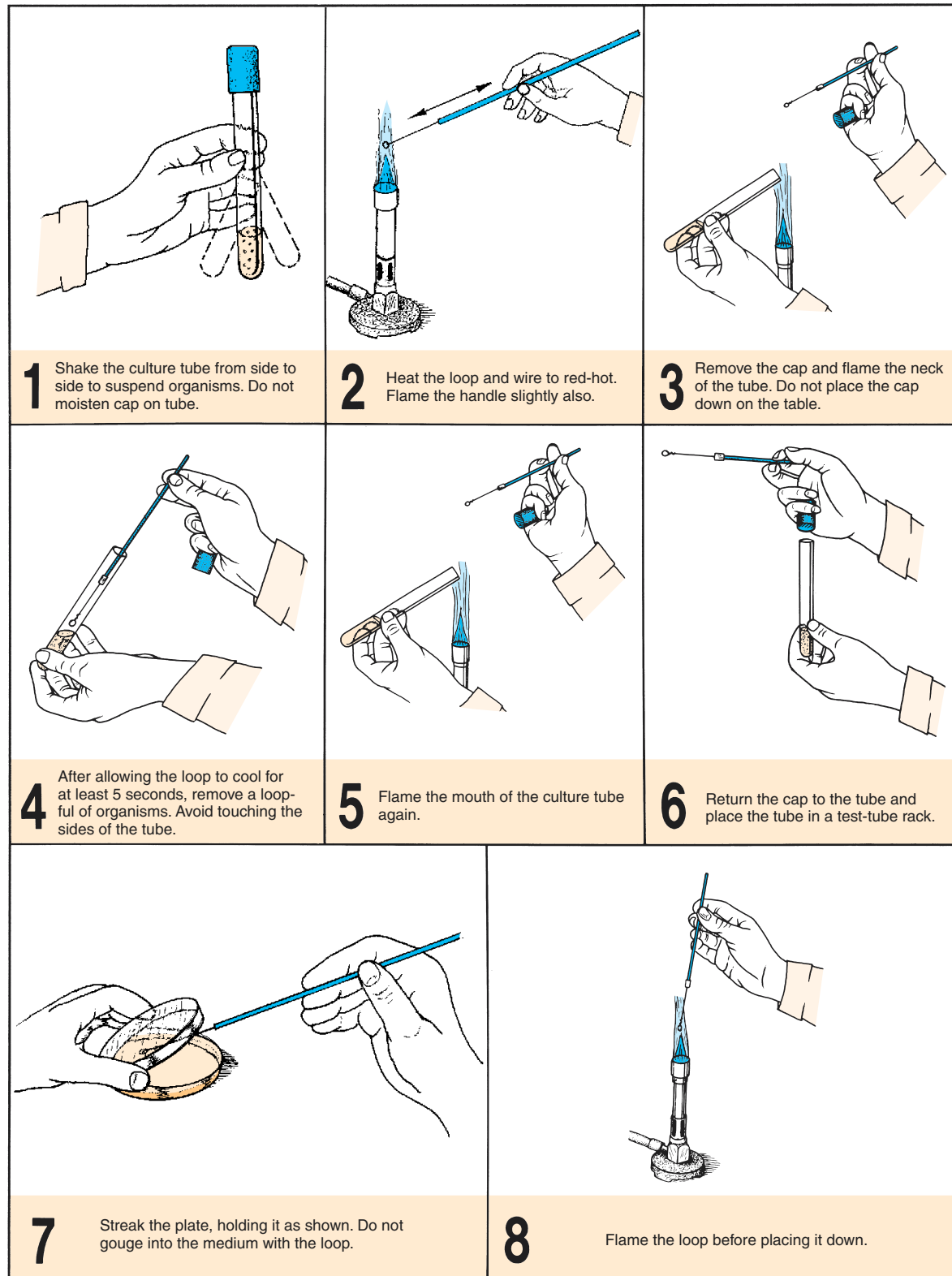


Figure 21.3 Routine for inoculating a Petri plate

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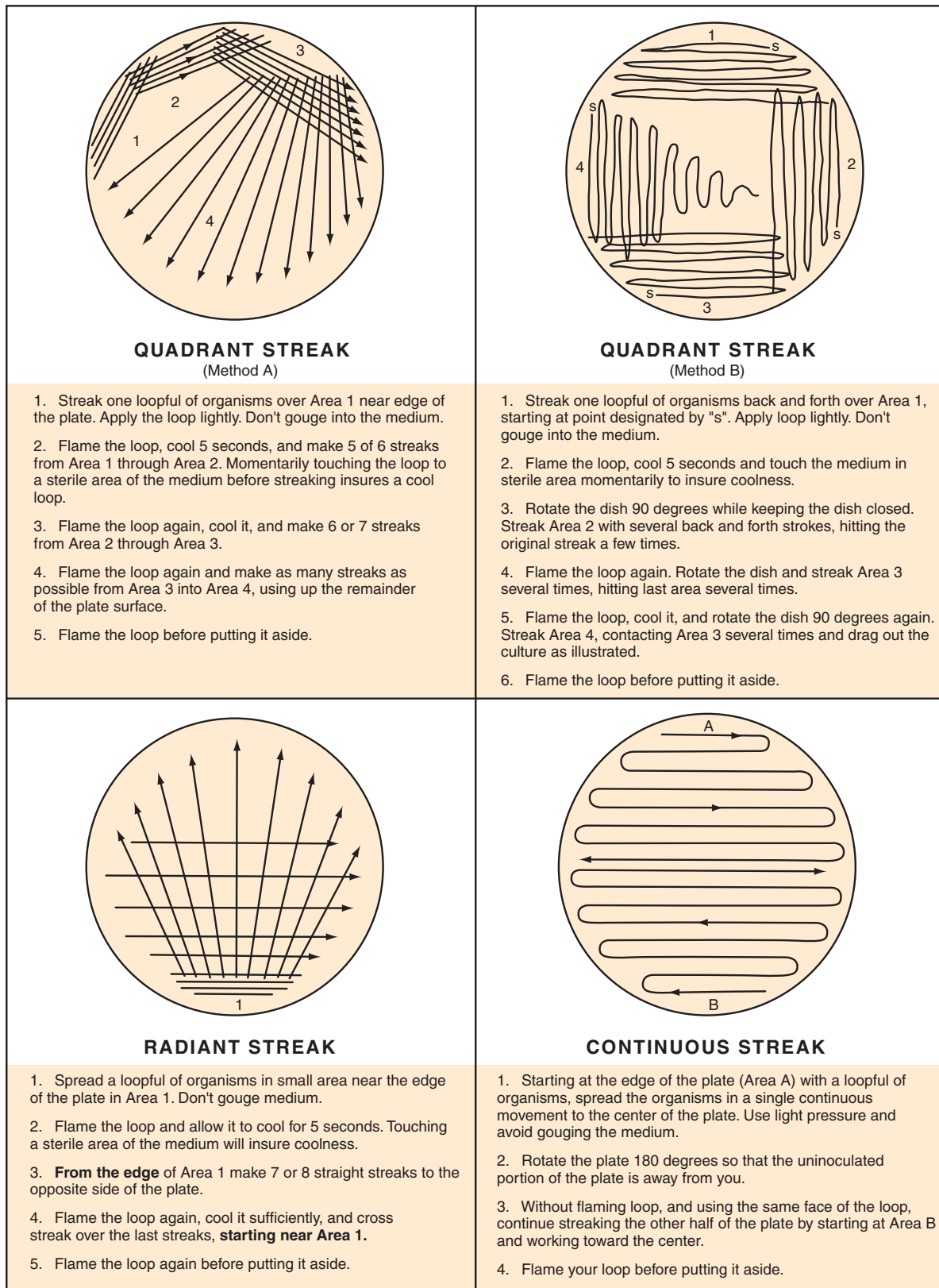


Figure 21.4 Four different streak techniques

Exercise 21 • Pure Culture Techniques

POUR PLATE METHOD**(Loop Dilution)**

This method of separating one species of bacteria from another consists of diluting out one loopful of organisms with three tubes of liquefied nutrient agar in such a manner that one of the plates poured will have an optimum number of organisms to provide good isolation. Figure 21.5 illustrates the general procedure. One advantage of this method is that it requires somewhat less skill than that required for a good streak plate; a disadvantage, however, is that it requires more media, tubes, and plates. Proceed as follows to make three dilution pour plates, using the same mixed culture you used for your streak plate.

Materials:

mixed culture of bacteria
3 nutrient agar pours
3 sterile Petri plates
electric hot plate
beaker of water
thermometer
inoculating loop and china marking pencil

1. Label the three nutrient agar pours **I**, **II**, and **III** with a marking pencil and place them in a beaker

of water on an electric hot plate to be liquefied. To save time, start with hot tap water if it is available.

2. While the tubes of media are being heated, label the bottoms of the three Petri plates **I**, **II**, and **III**.
3. Cool down the tubes of media to 50° C, using the same method that was used for the streak plate.
4. Following the routine in figure 21.5, inoculate tube I with one loopful of organisms from the mixed culture. Note the sequence and manner of handling the tubes in figure 21.6.
5. Inoculate tube II with one loopful from tube I after thoroughly mixing the organisms in tube I by shaking the tube from side to side or by rolling the tube vigorously between the palms of both hands. **Do not splash any of the medium up onto the tube closure.** Return tube I to the water bath.
6. Agitate tube II to completely disperse the organisms and inoculate tube III with one loopful from tube II. Return tube II to the water bath.
7. Agitate tube III, flame its neck, and pour its contents into plate III.
8. Flame the necks of tubes I and II and pour their contents into their respective plates.
9. After the medium has completely solidified, incubate the *inverted* plates at 25° C for 24–48 hours.

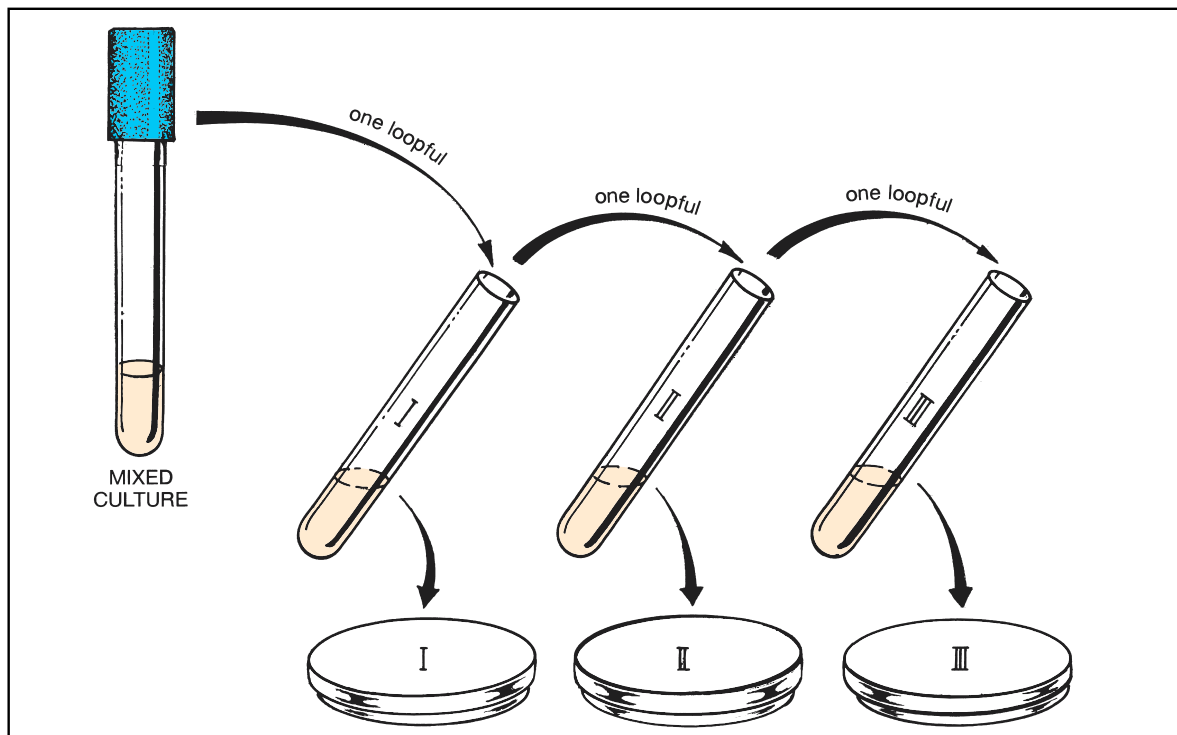


Figure 21.5 Three steps in the loop dilution technique for separating out organisms

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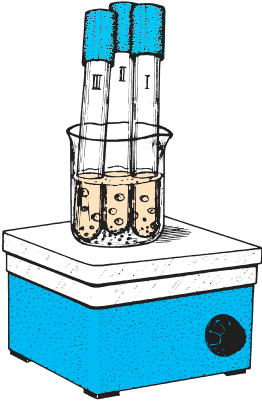
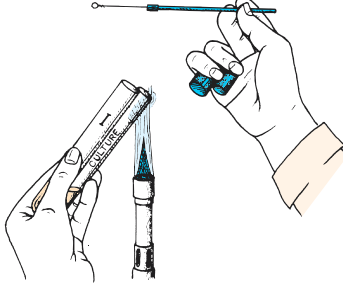
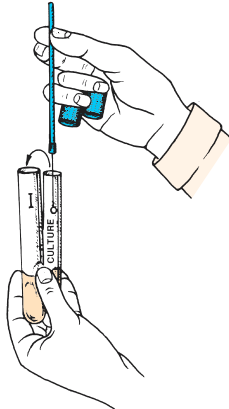
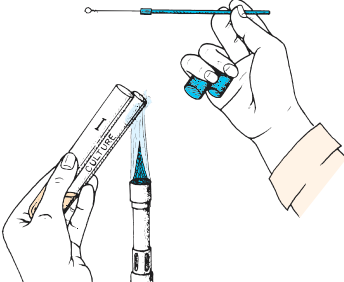
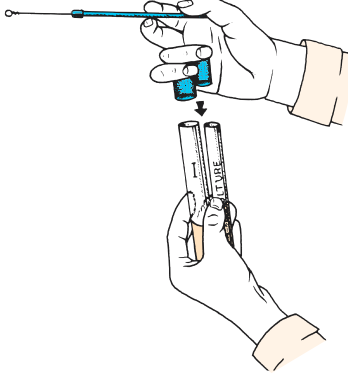
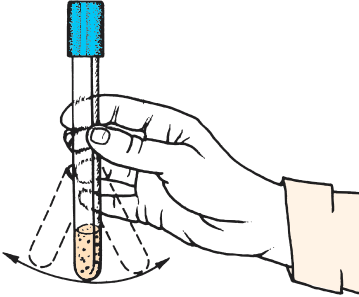
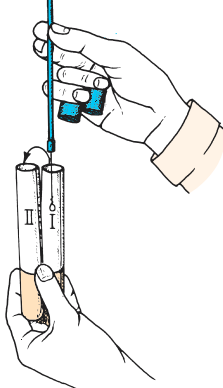
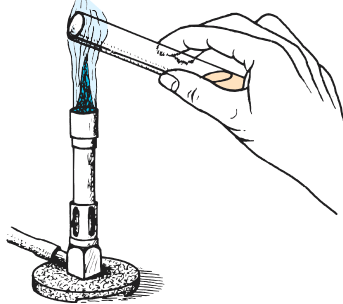
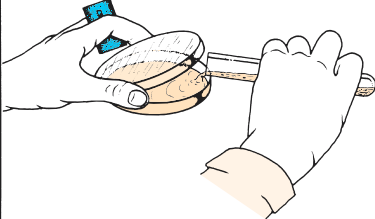
		
1 Liquefy three nutrient agar pours, cool to 50° C, and let stand for 10 minutes.	2 After shaking the culture to disperse the organisms, flame the loop and necks of the tubes.	3 Transfer one loopful of the culture to tube I.
		
4 Flame the loop and the necks of both tubes.	5 Replace the caps on the tubes and return the culture to the test-tube rack.	6 Disperse the organisms in tube I by shaking the tube or rolling it between the palms.
		
7 Transfer one loopful from tube I to tube II. Return tube I to the water bath.	8 After shaking tube II and transferring one loopful to tube III, flame the necks of each tube.	9 Pour the inoculated pours into their respective Petri plates.

Figure 21.6 Tube-handling procedure in making inoculations for pour plates

Exercise 21 • Pure Culture Techniques

EVALUATION OF THE TWO METHODS

After 24 to 48 hours of incubation examine all four Petri plates. Look for colonies that are well isolated from the others. Note how crowded the colonies appear on plate I as compared with plates II and III. Plate I will be unusable. Either plate II or III will have the most favorable isolation of colonies. Can you pick out three well-isolated colonies on your best pour plate that are white, yellow, and red?

Draw the appearance of your streak plate and pour plates on the Laboratory Report.

SUBCULTURING TECHNIQUES

The next step in the development of a pure culture is to transfer the organisms from the Petri plate to a tube of nutrient broth or a slant of nutrient agar. After this subculture has been incubated for 24 hours, a stained slide of the culture can be made to determine if a pure culture has been achieved. When transferring the organisms from the plate, an inoculating needle (straight wire) is used instead of the wire loop. The needle is inserted into the center of the colony where there is a greater probability of getting only one species of organism. Use the following routine in subculturing out the three different organisms.

Materials:

3 nutrient agar slants
inoculating needle
Bunsen burner

1. Label one tube *S. marcescens*, another *E. coli*, and the third *M. luteus* or *C. violaceum*.
2. Select a well-isolated red colony on either the streak plate or pour plate for your first transfer. Insert the inoculating needle into the center of the colony.

3. In the tube labeled *S. marcescens*, streak the slant by placing the needle near the bottom of the slant and drawing it up over its surface. One streak is sufficient.
4. Repeat this inoculating procedure on the other two slants for a white colony and a yellow (or purple) colony.
5. Incubate for 24 to 48 hours at 25° C.

EVALUATION OF SLANTS

After incubation, examine the slants. Is *S. marcescens* red? Is *E. coli* white? Is your third slant yellow or purple? If the incubation temperature has been too high, *S. marcescens* may appear white due to the fact that the red pigment forms only at a moderate temperature, such as 25° C. Draw the appearance of the slants with colored pencils on the Laboratory Report.

Although the colors of the growths on the slants may lead you to think that you have pure cultures, you cannot be absolutely certain until you have made a microscopic examination of each culture. For example, it is entirely possible that the yellow slant (*M. luteus*) may have some *E. coli* present that are masked by the yellow pigment.

To find out if you have a pure culture on each slant, make a gram-stained slide from each slant. Knowing that *S. marcescens*, *E. coli*, and *C. violaceum* are gram-negative rods, and that *M. luteus* is a gram-positive coccus, you should be able to evaluate your slants more precisely microscopically. Draw the organisms on the Laboratory Report.

LABORATORY REPORT

Complete the Laboratory Report for this exercise.

Cultivation of Anaerobes

22

The procedures for culturing bacteria that were used in the last exercise work well only if the organisms will grow in the presence of oxygen. Unfortunately, there are many bacteria that find oxygen toxic or at least inhibitory to their existence. For these organisms we need to create an anaerobic environment by using special media deficient or lacking in oxygen and containers that are oxygen-free. In this laboratory period we will learn how to find out what the oxygen requirements are for specific organisms and how to grow them in liquid and solid media. In doing so we will be inoculating special media with several organisms of different cultural requirements to evaluate their oxygen needs.

The oxygen requirements of bacteria range from **strict (obligate) aerobes** that cannot exist without this gas to the **strict (obligate) anaerobes** that die in its presence. In between these extremes are the facultatives, indifferents, and microaerophilics. The **facultatives** are bacteria that have enzyme systems enabling them to utilize free oxygen or some alternative oxygen source such as nitrate. If oxygen is present, they tend to utilize it in preference to the alternative. The **indifferents**, however, show no preference for either condition, growing equally well in aerobic and anaerobic conditions. **Microaerophiles**, on the other hand, are organisms that require free oxygen, but only in limited amounts. Figure 22.1 illustrates where these various types tend to grow with respect to the degree of oxygen tension in a medium.

In this experiment we will inoculate one liquid medium and two solid media with several organisms that have different oxygen requirements. The media are fluid thioglycollate medium (FTM), tryptone glucose yeast agar (TGYA), and Brewer's anaerobic agar. Each medium will serve a different purpose. A discussion of the function of each medium follows:

TGYA Shake This solid medium will be used in what is called a "shake tube." The medium is not primarily an anaerobic medium; instead it is a rich general purpose medium that favors the growth of a broad spectrum of organisms. It will be inoculated in the liquefied state, shaken to mix the organisms throughout the medium, and allowed to solidify. After incubation one determines the oxygen require-

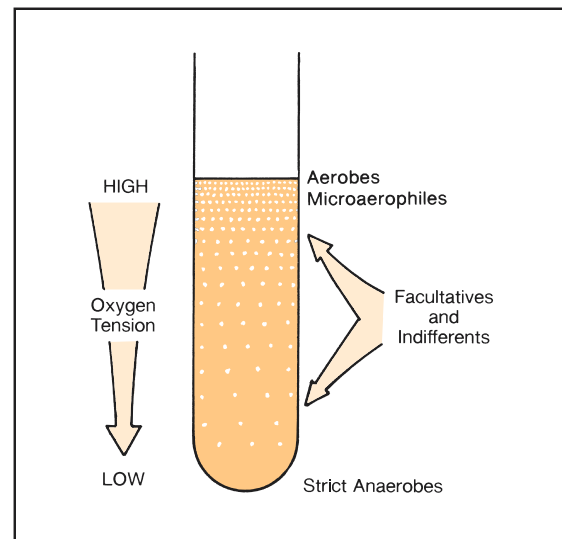


Figure 22.1 Oxygen needs of microorganisms

ments on the basis of where the growth occurs in the tube: top, middle, or bottom.

FTM Fluid thioglycollate medium is a rich liquid medium that supports the growth of both aerobic and anaerobic bacteria. It contains glucose, cystine, and sodium thioglycollate to reduce its oxidation-reduction (O/R) potential. It also contains the dye resazurin that is an indicator for the presence of oxygen. In the presence of oxygen the dye becomes pink. Since the oxygen tension is always higher near the surface of the medium, the medium will be pink at the top and colorless in the middle and bottom. The medium also contains a small amount of agar that helps to localize the organisms and favors anaerobiasis in the bottom of the tube.

Brewer's Anaerobic Agar This solid medium is an excellent medium for culturing anaerobic bacteria in Petri dishes. It contains thioglycollate as a reducing agent and resazurin as an O/R indicator. For strict anaerobic growth it is essential that plates be incubated in an oxygen-free environment.

To provide an oxygen-free incubation environment for the Petri plates of anaerobic agar we will

Exercise 22 • Cultivation of Anaerobes

use the **GasPak anaerobic jar**. Note in figure 22.2 that hydrogen is generated in the jar, which removes the oxygen by forming water. Palladium pellets catalyze the reaction at room temperature. The generation of hydrogen is achieved by adding water to a plastic envelope of chemicals. Note also that CO₂ is produced, which is a requirement for the growth of many fastidious bacteria. To make certain that anaerobic conditions actually exist in the jar, an indicator strip of methylene blue becomes colorless in the total absence of oxygen. If the strip is not reduced (decolorized) within 2 hours, the jar has not been sealed properly, or the chemical reaction has failed to occur.

In addition to doing a study of the oxygen requirements of six organisms in this experiment, an opportunity will be provided during the second period to do a microscopic study of the types of endospores formed by three spore-formers used in the inoculations. Proceed as follows:

FIRST PERIOD

(Inoculations and Incubation)

Since six microorganisms and three kinds of media are involved in this experiment, it will be necessary for economy of time and materials to have each student work with only three organisms. The materials

list for this period indicates how the organisms will be distributed.

During this period each student will inoculate three tubes of medium and only one Petri plate of Brewer's anaerobic agar. The tubes and all of the plates will be placed in a GasPak jar to be incubated in a 37° C incubator. Students will share results.

Materials:

per student:

- 3 tubes of fluid thioglycollate medium
- 3 TGYA shake tubes (liquefied)
- 1 Petri plate of Brewer's anaerobic agar

broth cultures for **odd-numbered students:**

Staphylococcus aureus, *Streptococcus faecalis*, and *Clostridium sporogenes*

broth cultures for **even-numbered students:**

Bacillus subtilis, *Escherichia coli*, and *Clostridium rubrum*

GasPak anaerobic jar, 3 GasPak generator envelopes, 1 GasPak anaerobic generator strip, scissors, and one 10 ml pipette
water baths at student stations (electric hot plate, beaker of water, and thermometer)

1. Set up a 45° C water bath at your station in which you can keep your tubes of TGYA shakes from solidifying. One water bath for you and your laboratory partner will suffice. (Note in the materials

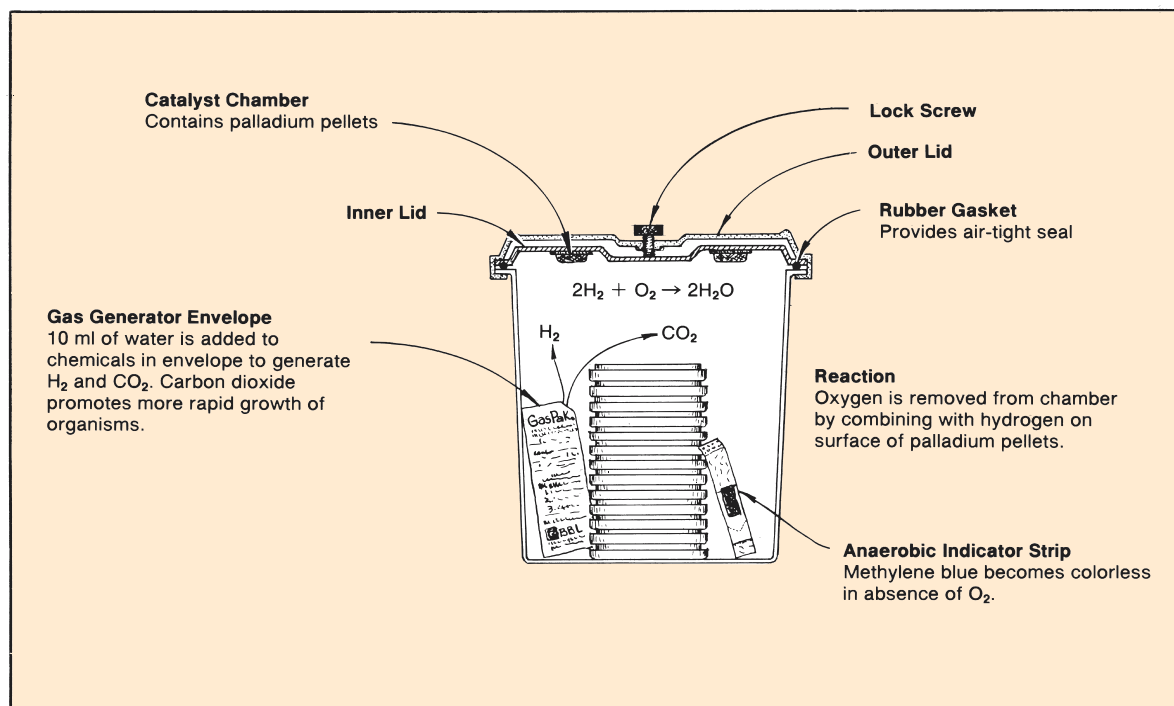


Figure 22.2 The GasPak anaerobic jar

Cultivation of Anaerobes • Exercise 22

list that the agar shakes have been liquefied for you prior to lab time.)

2. Label the six tubes with the organisms assigned to you (one organism per tube), your initials, and assignment number.

Note: Handle the tubes gently to avoid taking on any unwanted oxygen into the media. If the tubes of FTM are pink in the upper 30%, they must be boiled a few minutes to drive off the oxygen, then cooled to inoculate.

3. Heavily inoculate each of the TGYA shake tubes with several loopfuls of the appropriate organism for that tube. To get good dispersion of the organisms in the medium, roll each tube gently between the palms as shown in figure 22.3. To prevent oxygen uptake do not overly agitate the medium. Allow these tubes to solidify at room temperature.
4. Inoculate each of the FTM tubes with the appropriate organisms.
5. Streak your three organisms on the plate of anaerobic agar in the manner shown in figure 22.4. Note that only three straight-line streaks, well separated, are made. Place the Petri plate (inverted) in a cannister with the plates of other students that is to go into the GasPak jar.
6. Once all the students' plates are in cannisters, place the cannisters and tubes into the jar.
7. To activate and seal the GasPak jar, proceed as follows:
 - a. Peel apart the foil at one end of a GasPak indicator strip and pull it halfway down. The indicator will turn blue on exposure to the air. Place the indicator strip in the jar so that the wick is visible.
 - b. Cut off the corner of each of three GasPak gas generator envelopes with a pair of scissors. Place them in the jar in an upright position.

- c. Pipette 10 ml of tap or distilled water into the open corner of each envelope. Avoid forcing the pipette into the envelope.
- d. Place the inner section of the lid on the jar, making certain it is centered on top of the jar. Do not use grease or other sealant on the rim of the jar since the O-ring gasket provides an effective seal when pressed down on a clean surface.
- e. Unscrew the thumbscrew of the outer lid until the exposed end is completely withdrawn into the threaded hole. Unless this is done, it will be impossible to engage the lugs of the jar with the outer lid.
- f. Place the outer lid on the jar directly over the inner lid and rotate the lid slightly to allow it to drop in place. Now rotate the lid firmly to engage the lugs. The lid may be rotated in either direction.
- g. Tighten the thumbscrew by turning clockwise. If the outer lid raises up, the lugs are not properly engaged.
8. Place the jar in a 37° C incubator. After 2 or 3 hours check the jar to note if the indicator strip has lost its blue color. If decolorization has not occurred, replace the palladium pellets and repeat the entire process.
9. Incubate the tubes and plates for 24 to 48 hours.

SECOND PERIOD

(Culture Evaluations and Spore Staining)

Remove the lid from the GasPak jar. If vacuum holds the inner lid firmly in place, break the vacuum by sliding the lid to the edge. When transporting the plates

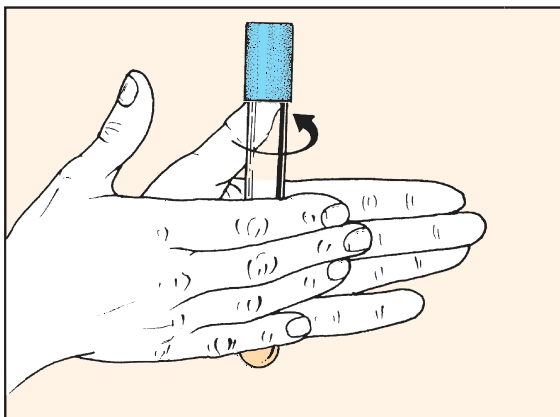


Figure 22.3 Organisms are dispersed in medium by rolling tube gently between palms.

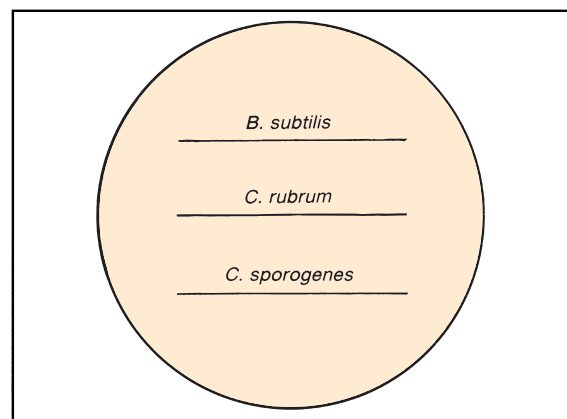


Figure 22.4 Three organisms are streaked on agar plate as straight-line streaks.

Exercise 22 • Cultivation of Anaerobes

and tubes to your desk *take care not to agitate the FTM tubes*. The position of growth in the medium can be easily changed if handled carelessly.

Materials:

tubes of FTM
shake tubes of TGYA
2 Brewer's anaerobic agar plates
spore-staining kits and slides

1. Compare the six FTM and TGYA shake tubes that you and your laboratory partner share with figure 22.5 to evaluate the oxygen needs of the six organisms.

2. Compare the growths (or lack of growth) on your Petri plate and the plate of your laboratory partner.
3. Record your results on the Laboratory Report.
4. If time permits, make a combined slide with three separate smears of the three spore-formers, using either one of the two spore-staining methods in Exercise 16. Draw the organisms in the circles provided on the Laboratory Report.

LABORATORY REPORT

Complete the Laboratory Report for this exercise.

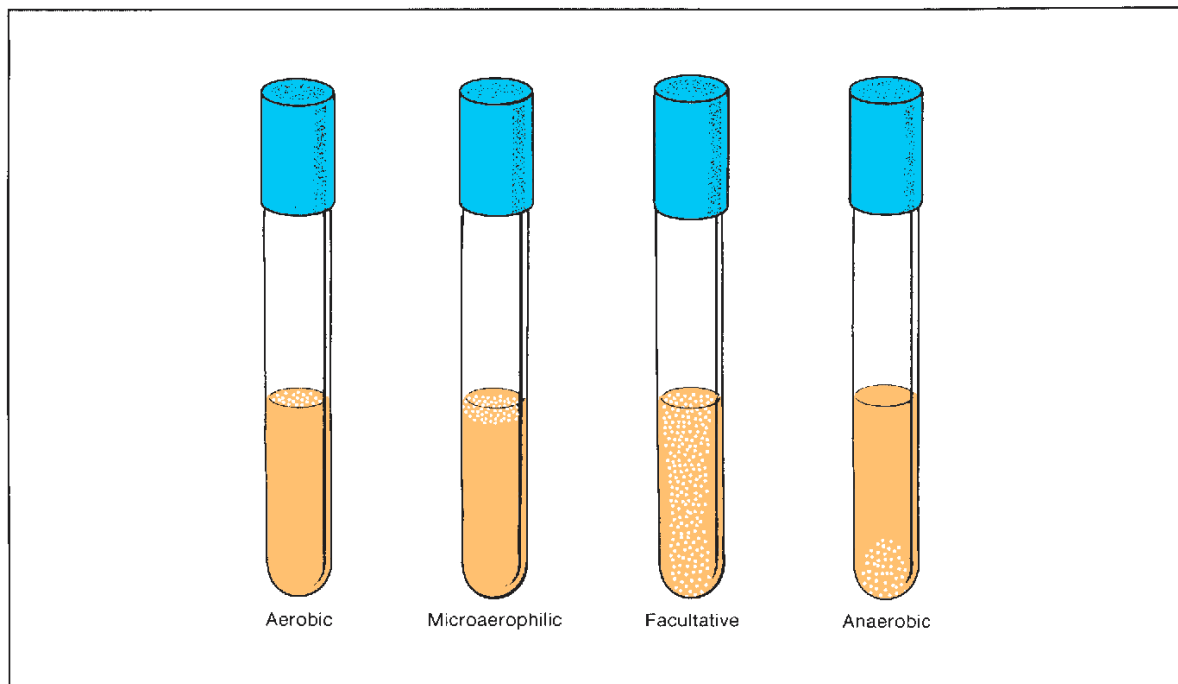


Figure 22.5 Growth patterns for different types of bacteria

Bacterial Population Counts

23

Many bacteriological studies require that we be able to determine the number of organisms that are present in a given unit of volume. Several different methods are available to us for such population counts. The method one uses is determined by the purpose of the study.

To get by with a minimum of equipment, it is possible to do a population count by diluting out the organisms and counting the organisms in a number of microscopic fields on a slide. Direct examination of milk samples with this technique can be performed very quickly, and the results obtained are quite reliable. A technique similar to this can be performed on a Petrof-Hauser counting chamber.

Bacterial counts of gas-forming bacteria can be made by inoculating a series of tubes of lactose broth and using statistical probability tables to estimate bacterial numbers. This method, which we will use in Exercise 63 to estimate numbers of coliform bacteria in water samples, is easy to use, works well in water testing, but is limited to water, milk, and food testing.

In this exercise we will use **quantitative plating** (Standard Plate Count, or SPC) and **turbidity measurements** to determine the number of bacteria in a culture sample. Although the two methods are somewhat parallel in the results they yield, there are distinct differences. For one thing, the SPC reveals information only as related to viable organisms; that is, colonies that are seen on the plates after incubation represent only living organisms, not dead ones. Turbidimetry results, on the other hand, reflect the presence of all organisms in a culture, dead and living.

QUANTITATIVE PLATING METHOD (Standard Plate Count)

In determining the number of organisms present in water, milk, and food, the **standard plate count** (SPC) is universally used. It is relatively easy to perform and gives excellent results. We can also use this basic technique to calculate the number of organisms in a bacterial culture. It is in this respect that this assignment is set up.

The procedure consists of diluting the organisms with a series of sterile water blanks as illustrated in

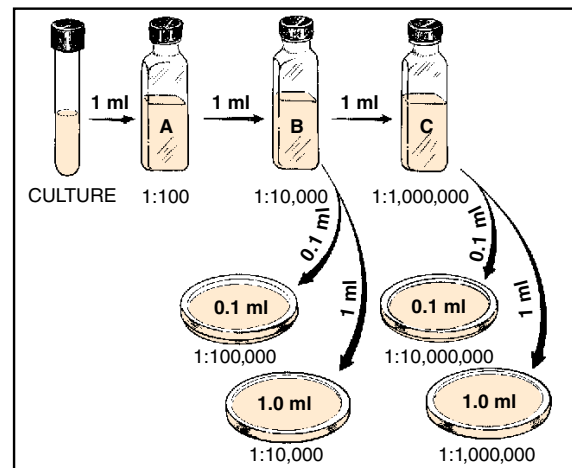


Figure 23.1 Quantitative plating procedure

figure 23.1. Generally, only three bottles are needed, but more could be used if necessary. By using the dilution procedure indicated here, a final dilution of 1:1,000,000 occurs in blank C. From blanks B and C, measured amounts of the diluted organisms are transferred into empty Petri plates. Nutrient agar, cooled to 50° C, is then poured into each plate. After the nutrient agar has solidified, the plates are incubated for 24 to 48 hours and examined. A plate that has between 30 and 300 colonies is selected for counting. From the count it is a simple matter to calculate the number of organisms per milliliter of the original culture. It should be pointed out that greater accuracy can be achieved by pouring two plates for each dilution and averaging the counts. Duplicate plating, however, has been avoided for obvious economic reasons.

Pipette Handling

Success in this experiment depends considerably on proper pipetting techniques. Pipettes may be available to you in metal cannisters or in individual envelopes; they may be disposable or reusable. In the distant past pipetting by mouth was routine practice. However, the hazards are obvious, and today it must be avoided. Your instructor will indicate the techniques that will

Exercise 23 • Bacterial Population Counts

prevail in this laboratory. If this is the first time that you have used sterile pipettes, consult figure 23.2, keeping the following points in mind:

- When removing a sterile pipette from a cannister, do so without contaminating the ends of the other pipettes with your fingers. This can be accomplished by *gently* moving the cannister from side to side in an attempt to isolate one pipette from the rest.
- After removing your pipette, replace the cover on the cannister to maintain sterility of the remaining pipettes.
- Don't touch the body of the pipette with your fingers or lay the pipette down on the table before or

after you use it. **Keep that pipette sterile** until you have used it, and don't contaminate the table or yourself with it after you have used it.

- Always use a mechanical pipetting device such as the one in illustration 3, figure 23.2. For safety reasons, deliveries by mouth are not acceptable in this laboratory.
- Remove and use only one pipette at a time; if you need 3 pipettes for the whole experiment and remove all 3 of them at once, there is no way that you will be able to keep 2 of them sterile while you are using the first one.
- When finished with a pipette, place it in the *discard cannister*. The discard cannister will have a

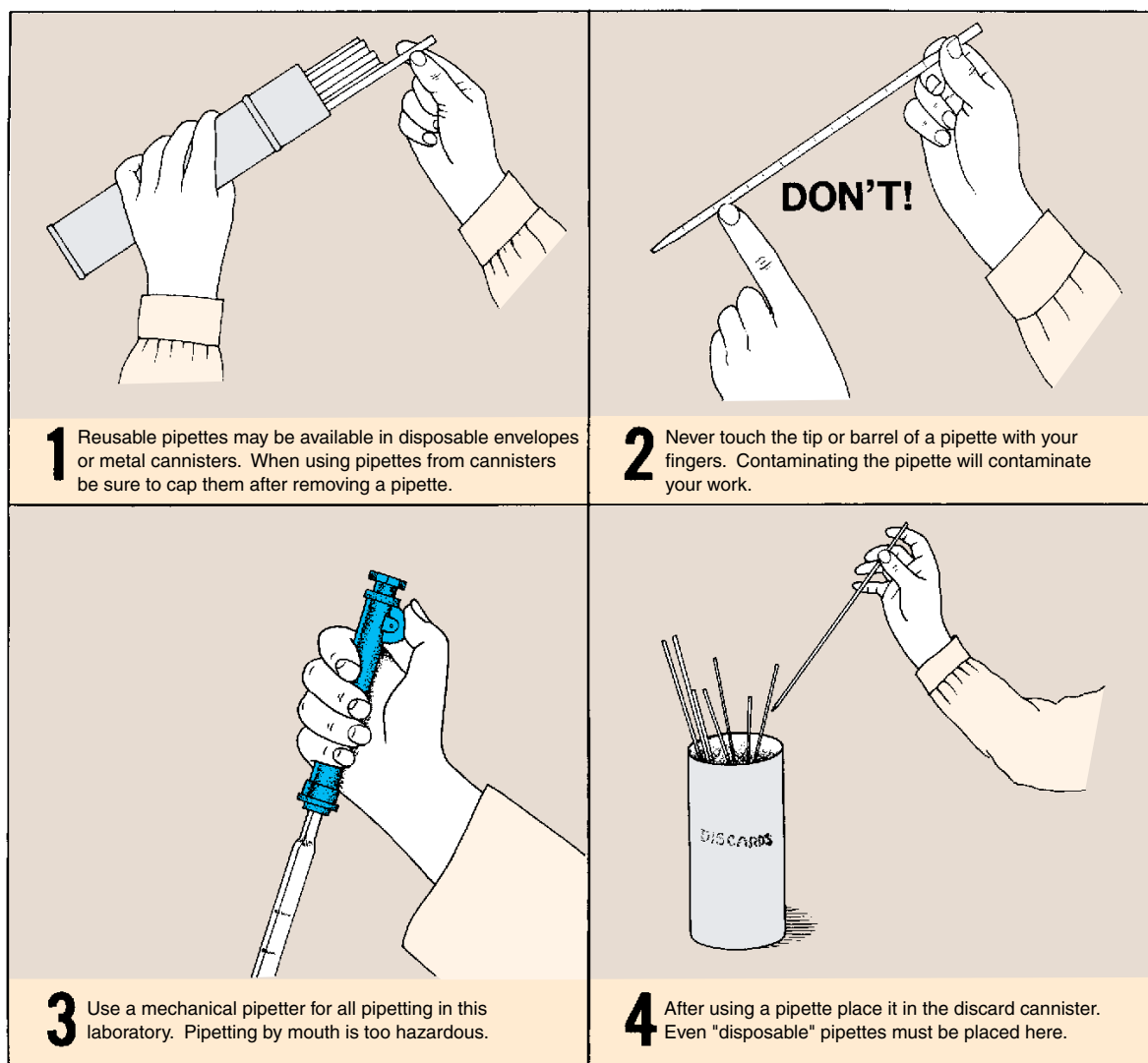


Figure 23.2 Pipette-handling techniques

disinfectant in it. At the end of the period reusable pipettes will be washed and sterilized by the laboratory assistant. Disposable pipettes will be discarded. Students have been known to absent-mindedly return used pipettes to the original sterile cannister, and, occasionally, even toss them into the wastebasket. We are certain that no one in this laboratory would *ever* do that!

Diluting and Plating Procedure

Proceed as follows to dilute out a culture of *E. coli* and pour four plates, as illustrated in figure 23.1.

Materials:

per 4 students:

- 1 bottle (40 ml) broth culture of *E. coli* per student:
- 1 bottle (80 ml) nutrient agar
- 4 Petri plates
- 1.1 ml pipettes
- 3 sterile 99 ml water blanks
- cannister for discarded pipettes

1. Liquefy a bottle of nutrient agar. While it is being heated, label three 99 ml sterile water blanks **A, B, and C**. Also, label the four Petri plates **1:10,000, 1:100,000, 1:1,000,000, and 1:10,000,000**. In addition, indicate with labels the amount to be pipetted into each plate (**0.1 ml** or **1.0 ml**).
2. Shake the culture of *E. coli* and transfer 1 ml of the organisms to blank A, using a sterile 1.1 ml pipette. After using the pipette, place it in the discard cannister.
3. Shake blank A 25 times in an arc of 1 foot for 7 seconds with your elbow on the table as shown in

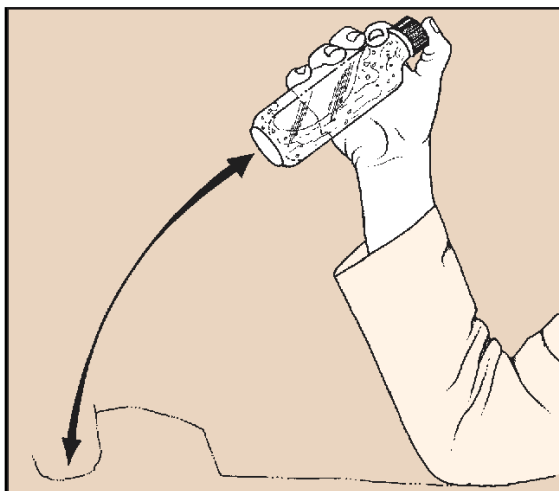


Figure 23.3 Standard procedure for shaking water blanks requires elbow to remain fixed on table

Bacterial Population Counts • Exercise 23

figure 23.3. Forceful shaking not only brings about good distribution, but it also breaks up clumps of bacteria.

4. With a different 1.1 ml pipette, transfer 1 ml from blank A to blank B.
5. Shake water blank B 25 times in same manner.
6. With another sterile pipette, transfer 0.1 ml from blank B to the 1:100,000 plate and 1.0 ml to the 1:10,000 plate. With the same pipette, transfer 1.0 ml to blank C.
7. Shake blank C 25 times.
8. With another sterile pipette, transfer from blank C 0.1 ml to the 1:10,000,000 plate and 1.0 ml to the 1:1,000,000 plate.
9. After the bottle of nutrient agar has boiled for 8 minutes, cool it down in a water bath at 50° C for **at least 10 minutes**.
10. Pour one-fourth of the nutrient agar (20 ml) into each of 4 plates. Rotate the plates **gently** to get adequate mixing of medium and organisms. **This step is critical!** Too little action will result in poor dispersion and too much action may slop inoculated medium over the edge.
11. After the medium has cooled completely, incubate at 35° C for 48 hours, inverted.

Counting and Calculations

Materials:

- 4 culture plates
- Quebec colony counter
- mechanical hand counter
- felt pen (optional)

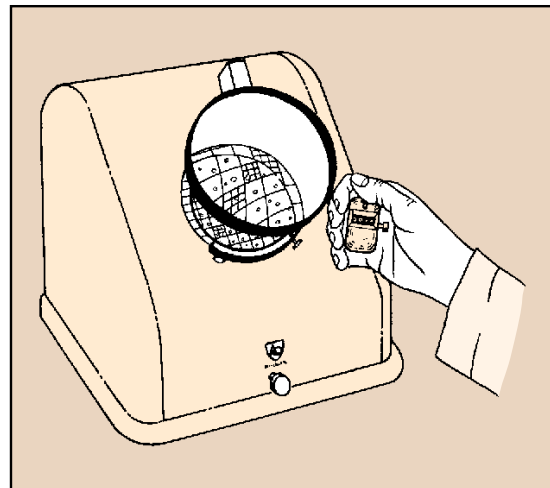


Figure 23.4 Colony counts are made on a Quebec counter, using a mechanical hand tally

Exercise 23 • Bacterial Population Counts

1. Lay out the plates on the table in order of dilution and compare them. *Select the plates that have no fewer than 30 nor more than 300 colonies for your count.* Plates with less than 30 or more than 300 colonies are statistically unreliable.
2. Place the plate on the Quebec colony counter with the lid removed. See figure 23.4. Start counting at the top of the plate, using the grid lines to prevent counting the same colony twice. Use a mechanical hand counter. Count every colony, regardless of how small or insignificant. Record counts on the table in section A of the Laboratory Report.

Alternative Counting Method: Another way to do the count is to remove the lid and place the plate upside down on the colony counter. Instead of using the grid to keep track, use a felt pen to mark off each colony as you do the count.

3. Calculate the number of bacteria per ml of undiluted culture using the data recorded in section A of the Laboratory Report. Multiply the number of colonies counted by the dilution factor (the reciprocal of the dilution).

Example: If you counted 220 colonies on the plate that received 1.0 ml of the 1:1,000,000 dilution: $220 \times 1,000,000$ (or 2.2×10^8) bacteria per ml. If 220 colonies were counted on the plate that received 0.1 ml of the 1:1,000,000 dilution, then the above results would be multiplied by 10 to convert from number of bacteria per 0.1 ml to number of bacteria per 1.0 ml (2,200,000,000, or 2.2×10^9).

Use only two significant figures. If the number of bacteria per ml was calculated to be 227,000,000, it should be recorded as 230,000,000, or 2.3×10^8 .

TURBIDIMETRY DETERMINATIONS

When it is necessary to make bacteriological counts on large numbers of cultures, the quantitative plate count method becomes a rather cumbersome tool. It not only takes a considerable amount of glassware and media, but it is also time-consuming. A much faster method is to measure the turbidity of the culture with a spectrophotometer and translate this into the number of organisms. To accomplish this, however, the plate count must be used to establish the count for one culture of known turbidity.

To understand how a spectrophotometer works, it is necessary, first, to recognize the fact that a culture of bacteria acts as a colloidal suspension, which will intercept the light as it passes through. Within certain limits the amount of light that is absorbed is directly proportional to the concentration of cells.

Figure 23.5 illustrates the path of light through a spectrophotometer. Note that a beam of white light passes through two lenses and an entrance slit into a diffraction grating that disperses the light into horizontal beams of all colors of the spectrum. Short wavelengths (violet and ultraviolet) are at one end and long wavelengths (red and infrared) are at the other end. The spectrum of light falls on a dark screen with a slit (exit slit) cut in it. Only that por-

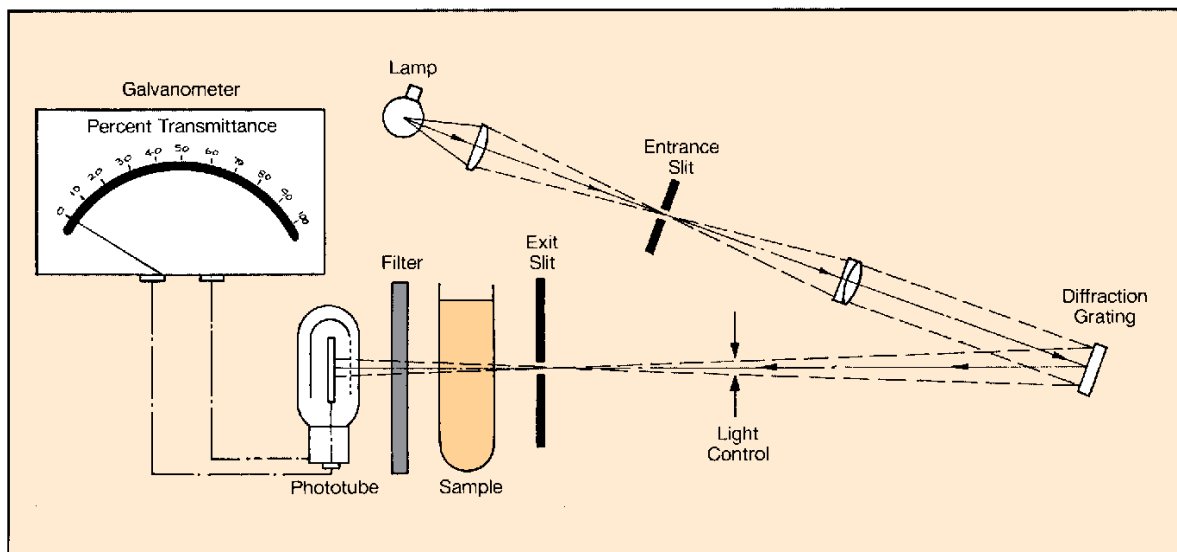


Figure 23.5 Schematic of a spectrophotometer

Bacterial Population Counts • Exercise 23

tion of the spectrum that happens to fall on the slit goes through into the sample. It will be a monochromatic beam of light. By turning a wavelength control knob on the instrument, the diffraction grating can be reoriented to allow different wavelengths to pass through the slit. The light that passes through the culture activates a phototube, which, in turn, registers **percent transmittance (% T)** on a galvanometer. The higher the percent transmittance, the fewer are the cells in suspension.

There should be a direct proportional relationship between the concentration of bacterial cells and the absorbance (optical density, O.D.) of the culture. To demonstrate this principle, you will measure the %T of various dilutions of the culture provided to you. These values will be converted to O.D. and plotted on a graph as a function of culture dilution. You may find that there is a linear relationship between concentration of cells and O.D. only up to a certain O.D. At higher O.D. values the relationship may not be linear. That is, for a doubling in cell concentration, there may be less than a doubling in O.D.

Materials:

- broth culture of *E. coli* (same one as used for plate count)
 - spectrophotometer cuvettes (2 per student)
 - 4 small test tubes and test-tube rack
 - 5 ml pipettes
 - bottle of sterile nutrient broth (20 ml per student)
1. Calibrate the spectrophotometer, using the procedure described in figure 23.7. These instructions are specifically for the Bausch and Lomb Spectronic 20. In handling the cuvettes, keep the following points in mind:
 - a. Rinse the cuvette several times with distilled water to get it clean before using.
 - b. Keep the lower part of the cuvette spotlessly clean by keeping it free of liquids, smudges, and fingerprints. Wipe it clean with Kimwipes or some other lint-free tissue. Don't wipe the cuvettes with towels or handkerchiefs.
 - c. Insert the cuvette into the sample holder with its index line registered with the index line on the holder.
 - d. After the cuvette is seated, line up the index lines exactly.
 - e. Handle these tubes with great care. They are expensive.
 2. Label a cuvette 1:1 (near top of tube) and four test tubes 1:2, 1:4, 1:8, and 1:16. These tubes will be used for the serial dilutions shown in figure 23.6.

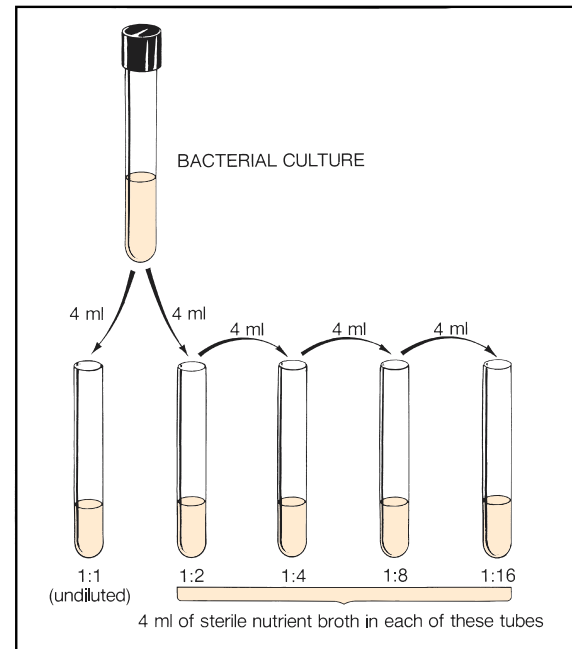


Figure 23.6 Dilution procedure for cuvettes

3. With a 5 ml pipette, dispense 4 ml of sterile nutrient broth into tubes 1:2, 1:4, 1:8, and 1:16.
4. Shake the culture of *E. coli* vigorously to suspend the organisms, and with the same 5 ml pipette, transfer 4 ml to the 1:1 cuvette and 4 ml to the 1:2 test tube.
5. Mix the contents in the 1:2 tube by drawing the mixture up into the pipette and discharging it into the tube three times.
6. Transfer 4 ml from the 1:2 tube to the 1:4 tube, mix three times, and go on to the other tubes in a similar manner. Tube 1:16 will have 8 ml of diluted organisms.
7. Measure the percent transmittance of each of the five tubes, starting with the 1:16 tube first. The contents of each of the test tubes must be transferred to a cuvette for measurement. Be sure to close the lid on the sample holder when making measurements. A single cuvette can be used for all the measurements.
8. Convert the percent transmittance values to optical density (O.D.) using the following formula:

$$\text{O.D.} = 2 - \log \text{ of percent transmittance}$$

Example: If the percent transmittance of one of your dilutions is 53.5, you would solve the problem in this way:

Exercise 23 • Bacterial Population Counts

$$\begin{aligned}\text{O.D.} &= 2 - \log \text{ of } 53.5 \\ &= 2 - 1.7284 \\ &= 0.272\end{aligned}$$

Table II of Appendix A is a log table. Of course, if you have a calculator, all this is much simpler.

Logarithm Refresher In case you have forgotten how to use logarithms, recall these facts:

Mantissa: The value you find in the log table (0.7284 in the above example) is the mantissa.

Characteristic: The number to the left of the decimal (1 in the example) is the characteristic.

This figure (the characteristic) is always one number less than the number of digits of the figure you are looking up.

Examples:

<i>number</i>	<i>characteristic</i>	<i>mantissa</i>
5.31	0	.7251
531	2	.7251

Although the galvanometer may show absorbance (O.D.) values, greater accuracy will result from calculating them from percent transmittance.

9. Record the O.D. values in the table of the Laboratory Report.
10. Plot the O.D. values on the graph of the Laboratory Report.

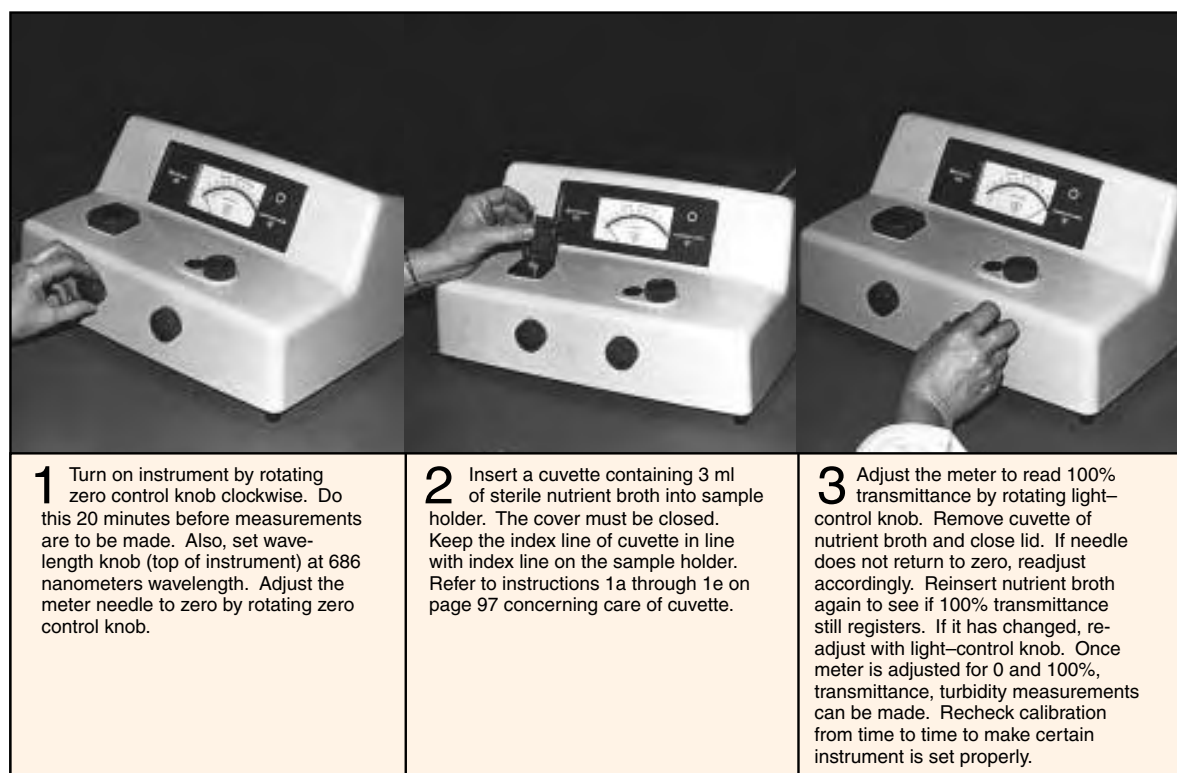


Figure 23.7 Calibration procedure for the B & L Spectronic 2 on page 97

Slide Culture: Autotrophs

24

There is probably no single medium or method that one can use to do a comprehensive population count of all living microorganisms in a specific biosphere. Those media that we categorize as being “general purpose” will, for various reasons, inhibit the growth of many organisms. To make comparative studies of free-living organisms in freshwater lakes, A. T. Henrici, in 1932, devised an immersed slide technique that revealed the presence of many organisms that did not show up by other methods. Although his original concern was with algal populations, the technique worked as well for bacteria and other microorganisms.

His method consists of suspending glass microscope slides in the body of water for a specified period of time. Microorganisms in the water adhere to the glass and multiply to form small colonies that are observable under the microscope. Although there is no guarantee that the organisms growing on the glass are autotrophs, many of them are.

Materials:

- adhesive tape (½" width)
- 2 microscope slides
- copper wire
- gummed labels
- acid-alcohol

1. Clean 2 microscope slides as follows:
 - a. Scrub with green soap or Bon Ami.
 - b. Dip them in acid-alcohol for 1 minute and dry with tissue.
 - c. Place them in a beaker of distilled water for 5 minutes to allow any residual solvent to dissipate.
2. Tape a piece of copper wire to one edge as illustrated in figure 24.1. Hold the slides back to back by their edges. Do not touch the flat surfaces with your fingers. Wrap all four edges with tape. For identification, attach a gummed label with your name to the wire.
3. Suspend the slide in an aquarium or container of water that is known to have a stabilized natural flora of bacteria.

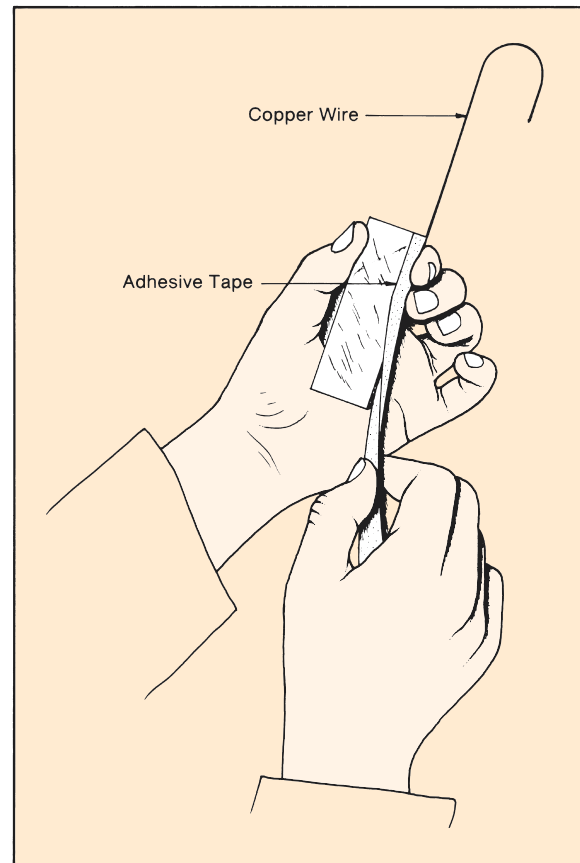


Figure 24.1 Preparation of slide for immersion

4. After 1 week remove the binding from the slides. Prepare one slide with Gram's stain and place a drop of water and cover glass on the other one.
5. Examine both slides under oil immersion and record your observations on the first portion of Laboratory Report 24, 25.

Reference: Henrici, A. T. 1933. Studies of fresh water bacteria. *J. Bact.* 25 (3): 277–286.

25

Slime Mold Culture

The classification system proposed by Alexopoulos and Mims places the slime molds in Division Gymnomycota of the Kingdom Myceteae. These heterotrophic microorganisms exist in cool, shady, moist places in the woods—on decaying logs, dead leaves, and other organic matter. Unlike the holophytic bacteria and other Myceteae, they ingest their food in a manner similar to the amoebas; that is, they are phagotrophic. In the vegetative stages, these microorganisms are unlike the other Myceteae in that the cells lack cell walls; when fruiting bodies are formed, however, cell walls are present.

The categorization of slime molds as protozoans or as fungi has always been problematical. Certainly, they are intermediate in that they have characteristics of both groups.

Figure 25.1 illustrates the life cycle of one type of slime mold, the plasmodial type. The genus *Physarum* is the one to be studied in this experiment. The assimilative stage of this organism is the **plasmodium**. This multinucleate structure is slimy in appearance and moves slowly by flowing its cytoplasm in amoeboid fashion over surfaces on which it feeds. Most species

feed on bacteria and possibly on other small organisms that they encounter.

Plasmodial growth continues as long as adequate food supply and moisture are available. Eventually, however, environmental changes may result in the formation of sclerotia or sporangia. A **sclerotium** is a hardened mass of irregular shape that forms from the plasmodium when moisture and temperature conditions become less than ideal. When conditions improve, the sclerotium reverts back to a plasmodium. Figure 25.2 is a photograph of two sclerotia that formed on a laboratory culture. **Sporangia** are fructifications that form under conditions similar to those required for sclerotia. Exactly why sporangia form instead of sclerotia is still not clearly understood. Sporangia form by the separation of the plasmodium into many rounded mounds of protoplasm that extend upward on stalks. The nuclei within the sporangia undergo meiosis to become haploid spores with tough cell walls. The sclerotia and sporangia of figures 25.2 and 25.3 were photographed on the same culture of laboratory-grown *Physarum*.

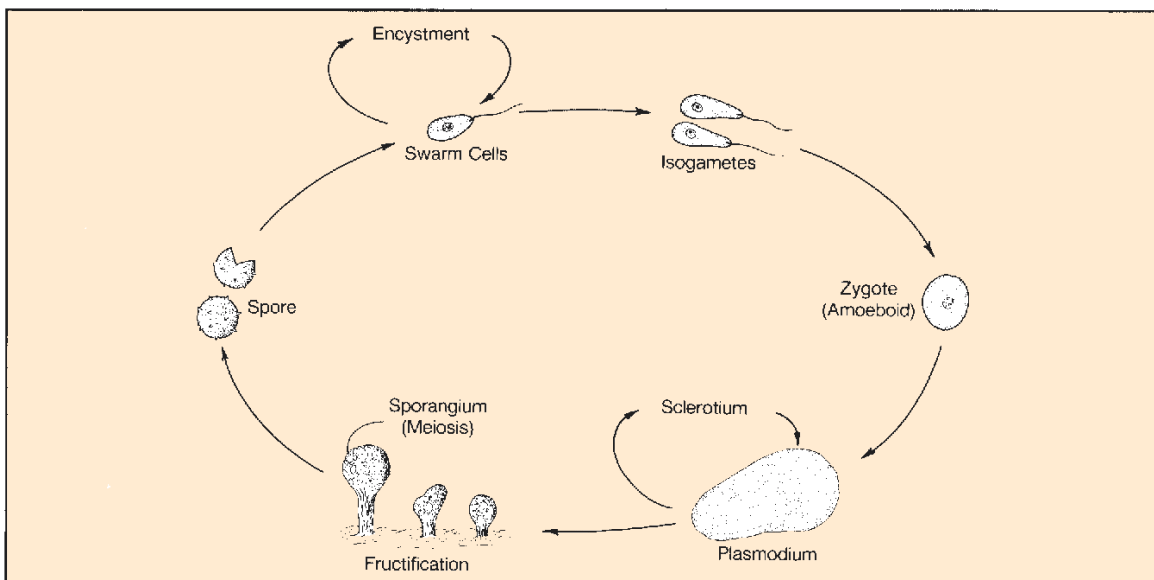


Figure 25.1 Life cycle of *Physarum polycephalum*

Slime Mold Culture • Exercise 25

Both sclerotia and spores may survive adverse environmental conditions for long periods of time. Once environmental conditions improve, the spores germinate to produce flagellated pear-shaped **swarm cells**. These swarm cells may do one of three things: (1) they may encyst if conditions suddenly become adverse, (2) they may divide one or more times to form isogametes, or (3) they may act as isogametes and unite directly to form a **zygote**. Once a zygote is formed, it takes on an amoeboid form and undergoes a series of mitotic divisions to produce a plasmodium. This completes the life cycle.

Three procedures will be described here for the study of *Physarum polycephalum*: (1) moist chamber culture, (2) agar culture method, and (3) spore germination technique. The techniques used will be determined by the availability of time and materials.

MOIST CHAMBER CULTURE

To grow large numbers of plasmodia, sclerotia, and sporangia that can be used for an entire class, one needs to create a rather large moisture chamber. Any covered glass or plastic container that is 10 to 12 inches square or round is suitable.

Materials:

sclerotia of *Physarum polycephalum*
container for culture (10½" dia Pyrex casserole
dish with cover or 10–12" square plastic box
with cover)
glass Petri dish cover
sharp scalpel
rolled oat flakes (long-cooking type)
10" dia filter paper or paper toweling

1. In the center of the container place a Petri dish cover, open end down. Lay a large piece of filter paper or paper toweling over the Petri dish and saturate with distilled water. The Petri dish pro-

vides a raised area above any excess water that may make the paper too wet.

2. With a sharp scalpel transfer a small fragment of sclerotium from the *Physarum* culture to the filter paper. A sclerotium may vary from dark orange to brown in color. See figure 25.2. Moisten the sclerotium with a drop of distilled water.
3. After a few hours the organism will be awakened to activity and begin to seek food. At this point, place a flake of rolled oats near the edge of the spreading growth for it to feed on.
4. Incubate the moist chamber in a dark place at room temperature. Add moisture (distilled water) and oat flakes periodically as needed. It is better to add a few fresh flakes daily than to overfeed by applying all flakes at once. Such a culture should keep for several weeks. To promote the formation of sclerotia, allow some of the water to evaporate away by leaving the lid partially open for a while. To bring about sporangia formation, withhold food while keeping the culture moist.

AGAR CULTURE METHOD (Plasmodial Study)

An actively metabolizing plasmodium is dark yellow and streaked with vessels. The streaming of protoplasm in these vessels is best observed under the microscope. To be able to study this unique structure, it is best to culture the organism on non-nutrient agar. Make such a culture as follows:

Materials:

rolled oat flakes
scalpel
Petri plate with 15 ml of nonsterile,
non-nutrient agar



Figure 25.2 Sclerotia of *Physarum polycephalum* (3X)



Figure 25.3 Sporangia of *Physarum polycephalum* (20X)

Exercise 25 • Slime Mold Culture

1. Lift some occupied oat flakes from the filter paper in the moist chamber and transfer to a plate of nonsterile, non-nutrient 1.5% agar. Maintain this culture by adding fresh oat flakes periodically, but don't add water.
2. After a well-developed plasmodium has formed, study the streaming protoplasm under low power of the microscope. Observation is made by transmitted light through the agar on the microscope stage. Look for periodical reversal of direction of flow.
3. Cut one of the vessels through in which the flow is active and observe the effect.
4. Transfer a piece of plasmodium to another part of the medium and watch it reconstitute itself.
5. Leave the cover slightly open on the Petri dish for several days and note any changes that might occur as time goes by.

SPORE GERMINATION

The observation of spore germination can be achieved with a hanging drop slide. Once sporangia are in abundance, one can make such a slide as follows:

Materials:

depression slides (sterile)
plain microscope slides (sterile)
cover glasses (sterile)

Vaseline
toothpicks
sporangia of *Physarum polycephalum*
Bunsen burner
70% alcohol

1. With a toothpick, place a small amount of Vaseline near each corner of the cover glass. (See figure 19.1, page 73.)
2. Saturate a sporangium with a drop of 70% alcohol on the center of a sterile plain microscope slide.
3. As soon as the alcohol has evaporated, add a drop of distilled water and place another sterile slide over the wet sporangium.
4. Crush the sporangium with thumb pressure on the upper slide. Separate the two slides to expose the crushed sporangium.
5. Transfer a few loopfuls of crushed sporangial material to a drop of distilled water on a sterile cover glass.
6. Place the depression slide over the cover glass, make contact, and quickly invert to produce a completed hanging drop slide.
7. Examine under low and high power.

LABORATORY REPORT

Complete all the answers on Laboratory Report 24, 25 that pertain to this exercise.

Slide Culture: Molds

26

The isolation, culture, and microscopic examination of molds require the use of suitable selective media and special microscopic slide techniques. If simple wet mount slides of molds were attempted in Exercise 10, it became apparent that wet mount slides made from mold colonies usually don't reveal the arrangement of spores that is so necessary in identification. The process of merely transferring hyphae to a slide breaks up the hyphae and sporangiophores in such a way that identification becomes very difficult. In this exercise a slide culture method will be used to prepare stained slides of molds. The method is superior to wet mounts in that the hyphae, sporangiophores, and spores remain more or less intact when stained.

When molds are collected from the environment, as in Exercise 10, Sabouraud's agar is most frequently used. It is a simple medium consisting of 1% peptone, 4% glucose, and 2% agar-agar. The pH of the medium is adjusted to 5.6 to inhibit bacterial growth.

Unfortunately, for some molds the pH of Sabouraud's agar is too low and the glucose content is too high. A better medium for these organisms is one suggested by C. W. Emmons that contains only 2% glucose, with 1% neopeptone, and an adjusted pH of

6.8–7.0. To inhibit bacterial growth, 40 mg of chloramphenicol is added to one liter of the medium.

In addition to the above two media, cornmeal agar, Czapek solution agar, and others are available for special applications in culturing molds.

Figure 26.2 illustrates the procedure that will be used to produce a mold culture on a slide that can be stained directly on the slide. Note that a sterile cube of Sabouraud's agar is inoculated on two sides with spores from a mold colony. Figure 26.1 illustrates how the cube is held with a scalpel blade as inoculation takes place. The cube is placed in the center of a microscope slide with one of the inoculated surfaces placed against the slide. On the other inoculated surface of the cube is placed a cover glass. The assembled slide is incubated at room temperature for 48 hours in a moist chamber (Petri dish with a small amount of water). After incubation the cube of medium is carefully separated from the slide and discarded.

During incubation the mold will grow over the glass surfaces of the slide and cover glass. By adding a little stain to the slide a semipermanent slide can be made by placing a cover glass over it. The cover glass

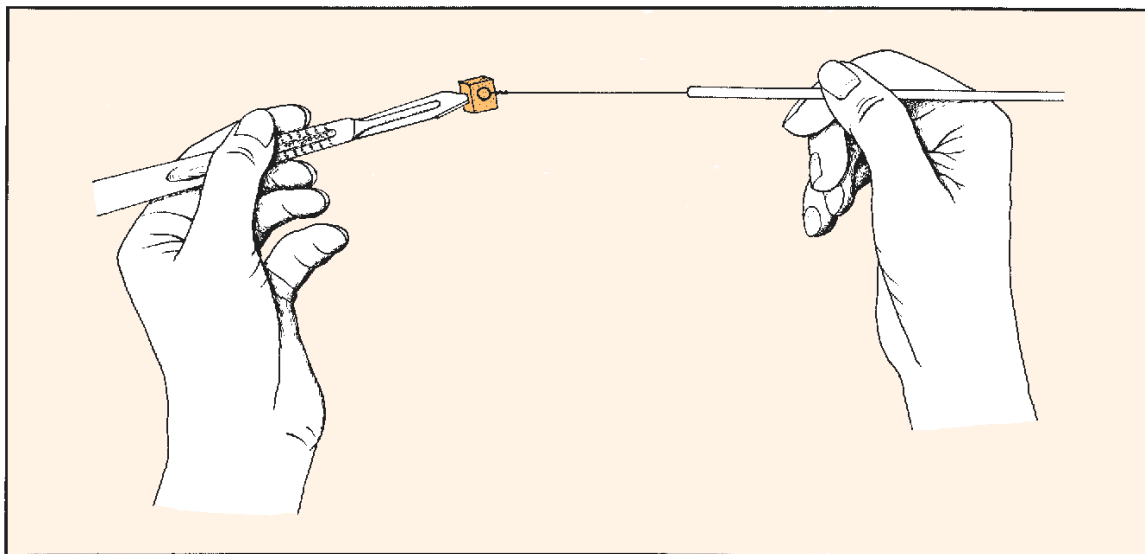


Figure 26.1 Inoculation technique

Exercise 26 • Slide Culture: Molds

can also be used to make another slide by placing it on another clean slide with a drop of stain on it. Before the stain (lactophenol cotton blue) is used, it is desirable to add to the hyphae a drop of alcohol, which acts as a wetting agent.

FIRST PERIOD (Slide Culture Preparation)

Proceed as follows to make slide cultures of one or more mold colonies.

Materials:

Petri dishes, glass, sterile
filter paper (9 cm dia, sterile)
glass U-shaped rods
mold culture plate (mixture)
1 Petri plate of Sabouraud's agar or Emmons'
medium per 4 students
scalpels
inoculating loop
sterile water
microscope slides and cover glasses (sterile)
forceps

1. Aseptically, with a pair of forceps, place a sheet of sterile filter paper in a Petri dish.
2. Place a sterile U-shaped glass rod on the filter paper. (Rod can be sterilized by flaming, if held by forceps.)
3. Pour enough sterile water (about 4 ml) on filter paper to completely moisten it.
4. With forceps, place a sterile slide on the U-shaped rod.
5. *Gently* flame a scalpel to sterilize, and cut a 5 mm square block of the medium from the plate of Sabouraud's agar or Emmons' medium.
6. Pick up the block of agar by inserting the scalpel into one side as illustrated in figure 26.1. Inoculate both top and bottom surfaces of the cube with spores from the mold colony. Be sure to flame and cool the loop prior to picking up spores.
7. Place the inoculated block of agar in the center of a microscope slide. Be sure to place one of the inoculated surfaces down.
8. Aseptically, place a sterile cover glass on the upper inoculated surface of the agar cube.

9. Place the cover on the Petri dish and incubate at room temperature for 48 hours.
10. After 48 hours examine the slide under low power. If growth has occurred you should see hyphae and spores. If growth is inadequate and spores are not evident, allow the mold to grow another 24–48 hours before making the stained slides.

SECOND PERIOD (Application of Stain)

As soon as there is evidence of spores on the slide, prepare two stained slides from the slide culture, using the following procedure:

Materials:

microscope slides and cover glasses
95% ethanol
lactophenol cotton blue stain
forceps

1. Place a drop of lactophenol cotton blue stain on a clean microscope slide.
2. Remove the cover glass from the slide culture and discard the block of agar.
3. Add a drop of 95% ethanol to the hyphae on the cover glass. As soon as most of the alcohol has evaporated place the cover glass, mold side down, on the drop of lactophenol cotton blue stain on the slide. This slide is ready for examination.
4. Remove the slide from the Petri dish, add a drop of 95% ethanol to the hyphae and follow this up with a drop of lactophenol cotton blue stain. Cover the entire preparation with a clean cover glass.
5. Compare both stained slides under the microscope; one slide may be better than the other one.

LABORATORY REPORT

There is no Laboratory Report for this exercise.

Slide Culture: Molds • Exercise 26

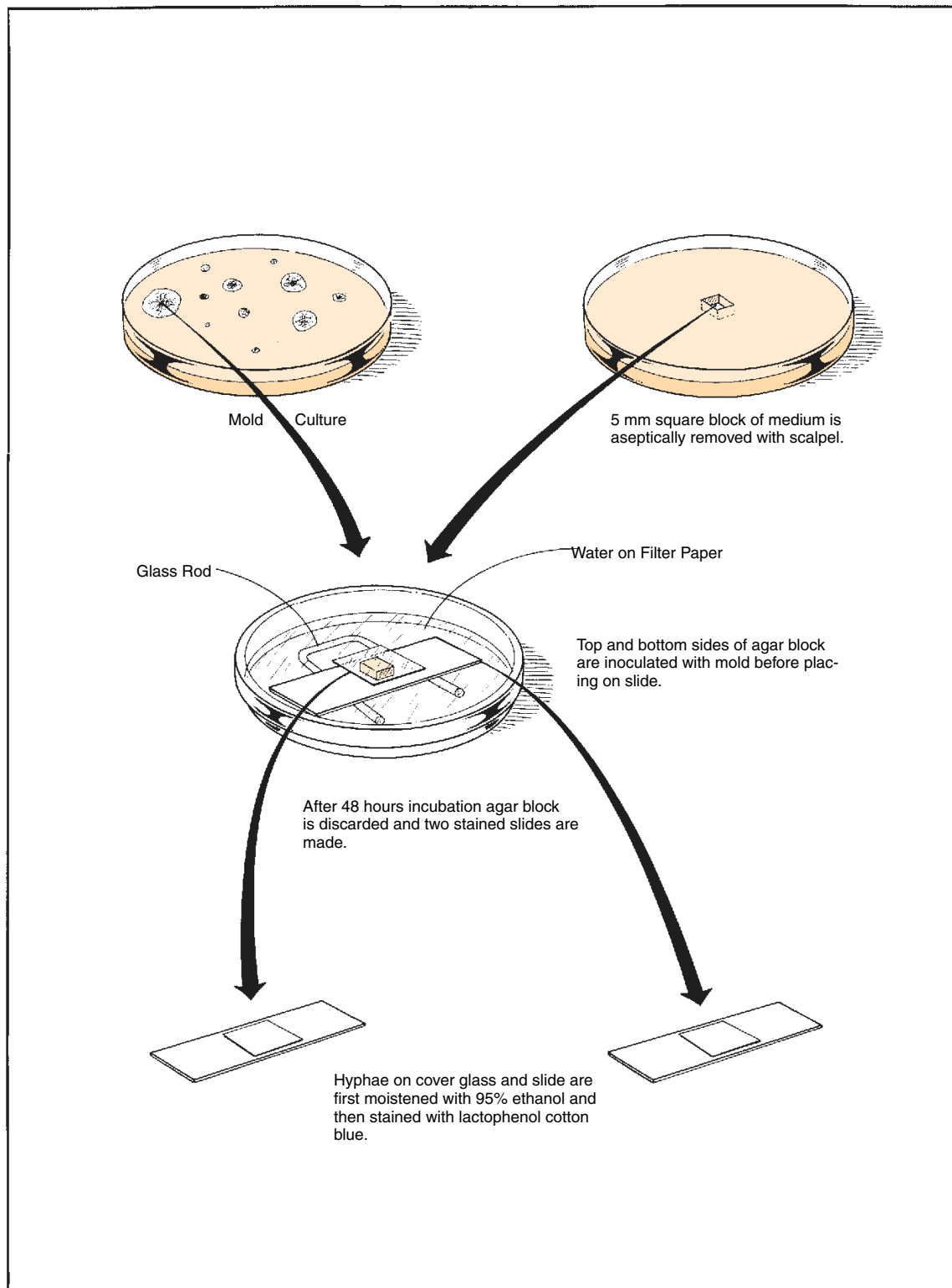


Figure 26.2 Procedure for making two stained slides from slide culture

27 Isolation of Anaerobic Phototrophic Bacteria: Using the Winogradsky Column

The culture of photosynthetic bacteria requires special culture methods to promote their growth. These prokaryotes contain photopigments, such as chlorophyll and carotenoids, which convert solar energy into cellular constituents. There are two groups of phototrophic bacteria: (1) the aerobic phototrophic **cyanobacteria**, which we studied in Exercise 6, and (2) the **anaerobic** phototrophic bacteria, which include the **purple** and **green bacteria**. It is this latter group that will be studied in this exercise.

As pointed out in Exercise 6, the cyanobacteria contain chlorophyll *a*, carotenoids, and phycobilisomes. The nonchlorophyll pigments in this group are accessory pigments for capturing light. They resemble higher plants in that they split water for a source of reducing power and evolve oxygen in the process.

The anaerobic phototrophic bacteria, on the other hand, differ in that they contain bacteriochlorophyll, which is chemically distinct from chlorophyll. Instead of utilizing water as a source of reducing power, the purple and green bacteria use sulfide or organic acids for the reduction of carbon dioxide. The purple bacteria that utilize organic acids instead of sulfide are essentially photoheterotrophic since they derive their carbon from organic acids rather than carbon dioxide.

These bacteria are ubiquitous in the ooze sediment of ditches, ponds, and lakes: i.e., mostly everywhere that freshwater lies relatively stagnant for long periods of time and subject to sunlight. In this environment, fermentation processes produce the sulfides and organic acids that are essential to their existence.

CHARACTERIZATION

According to *Bergey's Manual* (Section 18, Vol. 3), there are approximately 30 genera of anaerobic phototrophic bacteria. The purple bacteria belong to the family *Chromatiaceae*. The green ones are in the family *Chlorobiaceae*. The morphological, cultural, and physiological differences between the purple and green sulfur bacteria are as follows:

Purple Sulfur Bacteria Members of this group are all gram-negative, straight or slightly curved rods that are motile with polar flagellation. Colors of the various genera vary considerably—from orange-brown to

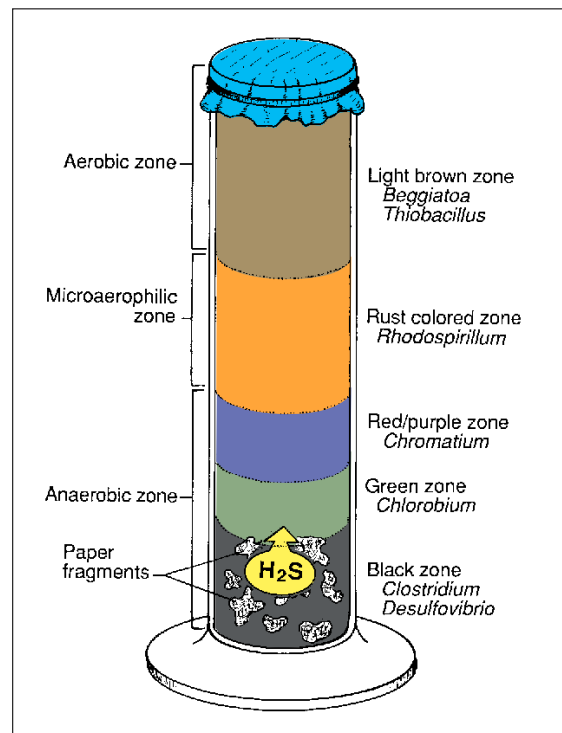


Figure 27.1 Winogradsky's Column

brown, brownish-red to pink, and purple-red to purple-violet. Color variability is due to the blend of bacteriochlorophyll with the type of carotenoid present. All species contain elemental-sulfur internally in the form of globules. Some species are able to fix nitrogen. Sulfides are required as electron donors; bicarbonate, acetate, and pyrovate are also required. They cannot utilize thiosulfate, sugars, alcohols, amino acids, or benzoates.

Green Sulfur Bacteria All of these bacteria are gram-negative, spherical to straight, or curved rods. Arrangement of cells may be in chains like streptococci. Some are motile by gliding, others are non-motile. Color may be grass-green or brown. *Sulfur by-product is excreted, not retained* in cells. Some are

Isolation of Anaerobic Phototrophic Bacteria: Using the Winogradsky Column • Exercise 27

able to utilize thiosulfate. Many are mixotrophic in that they can photoassimilate simple organic compounds in the presence of sulfide and bicarbonate.

WINOGRADSKY'S COLUMN

To create a small ecosystem that is suitable for the growth of these bacteria, one can set up a *Winogradsky column* as illustrated in figure 27.1. Sergii Winogradsky, a Russian microbiologist, developed this culture technique to study the bacteria that are involved in the sulfur cycle. From his studies he defined the chemoautotrophic bacteria.

This setup consists of a large test tube or graduated cylinder that is packed with pond ooze, sulfate, carbonate, and some source of cellulose (shredded paper or cellulose powder). It is incubated for a period of time (up to 8 weeks) while being exposed to incandescent light. Note that different layers of microorganisms develop, much in the same manner that is found in nature.

Observe that in the bottom of the column the cellulose is degraded to fermentation products by *Clostridium*. The fermentation products and sulfate are then acted upon by other bacteria (*Desulfovibrio*) to produce hydrogen sulfide, which diffuses upward toward the oxygenated zone, creating a stable hydrogen sulfide gradient. Note, also, that the *Chlorobium* species produce an olive-green zone deep in the column. A red to purple zone is produced by *Chromatium* a little farther up. Ascending the column farther where the oxygen gradient increases, other phototrophic bacteria such as *Rhodospirillum*, *Beggiatoa*, and *Thiobacillus* will flourish.

Once the column has matured, one can make subcultures from the different layers, using an enrichment medium. The subcultures can be used for making slides to study the morphological characteristics of the various types of organisms. Figure 27.2 illustrates the overall procedure to be used for subculturing. Proceed as follows:

FIRST PERIOD

You will set up your Winogradsky column in a 100 ml glass graduate. It will be filled with mud, sulfate, water, phosphate, carbonate, and a source of fermentable cellulose. The cellulose, in this case, will be in the form of a shredded paper slurry.

The column will be covered completely at first with aluminum foil to prevent the overgrowth of amoeba and then later uncovered and illuminated with incandescent light to promote the growth of phototrophic bacteria. The column will be examined at 2-week intervals to look for the development of

different-colored layers. Once distinct colored layers develop, subcultures will be made to tubes of enrichment medium with a pipette. The subcultures will be incubated at room temperature with exposure to incandescent light and examined periodically for color changes. Figure 27.2 illustrates the subculturing steps.

Materials:

graduated cylinder (100 ml size)
cellulose source (cellulose powder, newspaper, or filter paper)
calcium sulfate, calcium carbonate, dipotassium phosphate
mud from various sources (freshly collected)
water from ponds (freshly collected)
beaker (100 ml size)
glass stirring rod
aluminum foil
rubber bands
incandescent lamp (60–75 watt)

1. Using cellulose powder or some form of paper, prepare a thick slurry with water in a beaker. If you are using paper, tear the paper up into small pieces and macerate it in a small volume of water with a glass rod. If you are using cellulose powder, start with 1–2 g of powder in a small amount of water. The slurry should be thick but not a paste.
2. Fill the cylinder with the slurry until it is one-third full.
3. To 200 g of mud, add 1.64 g of calcium sulfate and 1.3 g each of calcium carbonate and dipotassium phosphate. Keep a record of the source of the mud you are using.
4. Add some “self water” (pond water collected with the mud) to the mud and chemical mixture and mix the ingredients well.
5. Pour the mud mixture into the cylinder on top of the cellulose slurry.
6. With a glass rod, gently mix and pack the contents of the cylinder. As packing occurs, you may find that you need to add more “self water” to bring the level up to two-thirds or three-fourths of the graduate. Make sure all trapped air bubbles are released.
7. Top off the cylinder by adding pond water until the graduate is 90% full.
8. Cap the cylinder with foil, using a rubber band to secure the cover.
9. Record on the Laboratory Report the initial appearance of the cylinder.
10. Wrap the sides of the cylinder completely with aluminum foil to exclude light.
11. Incubate the cylinder at room temperature for one and a half to two weeks.

Exercise 27 • Isolation of Anaerobic Phototrophic Bacteria: Using the Winogradsky Column

TWO WEEKS LATER

Remove the aluminum foil from the sides of the cylinder. Note the color of the mud, particularly in the bottom. Its black appearance will indicate sulfur respiration with the formation of sulfides by *Desulfovibrio* and other related bacteria. Record the color differences of different layers and the overall appearance of the entire cylinder on the Laboratory Report.

Place a lamp with a 75 watt bulb within a few inches of the cylinder and continue to incubate the cylinder at room temperature.

SUBSEQUENT EXAMINATIONS

Examine the cylinder periodically at each laboratory period, looking for the color changes that might occur. The presence of green, purple, red, or brown areas on the surface of the mud should indicate the presence of blooms of anaerobic phototrophic bacterial growth. Record your results on the Laboratory Report.

SUBCULTURING

After 6 to 8 weeks, make several subcultures from your Winogradsky column following the procedure shown in figure 27.2.

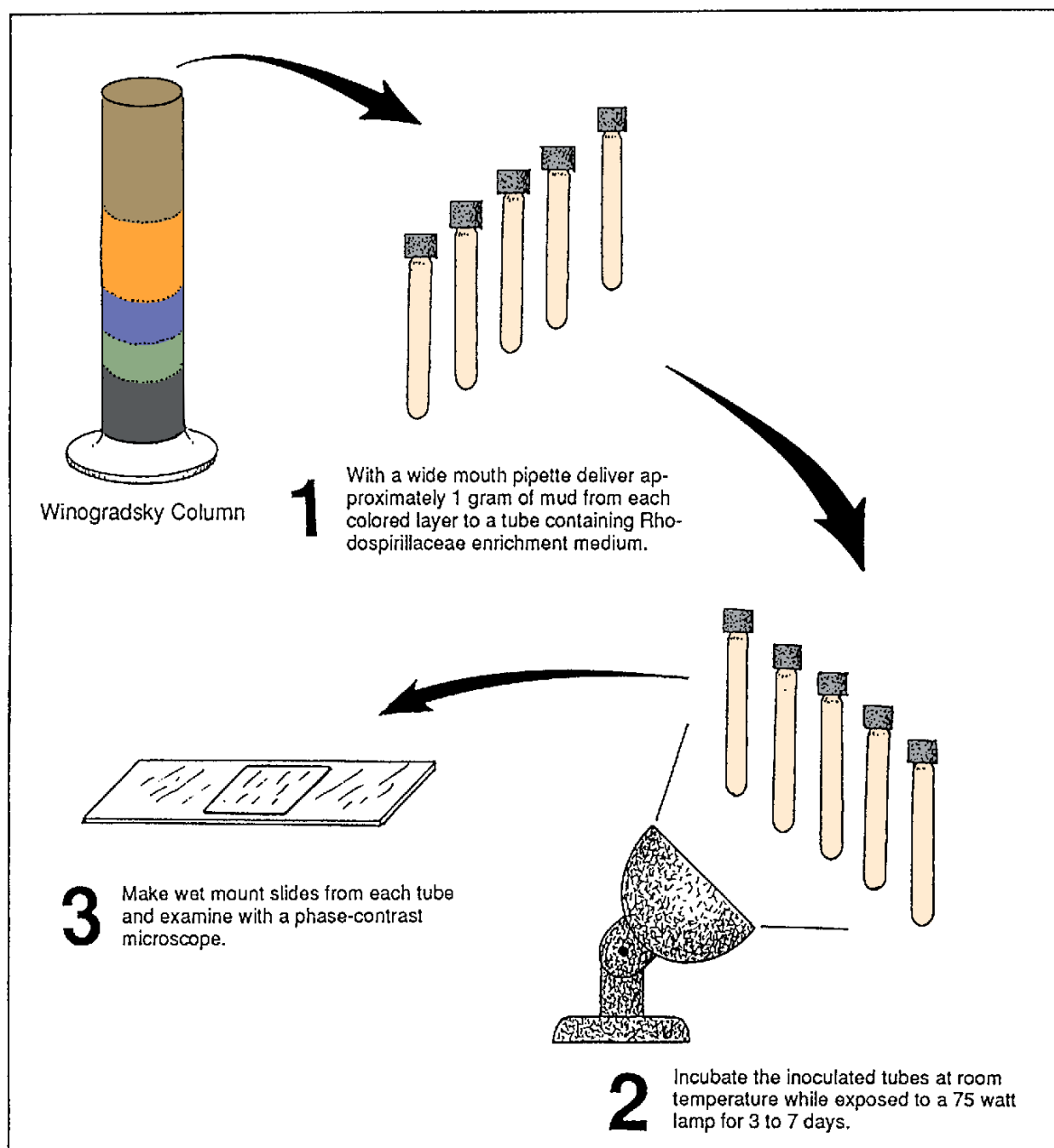


Figure 27.2 Procedure for subculturing and microscopic examination

Isolation of Anaerobic Phototrophic Bacteria: Using the Winogradsky Column • **Exercise 27**

Materials

- 5 screw-cap test tubes (13 × 200 mm size)
- 1 prescription bottle containing 200 ml of Rhodospirillaceae enrichment medium.
- 5 wide-mouth 10 ml pipettes
- 1. Label the screw-cap test tubes with the colors of the areas to be subcultured from your Winogradsky column. They may be brown, red, reddish-purple, or green. If such areas are not obvious, collect mud from areas that are black to grey.
- 2. With a pipette, deliver Rhodospirillaceae enrichment medium from the prescription bottle to each of the test tubes. Fill each tube about two-thirds full with the medium.
- 3. With a pipette, collect about 1 g of mud from each colored area of the column and deliver the mud to the properly labeled tube. Use a fresh pipette for each delivery.
- 4. After inoculating each tube, completely fill the tubes with additional enrichment medium.
- 5. Place screw caps on each tube and tighten each cap securely. Invert each tube several times to mix the mud and enrichment medium.
- 6. Place all the tubes in front of a 75 watt incandescent lamp and incubate at room temperature for several days to a week.
- 7. Observe the cultures at several intervals. When the cultures have developed a green, red-brown, or red-purple coloration, make wet mount slides and examine with a phase-contrast microscope. If phase-contrast microscopy is unavailable, make gram-stained slides. Record your results on the Laboratory Report.

LABORATORY REPORT

Complete the Laboratory Report for this exercise.

PART

5

Bacterial Viruses: Isolation and Propagation

The viruses differ from bacteria in being much smaller, noncellular and intracellular parasites. In addition, they cannot be grown on ordinary media. Despite these seemingly difficult obstacles to laboratory study, we are readily able to detect their presence by observing their effects upon the cells they parasitize.

Specific viruses are associated with all types of cells, eukaryotic and prokaryotic. Their dependence on other cells is due to their inability to synthesize enzymes needed for their own metabolism. By existing within cells, however, they are able to utilize the

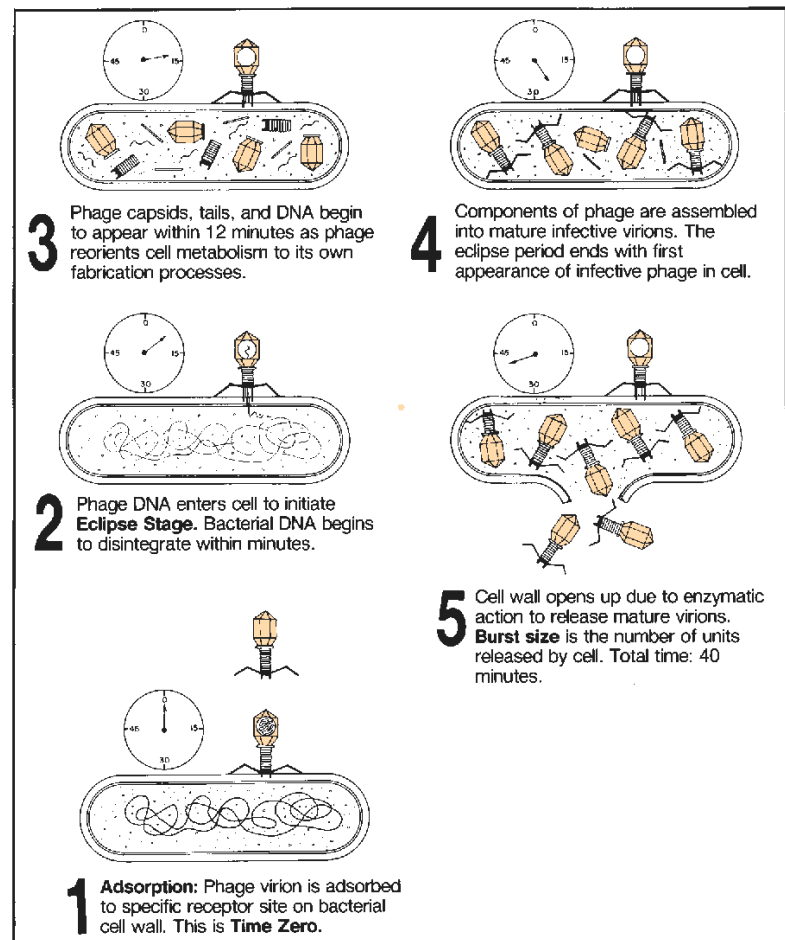


Figure V.1 The lytic cycle of a virulent bacteriophage

Part 5 • Bacterial Viruses: Isolation and Propagation

enzymes of their host. They may contain DNA or RNA, but never both of these nucleic acids.

The study of viruses that parasitize plant and animal cells is time-consuming and requires special tissue culture techniques. Viruses that parasitize bacteria, however, are relatively easy to study, utilizing ordinary bacteriological techniques. It is for this reason that bacterial viruses will be studied here. Principles learned from studying the viruses of bacteria apply to viruses of eukaryotic cells.

Viruses that parasitize bacteria are called **bacteriophage**, or **phage**. These viruses exist in many shapes and sizes. Some of the simplest ones exist as a single-stranded DNA virion. Most of them are tadpole-like, with “heads” and “tails” as seen in figure V.1 on the previous page. The head, or **capsid**, may be round, oval, or polyhedral and is composed of protein. It forms a protective envelope for the DNA of the organism. The tail structure is hollow and provides an exit for the DNA from the capsid into the cytoplasm of the bacterial cell. The extreme end of the tail has the ability to become attached to specific receptor sites on the surface of phage-sensitive bacteria. Once the tail of the virus attaches itself to a cell, it literally digests its way through the wall of the host cell.

With the invasion of a bacterial cell by the DNA, one of two things will occur: lysis or lysogeny. In the event that **lysis** occurs, as illustrated on the previous page, the metabolism of the bacterial cell becomes reoriented to the synthesis of new viral DNA and protein to produce mature phage particles. Once all the cellular material is used up, the cell bursts to release phage virions that, in turn, are prepared to invade other cells.

Phage that cause lysis are said to be *virulent*. If the phage does not cause lysis, however, it is termed *temperate* and establishes a relationship with the bacterial cell known as **lysogeny**. In these cells, the DNA of the phage becomes an integral part of the bacterial chromosome. Lysogenic bacteria grow normally, but their cultures always contain some phage. Periodically, however, phage virions are released by lysogenized cells in lytic bursts similar to that seen in the lytic cycle.

Visual evidence of lysis is demonstrated by mixing a culture of bacteria with phage and growing the mixture on nutrient agar. Areas where the phage are active will show up as clear spots called *plaques*.

The most thoroughly studied bacterial viruses are those that parasitize *Escherichia coli*. They are collectively referred to as the *coliphages*. They are readily isolated from raw sewage and coprophagous (dung-eating) insects. Exercises 28 and 29 pertain to these techniques. Exercise 30 provides a method for determining the burst size of a phage. Before attempting any of these experiments, be certain that you thoroughly understand the various stages in the phage lytic cycle as depicted here.

Isolation of Phage from Sewage

28

Establishing the presence of phage virions in sewage involves three steps. First, it is necessary to increase the phage numbers by *enrichment* with special media. Second, it is necessary to separate the phage from the bacteria by *filtration*. The final step is to produce plaque evidence by *seeding* a “lawn” of bacteria with phage in the filtrate. Figures 28.1 and 28.2 illustrate the three steps we will go through. Before beginning this experiment, however, here are a few comments about safety procedures that must be followed in handling sewage:

- When collecting sewage samples always wear latex gloves. Raw sewage is a potent source of bacterial, viral, and fungal pathogens.

- Raw sewage rich in bacteriophage is best collected at municipal sewage treatment plants. Usually, collection is made through manhole access.
- As emphasized in previous pages of this manual, no mouth pipetting permitted!

ENRICHMENT

To increase the number of phage virions in a raw sewage sample, it is necessary to add 5 ml of deca-strength phage broth (DSPB) and 5 ml of *E. coli* to 45 ml of raw sewage as in illustration 1 of figure 28.1. The DSPB medium is 10 times as strong as ordinary broth to accommodate dilution with 45 ml of sewage. This mixture is incubated at 37° C for 24 hours.

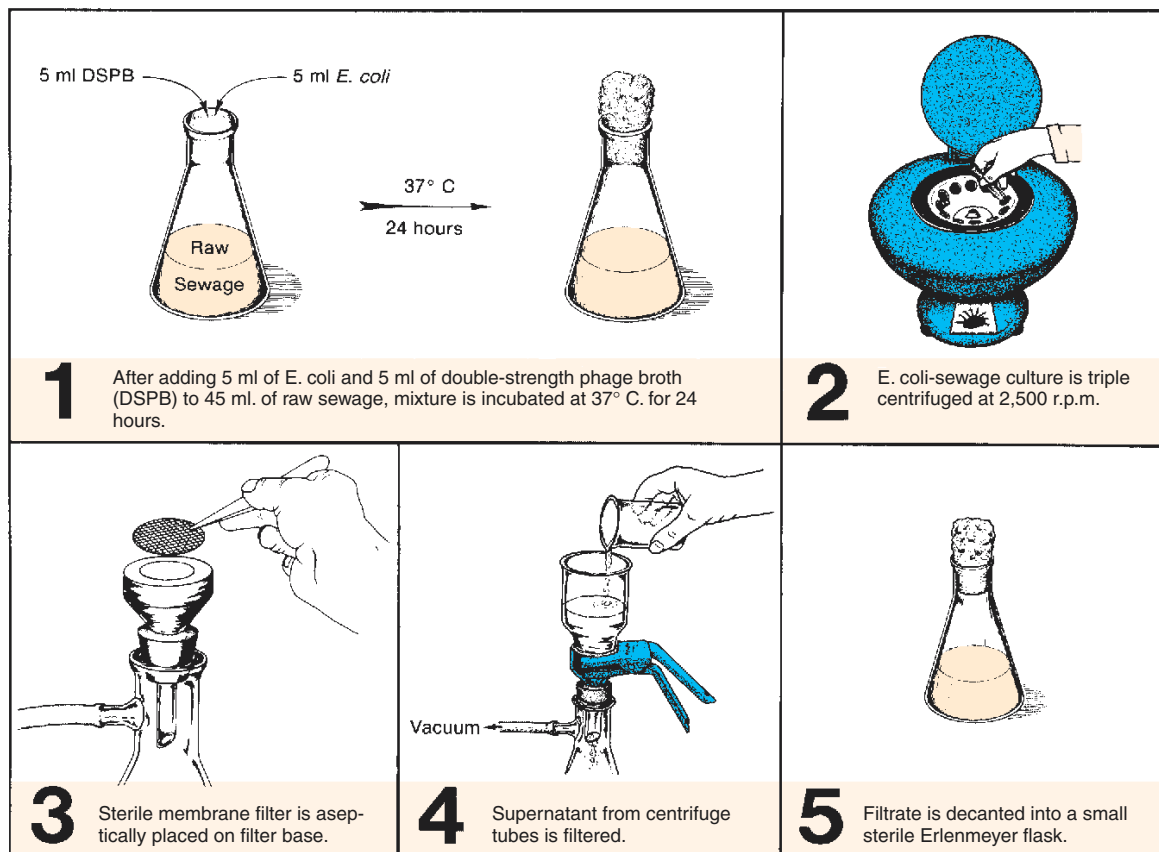


Figure 28.1 Enrichment and separation of phage from sewage

Exercise 28 • Isolation of Phage from Sewage

Materials:

flask of raw sewage
1 Erlenmeyer flask (125 ml size)
5 ml of DSPB medium
nutrient broth culture of *E. coli* (strain B)
graduate and 5 ml pipettes

1. With a graduate, measure out 45 ml of raw sewage and decant into an Erlenmeyer flask.
2. Pour 5 ml of DSPB medium and 5 ml of *E. coli* into the flask of sewage. If these constituents are not premeasured, use a 5 ml pipette. If the medium is pipetted first, the same pipette can be used for pipetting the *E. coli*.
3. Place the flask in the 37° C incubator for 24 hours.

FILTRATION

Rapid filtration to separate the phage from *E. coli* in the enrichment mixture requires adequate centrifugation first. If centrifugation is inadequate, the membrane filter will clog quickly and impair the rate of filtration. To minimize filter clogging, a **triple centrifugation** procedure will be used. To save time in the event that filter clogging does occur, an extra filter assembly and an adequate supply of membrane filters should be available. These membrane filters have a maximum pore size of 0.45 μm , which holds back all bacteria, allowing only the phage virions to pass through.

Materials:

centrifuge and centrifuge tubes (6–12)
2 sterile membrane filter assemblies (funnel,
glass base, clamp, and vacuum flask)
package of sterile membrane filters
sterile Erlenmeyer flask with cotton plug (125
ml size)
forceps and Bunsen burner
vacuum pump and rubber hose

1. Into 6 or 8 centrifuge tubes, dispense the sewage–*E. coli* mixture, filling each tube to within $\frac{1}{2}$ " of the top. Place the tubes in the centrifuge so that the load is balanced. Centrifuge the tubes at 2,500 rpm for 10 minutes.
2. Without disturbing the material in the bottom of the tubes, decant all material from the tubes to within 1" of the bottom into another set of tubes.
3. Centrifuge this second set of tubes at 2,500 rpm for another 10 minutes. While centrifugation is taking place, rinse out the first set of tubes.
4. When the second centrifugation is complete, pour off the top two-thirds of each tube into the clean set of tubes and centrifuge again in the same manner.
5. While the third centrifugation is taking place, aseptically place a membrane filter on the glass base of a sterile filter assembly (illustration 3, figure 28.1). Use flamed forceps. Note that the filter is a thin sheet with grid lines on it.

6. Place the glass funnel over the filter and fix the clamp in place.
7. Hook up a rubber hose between the vacuum flask and pump.
8. Carefully decant the top three-fourths of each tube into the filter funnel. **Do not disturb the material in the bottom of the tube.**
9. Turn on the vacuum pump. If centrifugation has removed all bacteria, filtration will occur almost instantly. If the filter becomes clogged and you have enough filtrate to complete the experiment, go on to step 10. (If this filtrate is to be used by the entire class, you will need 25–50 ml.)

If the filter clogs before you have enough filtrate, pour the unfiltered material from the funnel back into another set of centrifuge tubes and re-centrifuge for 10 minutes at 2,500 rpm.

While centrifugation is taking place, set up the other filter assembly and pour whatever filtrate you have from the first flask into the funnel of the new setup. After centrifugation, decant the top three-fourths of material from each tube into the funnel and turn on the vacuum pump. Filtration should take place rapidly now.

10. Aseptically transfer the final filtrate from the vacuum flask to a sterile 125 ml Erlenmeyer flask that has a sterile cotton plug. Putting the filtrate in a small flask is necessary to facilitate pipetting. *Be sure to flame the necks of both flasks while pouring from one to the other.*

SEEDING

Evidence of phage in the filtrate is produced by providing a "lawn" of *E. coli* and phage. The medium used is soft nutrient agar. Its jellylike consistency allows for better development of plaques. The soft agar is poured over the top of prewarmed hard nutrient agar. Prewarmed plates result in a smoother top agar surface. Figure 28.2 illustrates the general procedure.

Materials:

nutrient broth culture of *E. coli* (strain B)
flask of enriched sewage filtrate
4 metal-capped tubes of soft nutrient agar (5 ml
per tube)
4 Petri plates of nutrient agar (15 ml per plate,
preferably prewarmed at 37° C)
1 ml serological pipettes

1. Liquefy 4 tubes of soft nutrient agar and cool to 50° C. Keep the tubes in a 50° C water bath to prevent solidification.
2. Label the tubes 1, 2, 3, and 4. Label the plates 1, 2, 3, and control.
3. With a 1 ml pipette, transfer 1 drop of filtrate to tube 1, 3 drops to tube 2, and 6 drops to tube 3. Don't put any filtrate into tube 4.

Isolation of Phage from Sewage • Exercise 28

4. With a fresh 1 ml pipette, transfer 0.3 ml of *E. coli* to each of the four tubes of soft agar.
5. After flaming the necks of each of the soft agar tubes, pour the contents of each tube over the hard agar of similarly numbered agar plates. Note that tube 4 is poured over the control plate.
6. Once the agar is cooled completely, put the plates, inverted, into a 37° C incubator. If possible, ex-

amine the plates **3 hours later** to look for plaque formation. If some plaques are visible, measure them and record their diameters on the Laboratory Report. Plaque size should be checked **every hour** for changes.

LABORATORY REPORT

Record all results on Laboratory Report 28, 29.

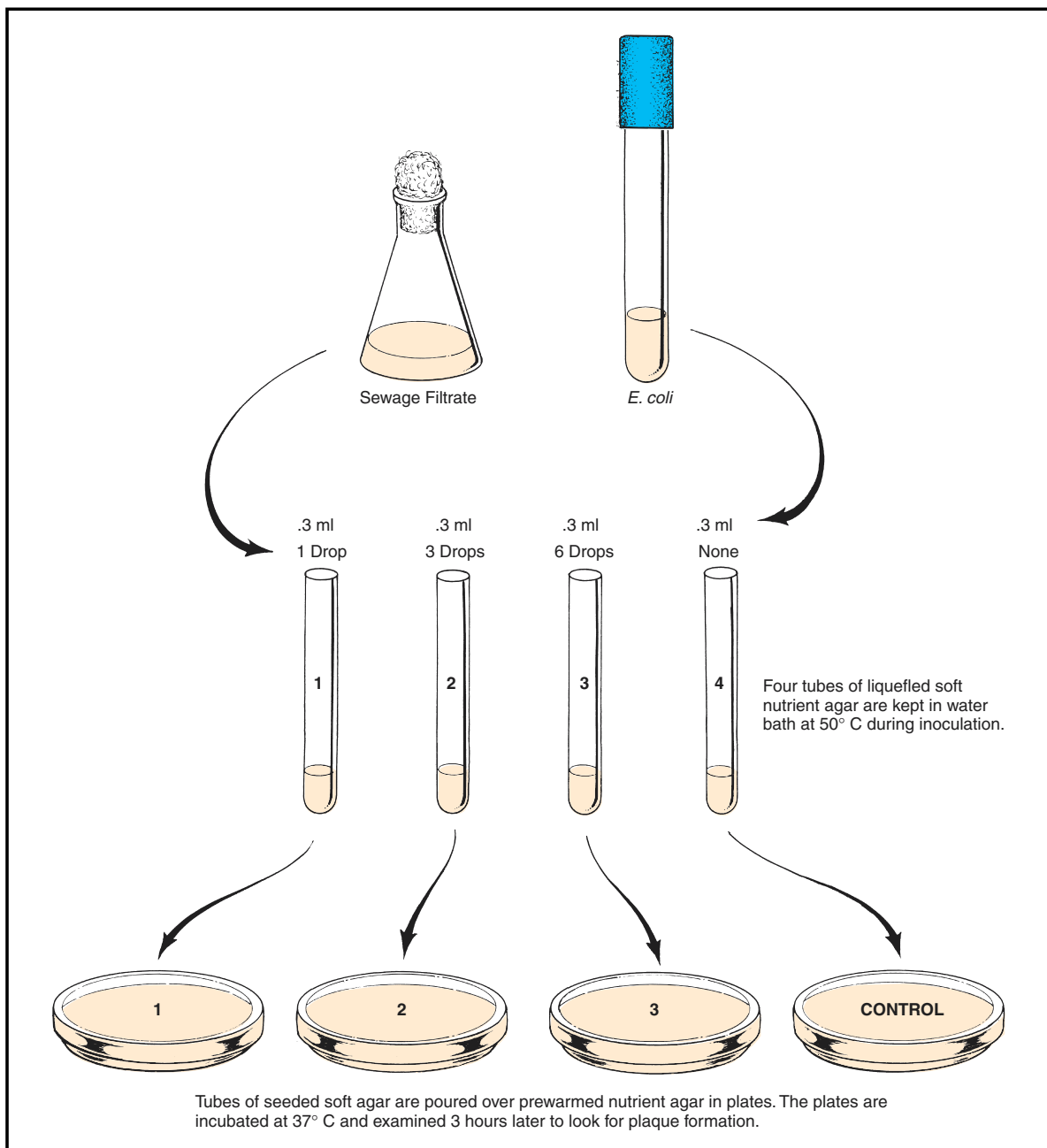


Figure 28.2 Overlay method of seeding *Escherichia coli* cultures with phage

29

Isolation of Phage from Flies

As stated earlier, coprophagous insects, as well as raw sewage, contain various kinds of bacterial viruses. Houseflies fall into the coprophagous category because they deposit their eggs in fecal material where the young larvae feed, grow, pupate, and emerge as adult flies. This type of environment is heavily populated by *E. coli* and its inseparable parasitic phages.

In this experiment we will follow a procedure that is quite similar to the one used in working with raw sewage. An enrichment medium, utilizing cyanide, will be substituted for the DSPB, however. Figures 29.1 and 29.2 illustrate the procedure.

FLY COLLECTION

To increase the probability of success in isolating phage, it is desirable that one use 20 to 24 houseflies. A smaller number might be sufficient; the larger number, however, increases the probability of initial success. Houseflies should not be confused with the smaller blackfly or the larger blowfly. An ideal spot for collecting these insects is a barnyard or riding stable. One should not use a cyanide killing bottle or any other chemical means. Flies should be kept alive until just prior to crushing and placing them in the growth medium. There are many ways that one might use to capture them—use your ingenuity!

ENRICHMENT

Within the flies' digestive tracts are several different strains of *E. coli* and bacteriophage. Our first concern is to enhance the growth of both organisms to ensure an adequate supply of phage. To accomplish this the flies must be ground up with a mortar and pestle and then incubated in a special growth medium for a total of 48 hours. During the last 6 hours of incubation, a lysing agent, sodium cyanide, is included in the growth medium to augment the lysing properties of the phage.

Materials:

- bottle of phage growth medium* (50 ml)
- bottle of phage lysing medium* (50 ml)
- Erlenmeyer flask (125 ml capacity) with cotton plug
- mortar and pestle (glass)

*see Appendix C for composition

1. Into a clean nonsterile mortar place 24 freshly killed houseflies. Pour half of the growth medium into the mortar and grind the flies to a fine pulp with the pestle.
2. Transfer this fly-broth mixture to an empty flask. Use the remainder of the growth medium to rinse out the mortar and pestle, pouring all the medium into the flask.
3. Wash the mortar and pestle with soap and hot water before returning them to the cabinet.
4. Incubate the fly-broth mixture for 42 hours at 37° C.
5. At the end of the 42-hour incubation period add 50 ml of lysing medium to the fly-broth mixture. Incubate this mixture for another 6 hours.

CENTRIFUGATION

Before attempting filtration, you will find it necessary to separate the fly fragments and miscellaneous bacteria from the culture medium. If centrifugation is incomplete, the membrane filter will clog quickly and filtration will progress slowly. To minimize filter clogging, a triple centrifugation procedure will be used. To save time in the event filter clogging does occur, an extra filter assembly and an adequate supply of membrane filters should be available. These filters have a maximum pore size of 0.45 μm , which holds back all bacteria, allowing only the phage virions to pass through.

Materials:

- centrifuge
- 6–12 centrifuge tubes
- 2 sterile membrane filter assemblies (funnel, glass base, clamp, and vacuum flask)
- package of sterile membrane filters
- sterile Erlenmeyer flask with cotton plug (125 ml size)
- vacuum pump and rubber hose

1. Into 6 or 8 centrifuge tubes, dispense the enrichment mixture, filling each tube to within $\frac{1}{2}$ " of the top. Place the tubes in the centrifuge so that the load is balanced. Centrifuge the tubes at 2,500 rpm for 10 minutes.
2. Without disturbing the material in the bottom of the tubes, decant all material from the tubes to within 1" of the bottom into another set of tubes.

Isolation of Phage from Flies • Exercise 29

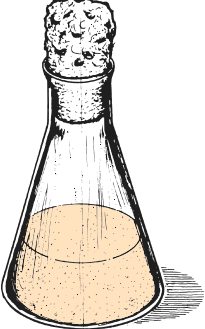
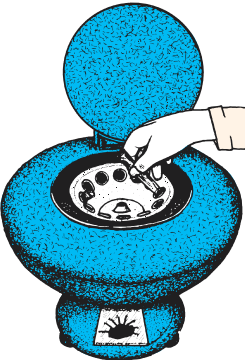
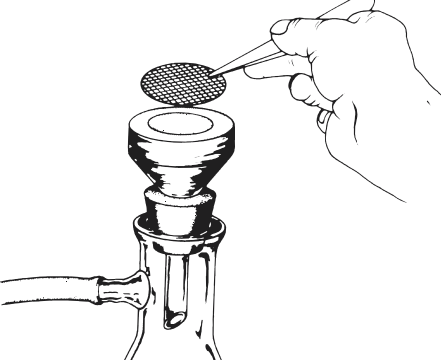
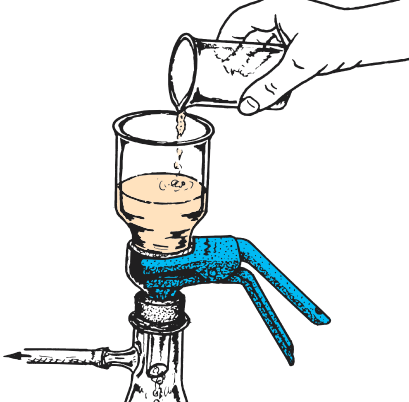
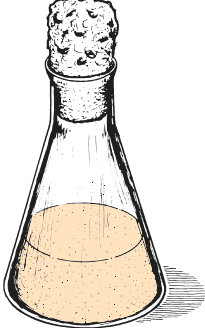
	
1 Twenty to twenty-four flies are ground up in phage growth medium with a mortar and pestle.	2 Crushed flies are incubated in growth medium for 42 hours at 37° C. After adding lysing medium it is incubated for another 6 hours.
	
3 Fly-broth culture is triple-centrifuged at 2,500 rpm.	4 Membrane filter assembly is set up for filtration. This step must be done aseptically.
	
5 Centrifuged supernatant is filtered to produce bacteria-free phage filtrate.	6 Phage filtrate is dispensed to a sterile Erlenmeyer flask from which layered plates will be made (Fig. 29.2).

Figure 29.1 Procedure for preparation of bacteriophage filtrate from houseflies

Exercise 29 • Isolation of Phage from Flies

3. Centrifuge this second set of tubes at 2,500 rpm for another 10 minutes. While centrifugation is taking place, rinse out the first set of tubes.
4. When the second centrifugation is complete, pour off the top two-thirds of each tube into the clean set of tubes and centrifuge again in the same manner.

FILTRATION

While the third centrifugation is taking place, aseptically place a membrane filter on the glass base of a sterile filter assembly (illustration 4, figure 29.1). Use flamed forceps. Note that the filter is a thin sheet with grid lines on it. Place the glass funnel over the filter and fix the clamp in place. Hook up a rubber hose between the vacuum flask and pump.

Now, carefully decant the top three-fourths of each tube into the filter funnel. Take care not to disturb the material in the bottom of the tube. Turn on the vacuum pump. If centrifugation and decanting have been performed properly, filtration will occur almost instantly. If the filter clogs before you have enough filtrate, recentrifuge all material and pass it through the spare filter assembly.

Aseptically, transfer the final filtrate from the vacuum flask to a sterile 125 ml Erlenmeyer flask that has a sterile cotton plug. Putting the filtrate in a small flask is necessary to facilitate pipetting. Be sure to flame the necks of both flasks while pouring from one to the other.

INOCULATION AND INCUBATION

To demonstrate the presence of bacteriophage in the fly-broth filtrate, a strain of phage-susceptible *E. coli* will be used. To achieve an ideal proportion of phage to bacteria, a proportional dilution method will be used. The phage and bacteria will be added to tubes of soft nutrient agar that will be layered over plates of hard nutrient agar. Soft nutrient agar contains only half as much agar as ordinary nutrient agar. This medium and *E. coli* provide an ideal “lawn” for phage growth. Its jelly-like consistency allows for better diffusion of phage particles; thus, more even development of plaques occurs.

Figure 29.2 illustrates the overall procedure. It is best to perform this inoculation procedure in the

morning so that the plates can be examined in late afternoon. As plaques develop, one can watch them increase in size with the multiplication of phage and simultaneous destruction of *E. coli*.

Materials:

nutrient broth cultures of *Escherchia coli* (ATCC #8677 phage host)
flask of fly-broth filtrate
10 tubes of soft nutrient agar (5 ml per tube) with metal caps
10 plates of nutrient agar (15 ml per plate, and prewarmed at 37° C)
1 ml serological pipettes, sterile

1. Liquefy 10 tubes of soft nutrient agar and cool to 50° C. Keep tubes in water bath to prevent solidification.
2. With a china marking pencil, number the tubes of soft nutrient agar 1 through 10. Keep the tubes sequentially arranged in the test-tube rack.
3. Label 10 plates of prewarmed nutrient agar 1 through 10. Also, label plate 10 **negative control**. Prewarming these plates will allow the soft agar to solidify more evenly.
4. With a 1 ml serological pipette, deliver 0.1 ml of fly-broth filtrate to tube 1, 0.2 ml to tube 2, etc., until 0.9 ml has been delivered to tube 9. Refer to figure 29.2 for sequence. **Note that no fly-broth filtrate is added to tube 10.** This tube will be your negative control.
5. With a fresh 1 ml pipette, deliver 0.9 ml of *E. coli* to tube 1, 0.8 ml to tube 2, etc., as shown in figure 29.2. **Note that tube 10 receives 1.0 ml of *E. coli*.**
6. After flaming the necks of each of the tubes, pour them into similarly numbered plates.
7. When the agar has cooled completely, put the plates, inverted, into a 37° C incubator.
8. **After about 3 hours** incubation, examine the plates, looking for plaques. If some are visible, measure them and record their diameters on the Laboratory Report.
9. If no plaques are visible, check the plates again in another **2 hours**.
10. Check the plaque size again at **12 hours**, if possible, recording your results. Incubate a total of 24 hours.
11. Complete Laboratory Report 28, 29.

Isolation of Phage from Flies • Exercise 29

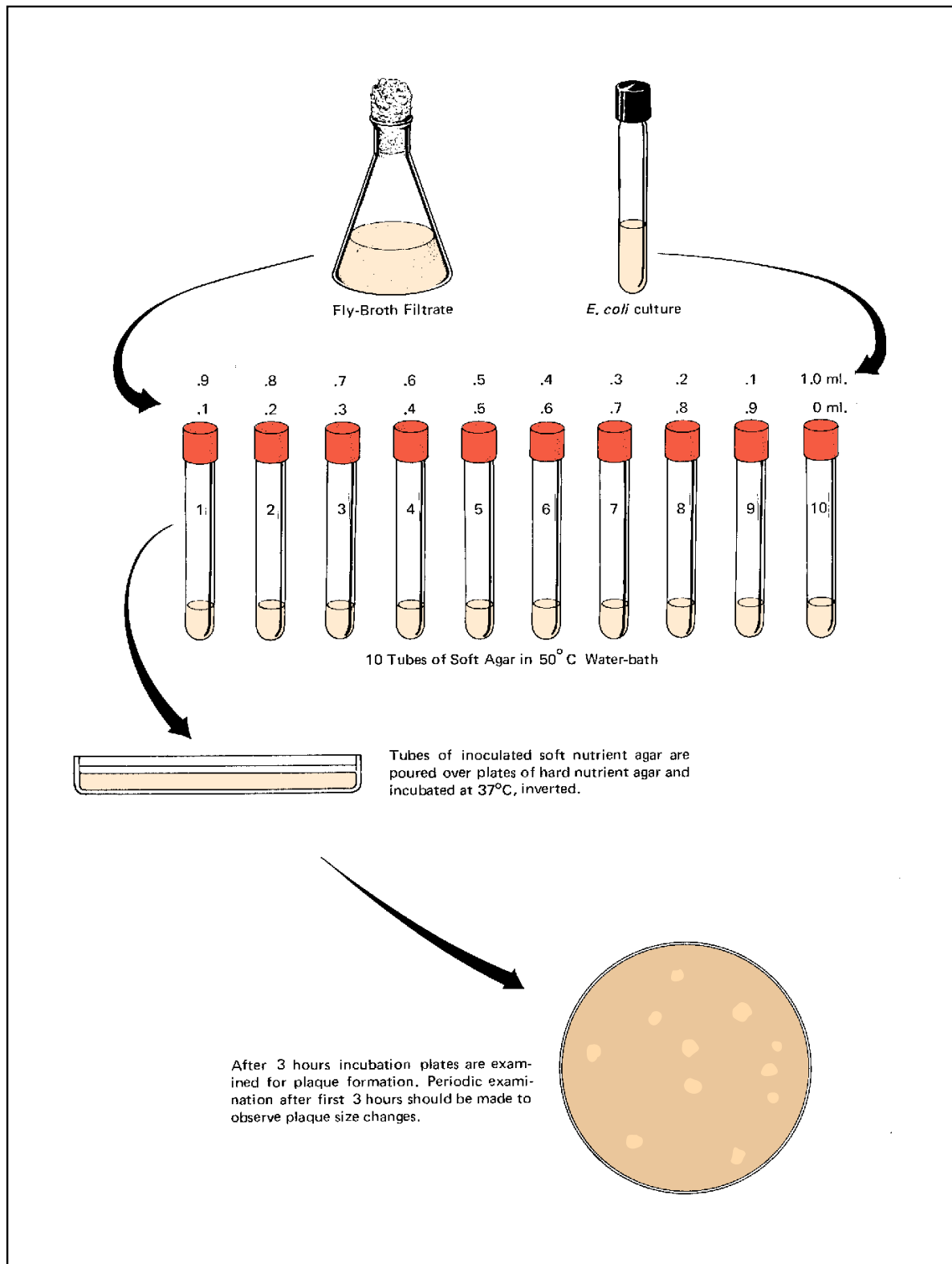


Figure 29.2 Inoculation of *Escherichia coli* with bacteriophage from fly-broth filtrate

30

Burst Size Determination: A One-Step Growth Curve

The average number of mature phage virions released by the lysis of a single bacterial cell is between 20 and 200. This number is called the **burst size**. It can be determined by adding a small amount of phage to a known quantity of bacteria and then lysing the cells at 5-minute intervals with chloroform. The chloroform-lysed cells, in turn, are mixed with bacteria, plated out, and incubated. By counting the plaques, it is possible to determine the burst size. In this experiment we will determine the burst size of coliphage T4 on host cells *E. coli*, strain B.

ADSORPTION

Figure 30.1 illustrates the procedure of this experiment. The first step is to add the phage to the susceptible bacteria. As soon as the two are mixed, adsorption begins. The phage collide in random fashion with the bacterial cells and attach their tails to specific receptor sites on the surfaces of host cells. The adsorption process can be stopped at any time by dilution. **Time zero** of adsorption is the time of mixture of phage and bacteria.

Note in figure 30.1 that 0.1 ml of coliphage T4 (2×10^8 /ml) and 2 ml of *E. coli* (5×10^8 /ml) are mixed in the first tube, which is labeled ADS. The ratio of phage to bacteria in this case is 0.02, which calculates out in this manner:

$$\frac{0.1 \times 2 \times 10^8}{2 \times 5 \times 10^8} = \frac{0.2 \times 10^8}{10 \times 10^8} = 0.02, \text{ or } 1/50$$

This ratio is called the **multiplicity of infection**, or **m.o.i.**

By referring back to figure V.1 on page 111, we can see what is occurring in this experiment. Note that during the adsorption stage, DNA in the capsid passes down through the tail into the host through a hole produced in the cell wall by enzymatic action at the tip of the phage tail.

ECLIPSE STAGE

As soon as the phage DNA gets inside the bacterial cell, the phage enters the *eclipse stage*. During this stage, which lasts approximately 12 minutes, the entire physiology of the host cell is reoriented toward

the production of phage components: capsids, tails, and DNA. If the cell is experimentally lysed with chloroform during this period of time, it will be seen that the incomplete components of phage are unable to infect new cells (no plaques are formed).

MATURATION STAGE

As phage components begin to assemble late in the eclipse stage to form mature infective virions, the phage enters the *maturation stage*. The lysing of cultures with chloroform beyond 12 minutes of time zero will reveal the presence of these mature units by producing plaques on poured plates. Lysis of a population of infected cells does not occur instantaneously, but instead follows a normal distribution curve, or **rise period**. The rise period, which lasts for several minutes, represents the growth in numbers of mature phage present. *The peak of the curve is the burst size*. It is this value that will be determined here.

TWO METHODS

To accommodate the availability of time and materials, there are two options for performing this experiment. The first option is for students to work in pairs to perform the entire experiment. Figure 30.1 illustrates the procedure for this method. The other option, which requires much less media and time, utilizes a team approach in which students, working in pairs, do just a portion of the experiment; in this case, data are pooled to complete the experiment. Figure 30.3 illustrates the procedure for this method. Your instructor will indicate which method will be used.

THE ENTIRE EXPERIMENT

To perform the experiment in its entirety, follow the procedures that are shown in figure 30.1.

Materials:

- 1 sterile serological tube (for ADS tube)
- 15 tubes tryptone broth (9.9 ml in each one)
- 8 tubes of nutrient soft agar (3 ml per tube)
- 8 Petri plates of tryptone agar
- 16 pipettes (1 ml size)

Burst Size Determination: A One-Step Growth Curve • Exercise 30

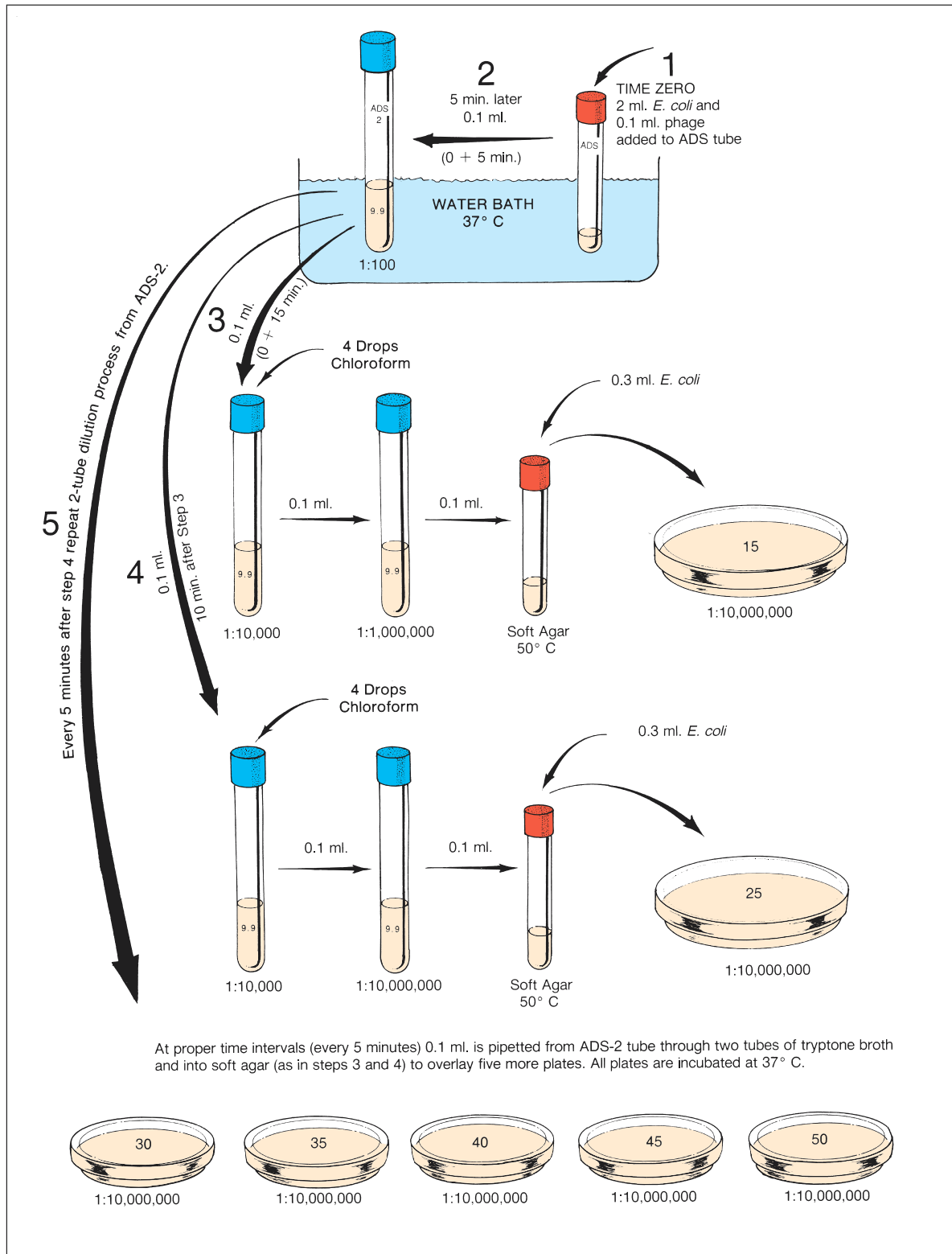


Figure 30.1 Procedure for entire experiment

Exercise 30 • Burst Size Determination: A One-Step Growth Curve

- 1 dropping bottle of chloroform
- 1 small wire basket to hold 7 tubes of soft agar in water bath
- 1 wire test-tube rack
- 2 water baths (37° C and 50° C)
- 1 culture of *E. coli*, strain B (5 ml) with concentration of 5×10^8 per ml
- 1 tube of T4 phage (2×10^8 per ml)

Preliminaries

1. Liquefy 8 tubes of soft nutrient agar by boiling in a beaker of water. Cool to 50° C and place in a wire basket or rack in 50° C water bath.
2. Label a sterile serological tube “ADS” to signify the adsorption tube.
3. Label 1 tube of tryptone broth “ADS-2.”
4. Label 8 tryptone agar plates: control, 15, 25, 30, 35, 40, 45, and 50.
5. Arrange the ADS, ADS-2, and the 14 tryptone broth tubes in a rack as shown in figure 30.2. Place the rack in a 37° C water bath.

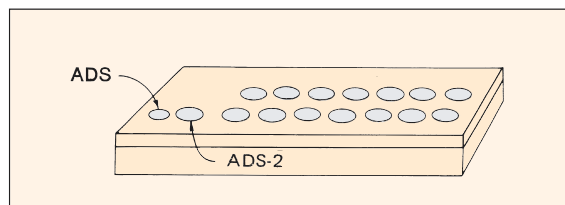


Figure 30.2 Tube arrangement

6. Dispense 3 to 4 drops of chloroform in each of the 7 tubes of tryptone broth that are in the front row.

Inoculations and Dilutions

1. Pipette 0.1 ml of *E. coli*, strain B, into a tube of liquefied soft nutrient agar and pour into the control plate. Swirl the plate gently to spread evenly. This plate will indicate whether any phage was in the original bacterial culture. Set this plate aside to harden.
2. With the same pipette as above, transfer 0.3 ml of *E. coli* into each of the tubes of soft nutrient agar. Keep the tubes in the 50° C water bath.
3. Still using the same pipette, transfer 2.0 ml of *E. coli* into the ADS tube.
4. With a fresh pipette, deliver 0.1 ml of T4 phage into the ADS tube and *immediately* record the time (*time zero*) of this mixing with *E. coli* in the following table. Mix gently and allow to remain in the 37° C water bath for 5 minutes.

	TIME	PLATE
Time zero	_____	none
Step 6 time (5 min later)	_____	none
Step 7 time (10 min later)	_____	15
Step 8 time (10 min later)	_____	25
Five minutes later	_____	30
Five minutes later	_____	35
Five minutes later	_____	40
Five minutes later	_____	45
Five minutes later	_____	50

5. While the mixture is incubating, fill in the table, recording all the projected times so that you will know when each step is to begin.
6. After the 5-minute incubation time, transfer 0.1 ml of the mixture to the ADS-2 tube, gently mix, and incubate at 37° C for another 10 minutes.
7. After 10 minutes, transfer 0.1 ml from the ADS-2 tube to the first front row tube of tryptone broth. Keep the ADS-2 tube in the water bath. Mix this dilution tube gently and transfer 0.1 ml to the adjacent tryptone broth tube in the second row. Mix this tube gently, also.
8. Transfer 0.1 ml from the second tube of tryptone broth to a tube of soft nutrient agar, mix gently, flame the tube neck, and pour the soft agar over the tryptone agar plate that is labeled “15.”
Swirl the plate carefully to disperse the soft agar mixture evenly.
9. Follow the above procedure 10 minutes later to produce a soft agar overlay plate on the plate labeled “25.”
10. Repeat at the allotted times for 30-, 35-, 40-, 45-, and 50-minute plates.
11. Invert and incubate all plates for 24–48 hours at 37° C.

Examination of the Plates

Once the plates have been incubated, count the plaques on all the plates, using a Quebec colony counter and hand tally counter. Record all counts on the Laboratory Report and determine burst size.

ABBREVIATED PROCEDURE (Team Method)

Performance of this experiment in teams will require a minimum of seven pairs of students. Each pair of students (team) will follow the procedure shown in

Burst Size Determination: A One-Step Growth Curve • Exercise 30

figure 30.3 to produce one soft agar overlay plate for a designated time.

Materials:

per team:

- 1 sterile serological tube
- 3 tubes of tryptone broth (9.9 ml per tube)
- 2 tubes of soft nutrient agar (3 ml per tube)
- 2 Petri plates of tryptone agar
- 4 1 ml pipettes
- 1 dropping bottle of chloroform
- 1 wire test-tube rack (small size)
- 1 small beaker (150 ml size)
- 1 tube of T4 phage (2×10^8 per ml)
- 1 culture of *E. coli*, strain B (5×10^8 per ml)
- water bath at 37° C (a small pan that will hold a test-tube rack)

2. Label a sterile serological tube “ADS” to signify the adsorption tube.
3. Label one tube of tryptone broth “ADS-2.”
4. Label the other tryptone tubes “I” and “II.”
5. Label one tryptone agar plate “control” and the other your designated time (15, 25, 30, 35, 40, 45, or 50). *Your instructor will assign you a specific time.* Put your names on both plates.
6. Arrange the ADS, ADS-2, and two tryptone tubes in a small test-tube rack in same order as shown in figure 30.3.
7. Place the rack of tubes in a pan of 37° C water. Although it is only necessary to incubate the ADS and ADS-2 tubes, it will be more convenient if they are all together.

Preliminaries

1. Liquefy two tubes of soft nutrient agar in boiling water. Use a small beaker. Cool the water to 50° C and keep the tubes of media at this temperature.

Inoculations and Dilutions

1. Pipette 0.1 ml of *E. coli*, strain B, into a tube of liquefied soft nutrient agar and pour it into the control plate. Swirl the plate gently to spread evenly.

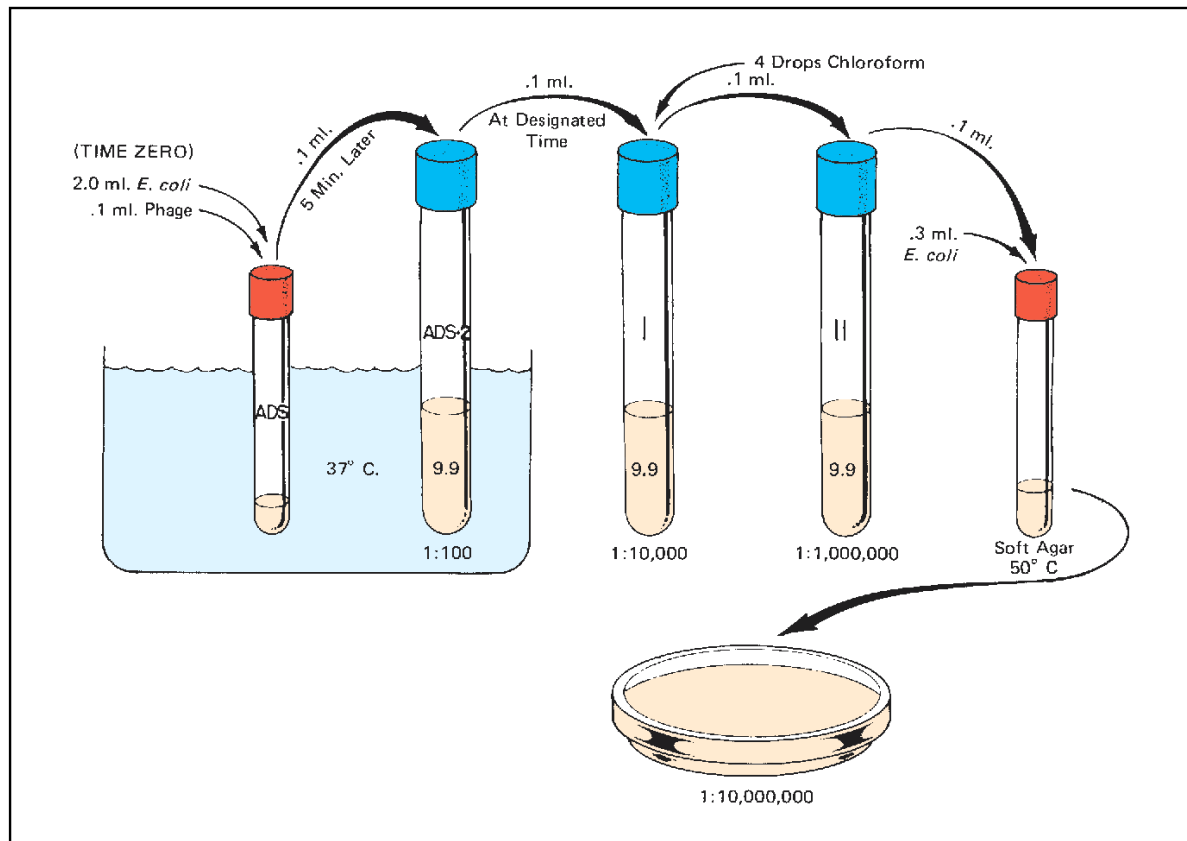


Figure 30.3 Abbreviated procedure (team) method

Exercise 30 • Burst Size Determination: A One-Step Growth Curve

This plate will indicate whether any phage was in the original bacterial culture. Set this plate aside to harden.

2. With the same pipette, transfer 0.3 ml of *E. coli* to the other tube of soft nutrient agar. Keep this tube in the beaker of water at 50° C.
3. Still using the same pipette, transfer 2.0 ml of *E. coli* into the ADS tube.
4. With a *fresh pipette*, deliver 0.1 ml of T4 phage into the ADS tube.

Record this time (time zero): _____

5. After 5 minutes, transfer 0.1 ml of the *E. coli*-phage mixture from the ADS tube to ADS-2 tube. Mix the ADS-2 tube gently.
6. After the designated time (time zero plus designated time), transfer 0.1 ml from ADS-2 tube to tryptone broth tube I. Mix gently.
7. Add 3 or 4 drops of chloroform to tube I.

8. With a *fresh pipette*, transfer 0.1 ml from tube I to tube II. Mix tube II gently.

9. With *another fresh pipette*, transfer 0.1 ml from tube II to the tube of soft agar.

10. After mixing the soft agar tube, pour it over the tryptone agar plate. Swirl the plate carefully to disperse the soft agar. Set aside to cool for a few minutes.

11. Incubate both plates at 37° C for 24–48 hours.

Examination of Plates

Once the plates have been incubated, examine both of them on a Quebec colony counter. The control plate should be free of plaques. Count the plaques on the other plate, using a hand tally counter if the number is great. Record your count on the Laboratory Report and on the chalkboard.

PART

6

Microbial Interrelationships

Populations within the microbial world relate to each other in various ways. Although many of them will be neutralistic toward each other by not interacting in any way, others will establish relationships that are quite different.

The three exercises in this unit reveal how certain organisms have developed relationships that are commensalistic, synergistic, and antagonistic. While most of these relationships are between bacteria, some are between bacteria and molds.

31

Bacterial Commensalism

There are many commensalistic relationships that exist between organisms in a mixed microbial population. The excretory products of one organism often become the nutrients of another. The oxygen usage of one species may produce the desired oxidation-reduction potential for another organism. In all cases of commensalism, the beneficiary contributes nothing in the way of benefit or injury to the other.

In this exercise we will culture two organisms separately and together to observe an example of commensalism. One of the organisms is *Staphylococcus aureus* and the other is *Clostridium sporogenes*. From your observations of the results, you are to determine which organism profits from the association and what controlling factor is changed when the two are grown together.

Materials:

3 tubes of nutrient broth
1 ml pipette
nutrient broth culture of *S. aureus*
fluid thioglycollate medium culture of *C. sporogenes*

FIRST PERIOD

1. Label one tube of nutrient broth *S. aureus*, a second tube *C. sporogenes*, and the third tube *S. aureus* and *C. sporogenes*.
2. Inoculate the first and third tubes with one loopful each of *S. aureus*.
3. With a 1 ml pipette, transfer 0.1 ml of *C. sporogenes* to tubes 2 and 3.
4. Incubate the three tubes at 37° C for 48 hours.

SECOND PERIOD

1. Compare the turbidity in the three tubes, noting which ones are most turbid. Record these results on the Laboratory Report.
2. After shaking the tubes for good dispersion, make a gram-stained slide of the organisms in each tube and record your observations on combined Laboratory Report 31–33.

Bacterial Synergism

32

Two or more organisms acting together to produce a substance that none can produce separately is a synergistic relationship. Such relationships are not uncommon among microorganisms. This phenomenon is readily demonstrated in the ability of some bacteria acting, synergistically, to produce gas by fermenting certain disaccharides.

In this exercise we will observe the fermentation capabilities of three organisms on two disaccharides. The two sugars, lactose and sucrose, will be inoculated with the individual organisms as well as with various combinations of the organisms to detect which organisms can act synergistically on which sugars. To conserve on media, the class will be divided into three groups (A, B, and C). Results of inoculations will be shared.

Materials:

per pair of students:

- 3 Durham* tubes of lactose broth with bromthymol blue indicator
- 3 Durham* tubes of sucrose broth with bromthymol blue indicator
- 1 nutrient broth culture of *S. aureus*
- 1 nutrient broth culture of *P. vulgaris*
- 1 nutrient broth culture of *E. coli*

*A Durham tube is a fermentation tube of sugar broth that has a small inverted vial in it. See figure 48.3, page 164.

FIRST PERIOD

Group A

1. Label one tube of each kind of broth *E. coli*.
2. Label one tube of each kind of broth *P. vulgaris*.

3. Label one tube of each kind of broth *E. coli* and *P. vulgaris*.
4. Inoculate each tube with one loopful of the appropriate organisms.
5. Incubate the six tubes at 37° C for 48 hours.

Group B

1. Label one tube of each kind of broth *E. coli*.
2. Label one tube of each kind of broth *S. aureus*.
3. Label one tube of each kind of broth *E. coli* and *S. aureus*.
4. Inoculate each tube with one loopful of the appropriate organisms.
5. Incubate the six tubes at 37° C for 48 hours.

Group C

1. Label one tube of each kind of broth *S. aureus*.
2. Label one tube of each kind of broth *P. vulgaris*.
3. Label one tube of each kind of broth *S. aureus* and *P. vulgaris*.
4. Inoculate each tube with one loopful of the appropriate organisms.
5. Incubate the six tubes at 37° C for 48 hours.

SECOND PERIOD

1. Look for acid and gas production in each tube, recording your results on the Laboratory Report.
2. Determine which organisms acted synergistically on which disaccharides.
3. Answer the questions for this exercise on combined Laboratory Report 31–33.

33

Microbial Antagonism

Microbial antagonisms, in which one organism is inhibited and the other is unaffected, are easily demonstrated. Usually, the inhibitor produces a substance that inhibits or kills one or more organisms. The substance may be specific in its action, affecting only a few species, or it may be nonspecific, affecting a large number of organisms.

In this exercise we will attempt to evaluate the antagonistic capabilities of three organisms on two test organisms. The antagonists are *Bacillus cereus* var. *mycoides*, *Pseudomonas fluorescens*, and *Penicillium notatum*. The test organisms are *Escherichia coli* (gram-negative) and *Staphylococcus aureus* (gram-positive).

Materials:

- 6 nutrient agar pours
- 6 sterile Petri plates
- nutrient broth cultures of *E. coli*, *S. aureus*, *B. cereus* var. *mycoides*, and *P. fluorescens*
- flask culture of *Penicillium notatum* (8–12 day old culture)

FIRST PERIOD

1. Liquefy six nutrient agar pours and cool to 50° C. Hold in 50° C water bath.

2. While the pours are being liquefied, label six plates as follows:

Test Organism	Antagonist
I <i>S. aureus</i>	<i>B. cereus</i> var. <i>mycoides</i>
II <i>S. aureus</i>	<i>P. fluorescens</i>
III <i>S. aureus</i>	<i>Penicillium notatum</i>
IV <i>E. coli</i>	<i>B. mycoides</i>
V <i>E. coli</i>	<i>P. fluorescens</i>
VI <i>E. coli</i>	<i>Penicillium notatum</i>

3. Label three liquefied pours *S. aureus*, and label the other three *E. coli*.
4. Inoculate each of the pours with a loopful of the appropriate organisms, flame their necks, and pour into their respective plates.
5. After the nutrient agar in the plates has hardened, streak each plate with the appropriate antagonist. Use a good isolation technique.
6. Invert and incubate the plates for 24 hours at 37° C.

SECOND PERIOD

1. Examine each plate carefully, looking for evidence of inhibition.
2. Record your results on combined Laboratory Report 31–33 and answer all the questions.

PART

7

Environmental Influences and Control of Microbial Growth

The 11 exercises of this unit are concerned with two aspects of microbial growth: promotion and control. On the one hand, the microbiologist is concerned with providing optimum growth conditions to favor maximization of growth. The physician, nurse, and other members of the medical arts profession, on the other hand, are concerned with the limitation of microbial populations in disease prevention and treatment. An understanding of one of these facets of microbial existence enhances the other.

In Part 4 we were primarily concerned with providing media for microbial growth that contain all the essential nutritional needs. Very little emphasis was placed on other limiting factors such as temperature, oxygen, or hydrogen ion concentration. An organism provided with all its nutritional needs may fail to grow if one or more of these essentials are not provided. The total environment must be sustained to achieve the desired growth of microorganisms.

Microbial control by chemical and physical means involves the use of antiseptics, disinfectants, antibiotics, ultraviolet light, and many other agents. The exercises of this unit that are related to these aspects are intended, primarily, to demonstrate methods of measurement; no attempt has been made to make in-depth evaluation.

34

Temperature:
Effects on Growth

Temperature is one of the most important factors influencing the activity of bacterial enzymes. Unlike warm-blooded animals, the bacteria lack mechanisms that conserve or dissipate heat generated by metabolism, and consequently their enzyme systems are directly affected by ambient temperatures. Enzymes have minimal, optimal, and maximal temperatures. At the **optimum** temperature the enzymatic reactions progress at maximum speed. Below the **minimum** and above the **maximum** temperatures the enzymes become inactive. At some point above the maximum temperature, destruction of a specific enzyme will occur. Low temperatures are less deleterious in most cases.

Microorganisms grow in a broad temperature range that extends from approximately 0° C to above 90° C. They are divided into three groups: **mesophiles** that grow between 10° C and 47° C, **psychrophiles** that are able to grow between 0° C and 5° C, and **thermophiles** that grow at high temperatures (above 50° C).

The psychrophiles and thermophiles are further subdivided into obligate and facultative groups. Obligate psychrophiles seldom grow above 22° C and facultative psychrophiles (psychrotrophs) grow very well above 25° C. Thermophiles that thrive *only* at high temperatures (above 50° C and not below 40° C) are considered to be obligate thermophiles; those that will grow below 40° C are considered to be facultative thermophiles.

In this experiment we will attempt to measure the effects of various temperatures on two physiological reactions: pigment production and growth rate. Nutrient broth and nutrient agar slants will be inoculated with three different organisms that have different optimum growth temperatures. One organism, *Serratia marcescens*, produces a red pigment called *prodigiosin* that is produced only in a certain temper-

ature range. It is our goal here to determine the optimum temperature for prodigiosin production and the approximate optimum growth temperatures for all three microorganisms. To determine optimum growth temperatures we will be incubating cultures at five different temperatures. A spectrophotometer will be used to measure turbidity densities in the broth cultures after incubation.

FIRST PERIOD
(Inoculations)

To economize on time and media it will be necessary for each student to work with only two organisms and seven tubes of media. Refer to table 34.1 to determine your assignment. Figure 34.1 illustrates the procedure.

Materials:

nutrient broth cultures of *Serratia marcescens*,
Bacillus stearothermophilus, and
Escherichia coli

per student:

2 nutrient agar slants
5 tubes of nutrient broth

1. Label the tubes as follows:
Slants: Label both of them *S. marcescens*; label one tube 25° C and the other tube 38° C.
Broths: Label each tube of nutrient broth with your other organism and one of the following five temperatures: 5° C, 25° C, 38° C, 42° C, or 55° C.
2. Inoculate each of the tubes with the appropriate organisms. Use a wire loop.
3. Place each tube in one of the five baskets that is labeled according to incubation temperature.

Table 34.1 Inoculation Assignments

Student Number	<i>S. marcescens</i>	<i>B. stearothermophilus</i>	<i>E. coli</i>
1, 4, 7, 10, 13, 16, 19, 22, 25	2 slants and 5 broths		
2, 5, 8, 11, 14, 17, 20, 23, 26	2 slants	5 broths	
3, 6, 9, 12, 15, 18, 21, 24, 27	2 slants		5 broths

Note: The instructor will see that the 5° C basket is placed in the refrigerator and the other four are placed in incubators that are set at the proper temperatures.

SECOND PERIOD (Tabulation of Results)

Materials:

slants and broth cultures that have been
incubated at various temperatures
spectrophotometer and cuvettes
tube of sterile nutrient broth

1. Compare the nutrient agar slants of *S. marcescens*. Using colored pencils, draw the appearance of the growths on the Laboratory Report.
2. Shake the broth cultures and compare them, noting the differences in turbidity. Those tubes that

Temperature: Effects on Growth • Exercise 34

appear to have no growth should be compared with a tube of sterile nutrient broth.

3. If a spectrophotometer is available, determine the turbidity of each tube following the instructions on the Laboratory Report.
4. If no spectrophotometer is available, record turbidity by visual observation. The Laboratory Report indicates how to do this.
5. Exchange results with other students to complete data collection for experiment.

LABORATORY REPORT

After recording all data, answer the questions on the Laboratory Report for this exercise.

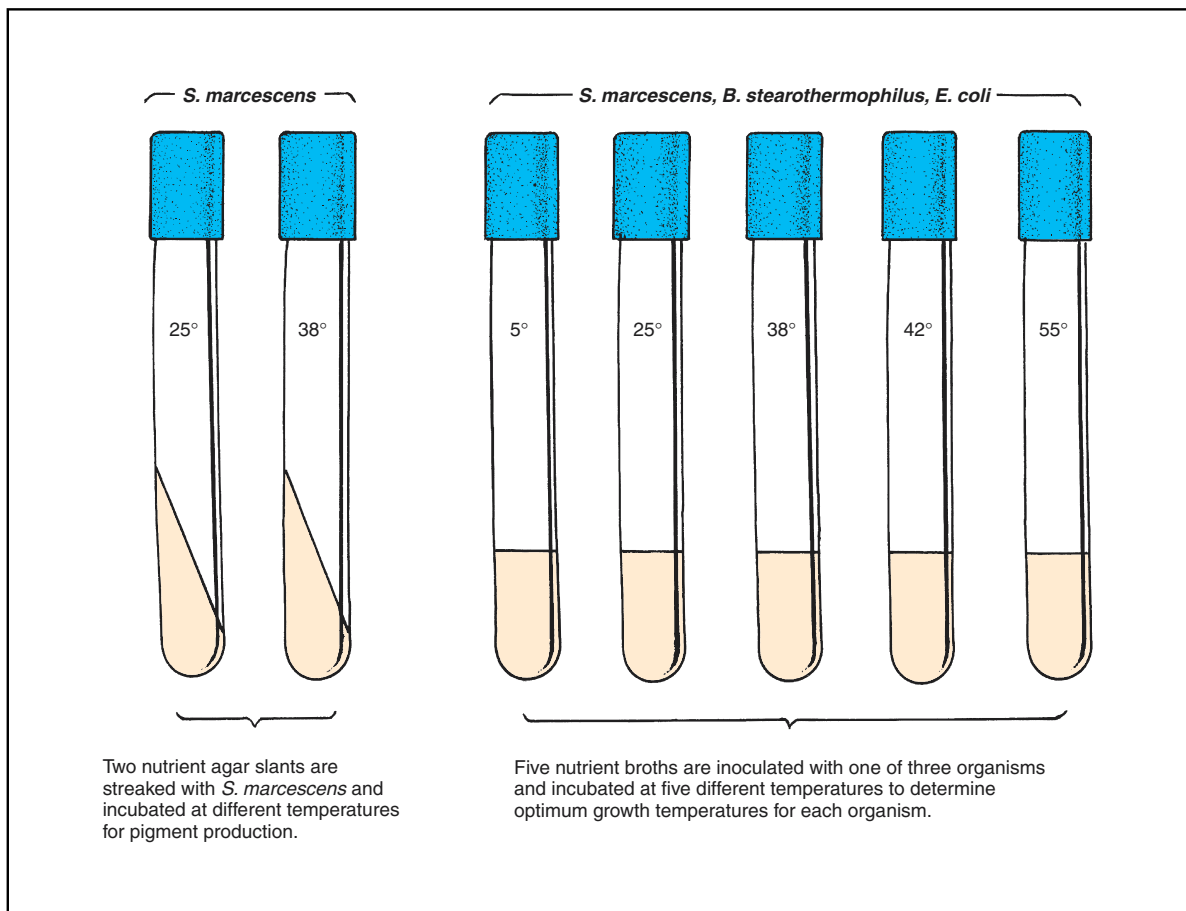


Figure 34.1 Inoculation procedure

35

Temperature:
Lethal Effects

In attempting to compare the susceptibility of different organisms to elevated temperatures, it is necessary to use some yardstick of measure. Two methods of comparison are used: the thermal death point and the thermal death time. The **thermal death point** (TDP) is the temperature at which an organism is killed in 10 minutes. The **thermal death time** (TDT) is the time required to kill a suspension of cells or spores at a given temperature. Since various factors such as pH, moisture, composition of medium, and age of cells will greatly influence results, these variables must be clearly stated.

In this exercise we will subject cultures of three different organisms to temperatures of 60°, 70°, 80°, 90°, and 100° C. At intervals of 10 minutes organisms will be removed and plated out to test their viability. The spore-former *Bacillus megaterium* will be compared with the non-spore-formers *Staphylococcus aureus* and *Escherichia coli*. The overall procedure is illustrated in figure 35.1.

Note in figure 35.1 that *before* the culture is heated a **control plate** is inoculated with 0.1 ml of the organism. When the culture is placed in the water bath, a tube of nutrient broth with a thermometer inserted into it is placed in the bath at the same time. Timing of the experiment starts when the thermometer reaches the test temperature.

Due to the large number of plates that have to be inoculated to perform the entire experiment, it will be necessary for each member of the class to be assigned a specific temperature and organism to work with. Table 35.1 provides assignments by student number. After the plates have been incubated, each student's results will be tabulated on a Laboratory Report chart at the demonstration table. The instructor will have

copies made of it to give each student so that everyone will have all the pertinent data needed to draw the essential conclusions.

Although this experiment is not difficult, it often fails to turn out the way it should because of student error. Common errors are (1) omission of the control plate inoculation, (2) putting the thermometer in the culture tube instead of in a tube of sterile broth, and (3) not using fresh sterile pipettes when instructed to do so.

Materials:*per student:*

- 5 Petri plates
- 5 pipettes (1 ml size)
- 1 tube of nutrient broth
- 1 bottle of nutrient agar (60 ml)
- 1 culture of organisms

class equipment:

- water baths set up at 60°, 70°, 80°, 90°, and 100° C

broth cultures:

- Staphylococcus aureus*, *Escherichia coli*, and *Bacillus megaterium* (minimum of 5 cultures of each species per lab section)

1. Consult table 35.1 to determine what organism and temperature has been assigned to you. If several thermostatically controlled water baths have been provided in the lab, locate the one that you will use. If a bath is not available for your temperature, set up a bath on an electric hot plate or over a tripod and Bunsen burner.

If your temperature is 100° C, a hot plate and beaker of water are the only way to go. When set-

Table 35.1 Inoculation Assignments

Organism	Student Number				
	60° C	70° C	80° C	90° C	100° C
<i>Staphylococcus aureus</i>	1, 16	4, 19	7, 22	10, 25	13, 28
<i>Escherichia coli</i>	2, 17	5, 20	8, 23	11, 26	14, 29
<i>Bacillus megaterium</i>	3, 18	6, 21	9, 24	12, 27	15, 30f

Temperature: Lethal Effects • Exercise 35

- ting up a water bath use hot tap water to start with to save heating time.
2. Liquefy a bottle of 60 ml of nutrient agar and cool to 50° C. This can be done while the rest of the experiment is in progress.
3. Label five Petri plates: **control, 10 min, 20 min, 30 min, and 40 min.**
4. Shake the culture of organisms and transfer 0.1 ml of organisms with a 1 ml pipette to the control plate.
5. Place the culture and a tube of sterile nutrient broth into the water bath. Remove the cap from the tube of nutrient broth and insert a thermometer into the tube. *Don't make the mistake of inserting the thermometer into the culture of organisms!*
6. As soon as the temperature of the nutrient broth reaches the desired temperature, record the time here: _____. Watch the temperature carefully to make sure it does not vary appreciably.

7. After 10 minutes have elapsed, transfer 0.1 ml from the culture to the 10-minute plate with a fresh 1 ml pipette. Repeat this operation at 10-minute intervals until all the plates have been inoculated. *Use fresh pipettes each time and be sure to shake the culture before each delivery.*
8. Pour liquefied nutrient agar (50° C) into each plate, rotate, and cool.
9. Incubate at 37° C for 24 to 48 hours. After evaluating your plates, record your results on the chart on the Laboratory Report and on the chart on the demonstration table.

LABORATORY REPORT

Complete the Laboratory Report once you have a copy of the class results.

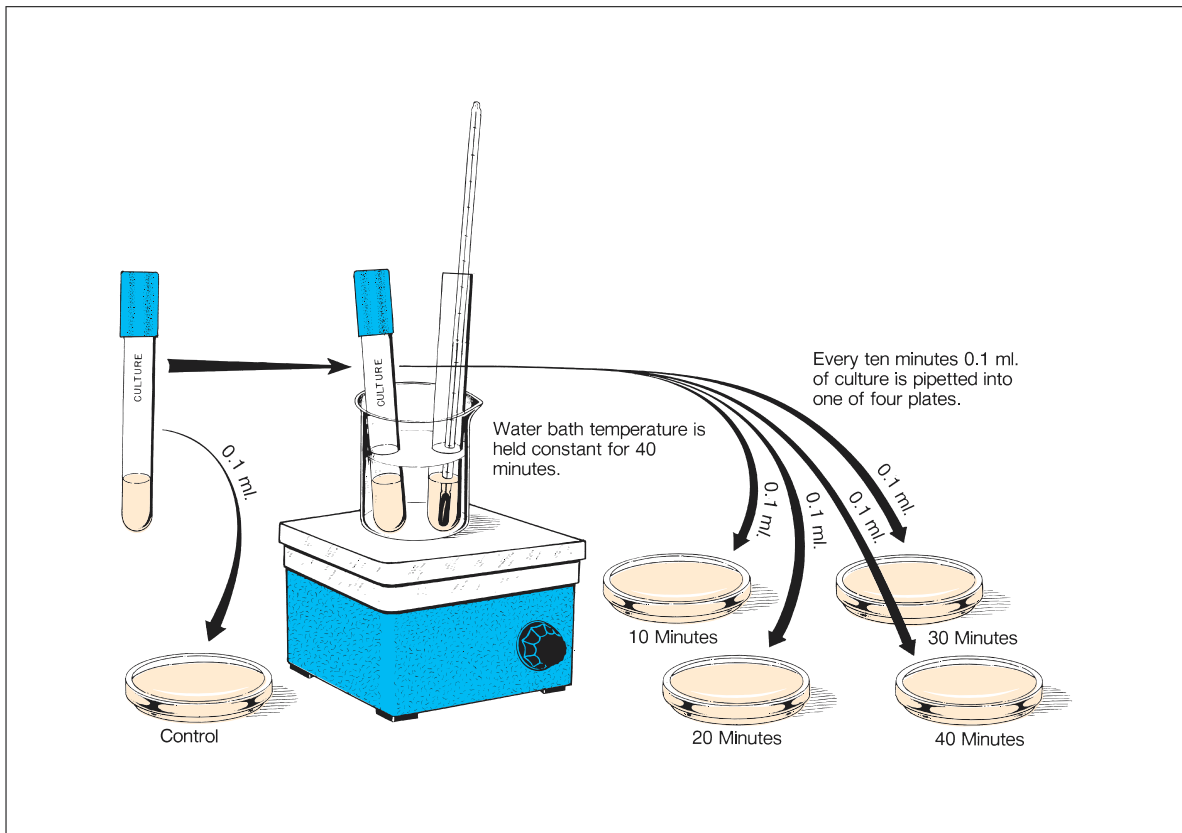


Figure 35.1 Procedure for determining thermal endurance

36

pH and Microbial Growth

Aside from temperature, the hydrogen ion concentration of an organism's environment exerts the greatest influence on its growth. The concentration of hydrogen ions, which is customarily designated by the term **pH** ($-\log 1/H^+$), limits the activity of enzymes with which an organism is able to synthesize new protoplasm. As in the case of temperature, there exists for each organism an optimum concentration of hydrogen ions in which it grows best. The pH values above and below which an organism fails to grow are, respectively, referred to as the minimum and maximum hydrogen ion concentrations. These values hold only when other environmental factors remain constant. If the composition of the medium, incubation temperature, or osmotic pressure is varied, the hydrogen ion requirements become different.

In this exercise we will test the degree of inhibition of microorganisms that results from media containing different pH concentrations. Note in the materials list that tubes of six different hydrogen concentrations are listed. Your instructor will indicate which ones, if not all, will be tested.

FIRST PERIOD

Materials:*per student:*

- 1 tube of nutrient broth of pH 3.0
- 1 tube of nutrient broth of pH 5.0
- 1 tube of nutrient broth of pH 7.0
- 1 tube of nutrient broth of pH 8.0
- 1 tube of nutrient broth of pH 9.0
- 1 tube of nutrient broth of pH 10.0

class materials:

- broth cultures of *Escherichia coli*
- broth cultures of *Staphylococcus aureus*
- broth cultures of *Alcaligenes faecalis**
- broth cultures of *Saccharomyces cerevisiae***

**Sporosarcina ureae* can be used as a substitute for *Alcaligenes faecalis*.

***Candida glabrata* is a good substitute for *Saccharomyces cerevisiae*.

1. Inoculate a tube of each of these broths with one organism. Use the organism following your assigned number from the table below:

Student Number	Organism
1, 5, 9, 13, 17, 21, 25	<i>Escherichia coli</i>
2, 6, 10, 14, 18, 22, 26	<i>Staphylococcus aureus</i>
3, 7, 11, 15, 19, 23, 27	<i>Alcaligenes faecalis</i> *
4, 8, 12, 16, 20, 24, 28	<i>Saccharomyces cerevisiae</i> **

2. Incubate the tubes of *E. coli*, *S. aureus*, and *A. faecalis* at 37° C for 48 hours. Incubate the tubes of *S. ureae*, *C. glabrata*, and *S. cerevisiae* at 20° C for 48 to 72 hours.

SECOND PERIOD

Materials:

- spectrophotometer
- 1 tube of sterile nutrient broth
- tubes of incubated cultures at various pHs

1. Use the tube of sterile broth to calibrate the spectrophotometer and measure the %T of each culture (page 98, Exercise 23). Record your results in the tables on the Laboratory Report.
2. Plot the O.D. values in the graph on the Laboratory Report and answer all the questions.

Osmotic Pressure and Bacterial Growth

37

Growth of bacteria can be profoundly affected by the amount of water entering or leaving the cell. When the medium surrounding an organism is **hypotonic** (low solute content), a resultant higher osmotic pressure occurs in the cell. Except for some marine forms, this situation is not harmful to most bacteria. The cell wall structure of most bacteria is so strong and rigid that even slight cellular swelling is generally inapparent.

In the reverse situation, however, when bacteria are placed in a **hypertonic** solution (high solute content), their growth may be considerably inhibited. The degree of inhibition will depend on the type of solute and the nature of the organism. In media of growth-inhibiting osmotic pressure, the cytoplasm becomes dehydrated and shrinks away from the cell wall. Such **plasmolyzed** cells are often simply inhibited in the absence of sufficient cellular water and return to normal when placed in an **iso-tonic** solution. In other instances, the organisms are irreversibly affected due to permanent inactivation of enzyme systems.

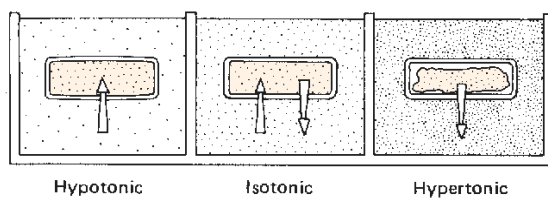


Figure 37.1 Osmotic variabilities

Organisms that thrive in hypertonic solutions are designated as halophiles or osmophiles. If they require minimum concentrations of salt (NaCl and other cations and anions) they are called **halophiles**. Obligate halophiles require a minimum of 13% sodium chloride. **Osmophiles**, on the other hand, require high concentrations of an organic solute, such as sugar.

In this exercise we will test the degree of inhibition of organisms that results with media containing different concentrations of sodium chloride. To accomplish this, you will streak three different organisms on four plates of media. The specific organisms

used differ in their tolerance of salt concentrations. The salt concentrations will be 0.5, 5, 10, and 15%. After incubation for 48 hours and several more days, comparisons will be made of growth differences to determine their degrees of salt tolerances.

Materials:

per student:

- 1 Petri plate of nutrient agar (0.5% NaCl)
- 1 Petri plate of nutrient agar (5% NaCl)
- 1 Petri plate of nutrient agar (10% NaCl)
- 1 Petri plate of milk salt agar (15% NaCl)

cultures:

- Escherichia coli* (nutrient broth)
- Staphylococcus aureus* (nutrient broth)
- Halobacterium salinarium* (slant culture)

1. Mark the bottoms of the four Petri plates as indicated in figure 37.2.
2. Streak each organism in a straight line on the agar, using a wire loop.
3. Incubate all the plates for 48 hours at room temperature with exposure to light (the pigmentation of *H. salinarium* requires light to develop). Record your results on the Laboratory Report.
4. Continue the incubation of the milk salt agar plate for several more days in the same manner, and record your results again on the first portion of Laboratory Report 37, 38.

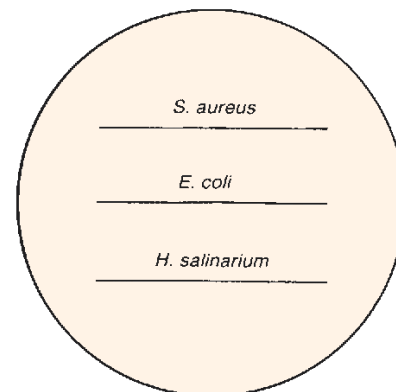


Figure 37.2 Streak pattern

38

Oligodynamic Action

The ability of small amounts of heavy metals to exert a lethal effect on bacteria is designated as **oligodynamic action** (Greek: *oligos*, small; *dynamis*, power). The effectiveness of these small amounts of metal is probably due to the high affinity of cellular proteins for the metallic ions. Although the concentration of ions in solution may be miniscule (a few parts per million), cells die due to the cumulative effects of ions within the cell.

The success of silver amalgam fillings to prevent secondary dental decay in teeth over long periods of time is due to the small amounts of silver and mercury ions that diffuse into adjacent tooth dentin. Its success in this respect has led to much debated concern that its toxicity may cause long-term injury to patients. In addition to its value (or harm) as a dental restoration material, oligodynamic action of certain other heavy metals has been applied to water purification, ointment manufacture, and the treatment of bandages and fabrics.

In this exercise we will compare the oligodynamic action of three metals (copper, silver, and aluminum) to note the differences.

Materials:

1 Petri plate
1 nutrient agar pour
forceps and Bunsen burner
acid-alcohol

broth culture of *E. coli* and *S. aureus*

3 metallic disks (copper, silver, aluminum)
water bath at student station (beaker of water
and electric hot plate)

1. Liquefy a tube of nutrient agar, cool to 50° C, and inoculate with either *E. coli* or *S. aureus* (odd: *E. coli*; even: *S. aureus*).
2. Pour half of the medium from each tube into a sterile Petri plate and leave the other half in a water bath (50° C). Allow agar to solidify in the plate.
3. Clean three metallic disks, one at a time, and place them on the agar, evenly spaced, as soon as they are cleaned. Use this routine:
 - Wash first with soap and water; then rinse with water.
 - With flamed forceps dip in acid-alcohol and rinse with distilled water.
4. Pour the remaining seeded agar from the tube over the metal disks. Incubate for 48 hours at 37° C.

LABORATORY REPORT

After incubation compare the zones of inhibition and record your results on the last portion of Laboratory Report 37, 38.

Ultraviolet Light: Lethal Effects

39

Except for the photosynthetic bacteria, most bacteria are harmed by ultraviolet radiation. Those that contain photosynthetic pigments require exposure to sunlight in order to synthesize substances needed in their metabolism. Although sunlight contains the complete spectrum of short to long wavelengths of light, it is only the short, invisible ultraviolet wavelengths that are injurious to the nonphotosynthetic bacteria.

Wavelengths of light may be expressed in nanometers (nm) or angstrom units ($\text{\AA}$). The angstrom unit is equal to 10^{-8} cm. In terms of nanometers, $10\text{\AA}$ equal one nanometer. Thus, a wavelength of 4500×10^{-8} cm would be expressed as $4500\text{\AA}$, 450 nm, or $0.45\text{ }\mu\text{m}$.

Figure 39.1 illustrates the relationship of ultraviolet to other types of radiations. By definition, ultraviolet light includes electromagnetic radiations that fall in the wavelength band between 40 and $4000\text{\AA}$. It bridges the gap between the X rays and the shortest wavelengths of light visible to the human eye. The visible range is approximately between 4000 and $7800\text{\AA}$. Actually, the practical range of ultraviolet, as far as we are concerned, lies between 2000 and $4000\text{\AA}$. The “extreme” range (40 – $2000\text{\AA}$) includes radiations that are absorbed by air and consequently function only in a vacuum. This region is also referred to as *vacuum ultraviolet*.

Ultraviolet is not a single entity, but is a very wide band of wavelengths. This fact is often not realized. Extending from 40 to $4000\text{\AA}$, it encompasses a span

of 1:100; visible wavelengths (4000 – $7800\text{\AA}$), on the other hand, represent only a twofold spread.

The germicidal effects of the ultraviolet are limited to only a specific region of the ultraviolet spectrum. As indicated in figure 39.1, the most effective wavelength is $2650\text{\AA}$. Low-pressure mercury vapor lamps, which have a high output (90%) of $2537\text{\AA}$, make very effective bactericidal lamps.

In this exercise organisms that have been spread on nutrient agar will be exposed to ultraviolet radiation for various lengths of time to determine the minimum amount of exposure required to effect a 100% kill. One-half of each plate will be shielded from the radiation to provide a control comparison. *Bacillus megaterium*, a spore-former, and *Staphylococcus aureus*, a non-spore-former, will be used to provide a comparison of the relative resistance of vegetative and spore types.

Exposure to ultraviolet light may be accomplished with a lamp as shown in figure 39.2 or with a UV box that has built-in ultraviolet lamps. The UV exposure effectiveness varies with the type of setup used. The exposure times given in table 39.1 work well for a specific type of mercury arc lamp. Note in the table that space is provided under the times for adding in different timing. Your instructor will inform you as to whether you should write in new times that will be more suited to the equipment in your lab. Proceed as follows to do this experiment.

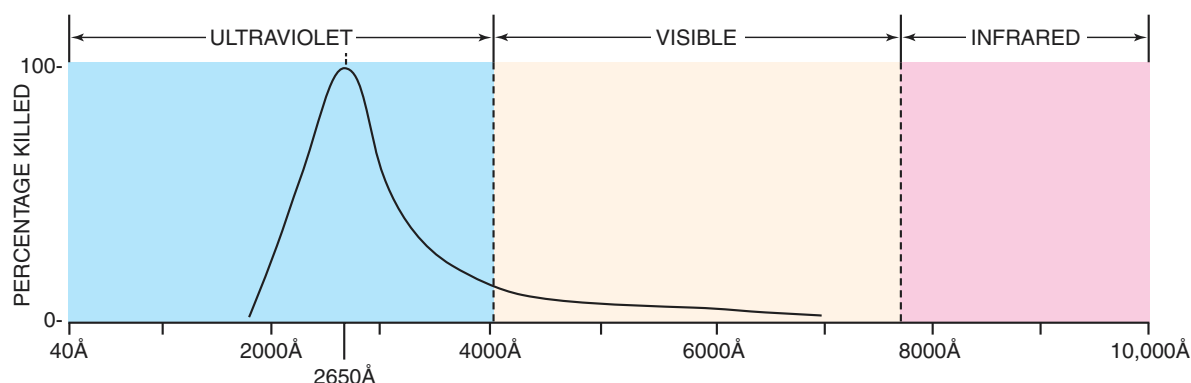


Figure 39.1 Lethal effectiveness of ultraviolet light

Exercise 39 • Ultraviolet Light: Lethal Effects**Materials:**

Petri plates of nutrient agar (one or more per student)
ultraviolet lamp or UV exposure box
timers (bell type)
cards (3" × 5")
nutrient broth cultures of *S. aureus* with swabs
saline suspensions of *B. megaterium* with swabs

1. Refer to table 39.1 to determine which organism you will work with. You may be assigned more than one plate to inoculate. If different times are to be used, your instructor will inform you what times to write in. Since there are only 16 assignment numbers in the table, more student assignment numbers can be written in as designated by your instructor.
2. Label the bottoms of the plates with your assignment number and your initials.
3. Using a cotton-tipped swab that is in the culture tube, swab the entire surface of the agar in each

plate. Before swabbing, express the excess culture from the swab against the inner wall of the tube.

4. Place the plates under the ultraviolet lamp *with the lids removed*. Cover half of each plate with a 3" × 5" card as shown in figure 39.2. Note that if your number is 8 or 16, you will not remove the lid from your plate. The purpose of this exposure is to see to what extent, if any, UV light can penetrate plastic.

CAUTION

Avoid looking directly into the ultraviolet lamp. These rays can cause cataracts and other eye injury.

5. After exposing the plates for the correct time durations, re-cover them with their lids, and incubate them inverted at 37° C for 48 hours.

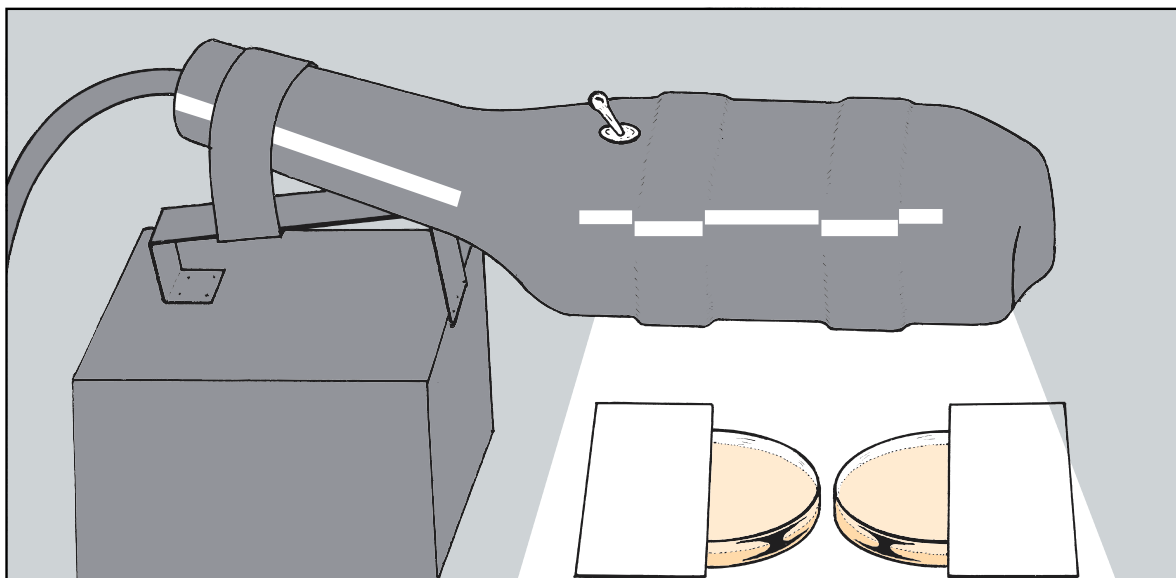
LABORATORY REPORT

Record your observations on the Laboratory Report and answer all the questions.

Table 39.1 Student Inoculation Assignments

	Exposure Times (Student Assignments)							
	1	2	3	4	5	6	7	8
<i>S. aureus</i>	10 sec	20 sec	40 sec	80 sec	2.5 min	5 min	10 min	20 min*
<i>B. megaterium</i>	9	10	11	12	13	14	15	16
	1 min	2 min	4 min	8 min	15 min	30 min	60 min	60 min*

*These Petri plates will be covered with dish covers during exposure.

**Figure 39.2** Plates are exposed to UV light with 50% coverage

Evaluation of Disinfectants: The Use-Dilution Method

40

When considering the relative effectiveness of different chemical agents against bacteria, some yardstick of comparison is necessary. Many different methods have been developed over the years since Robert Koch, in 1881, worked out the first scientific procedure by measuring the killing power of various germicides on silk threads that were impregnated with spores of *Bacillus anthracis*. Koch's method and many that followed proved unreliable for various reasons.

Finally, in 1931, the United States Food and Drug Administration adopted a method that was a modification of a test developed in England in 1903 by Rideal and Walker. In 1950 the Association of Official Agricultural Chemists adopted it as the official method of testing disinfectants. This method compares the effectiveness of various agents with phenol. A value called the **phenol coefficient** is arrived at that has significant meaning with certain limitations. The restrictions are that the test should be used only for phenol-like compounds that do not ex-

ert bacteriostatic effects and are not neutralized by the subculture media used. Many excellent disinfectants cannot be evaluated with this test. Disinfectants such as bichloride of mercury, iodine, metaphen, and quaternary detergents are unlike phenol in their germicidal properties and should not be evaluated in terms of phenol coefficients. Notwithstanding, however, many pharmaceutical companies have applied this test to such disinfectants with misleading results. A more suitable test for these nonphenolic disinfectants is the use-dilution method.

The **use-dilution method** makes use of small glass rods on which test organisms are dried for 30 minutes. The seeded rods are then exposed to the test solutions at 20° C for 1, 5, 10, and 30 minutes, rinsed with water or neutralizing solution, and transferred to the tubes of media. After incubation at 37° C for 48 hours, the tubes are examined for growth. When the results of this test are applied to practical conditions of use, they are found to be completely reliable.

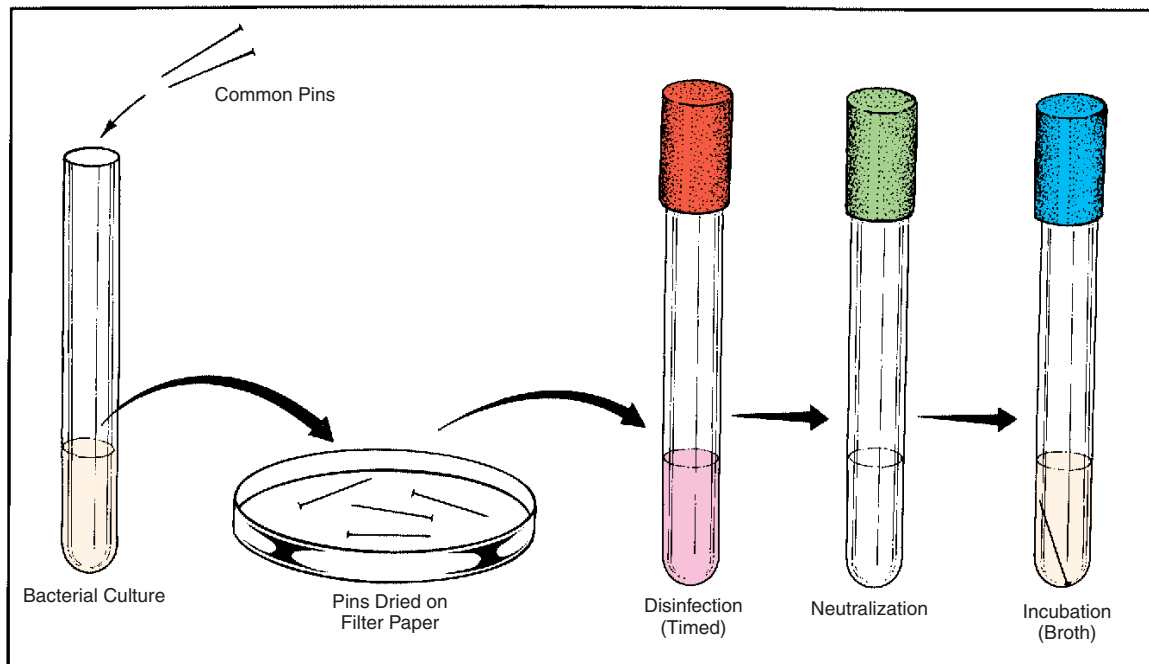


Figure 40.1 Procedure for use-dilution evaluation of a disinfectant

Exercise 40 • Evaluation of Disinfectants: The Use-Dilution Method

In this experiment you will follow a modified procedure of the use-dilution method to compare the relative merits of three different disinfectants on two kinds of bacteria: a spore-former, *Bacillus megaterium*, and a non-spore-former, *Staphylococcus aureus*. Instead of glass rods, we will use rustproof common pins.

Since each student will be performing only a small portion of the entire experiment it will be necessary to make student assignments with respect to agents used and timing. Table 40.1 indicates, according to student numbers, which agent each student will be working with, and the length of time to apply the agent to the pin. Note that a blank column is provided for write-in substitutions. Proceed as follows:

Materials:

per student:

- 2 tubes of one of the following agents:
 - 1:750 Zephiran for students 1, 4, 7, 10, 13, 16, 19, 22, 25, 28
 - 5% phenol for students 2, 5, 8, 11, 14, 17, 20, 23, 26, 29
 - 8% formaldehyde for students 3, 6, 9, 12, 15, 18, 21, 24, 27, 30
- 2 tubes of sterile water (about 7 ml each)
- 2 tubes of nutrient broth (about 7 ml each)
- forceps

on demonstration table:

- 1 nutrient broth culture of *S. aureus*
 - 1 physiological saline suspension of a 48 hour nutrient agar slant culture of *B. megaterium*
 - 2 sterile Petri plates with filter paper in bottom
 - several forceps and Bunsen burner
 - 2 test tubes containing 36 sterile common pins in each one (pins must be plated brass, which are rustproof)
1. Consult table 40.1 or the materials list to determine which disinfectant you are to use.
 2. Get two tubes of the disinfectant, two tubes of sterile water, and two tubes of nutrient broth from

the table. Label one of each pair *B. megaterium* and the other *S. aureus*.

Instructor: While the students are getting their supplies together, you can start the experiment by pouring the broth culture of *S. aureus* into one of the tubes of pins and the saline suspension of *B. megaterium* into the other tube of pins. After decanting the organisms into a beaker of disinfectant, the pins are deposited onto filter paper in separate Petri plates to dry. Plates should be clearly labeled as to contents. Allow a few minutes for the pins to dry before allowing students to take them. Make certain, also, that a Bunsen burner and forceps are set up near the two dishes of pins.

3. Gently flame a pair of forceps, let cool, and transfer one pin from each Petri plate to the separate tubes of disinfectant. Be sure to put them into the right tubes.
4. Leave the pins in the disinfectant for the length of time indicated in table 40.1. Find your number under the time indicated for your disinfectant.
5. At the end of the assigned time, flame the mouths of the tubes of disinfectant and *carefully pour the disinfectant into the sink without discarding the pins*. Then, transfer the pins into separate tubes of sterile water. Avoid transferring any of the disinfectant to the water tubes with the pins.
6. After 1 minute in the tubes of water, flame the mouths of the water and broth tubes, pour off the water, and shake the pins out of the emptied tubes into separate, labeled tubes of nutrient broth.

Instructor: At this point the instructor, or a designated class member, should put one pin from each of the Petri plates into separate labeled tubes of nutrient broth to be used as positive **controls** for each organism.

7. Incubate all nutrient broth tubes with pins for 48 hours at 37° C. Examine them and record your results on the Laboratory Report.

Table 40.1 Student Assignments for Agents and Timing

Disinfectant		Time in Minutes				
	Substitution	1	5	10	30	60
1:750 Zephiran		1, 16	4, 19	7, 22	10, 25	13, 28
5% Phenol		2, 17	5, 20	8, 23	11, 26	14, 29
8% Formaldehyde		3, 18	6, 21	9, 24	12, 27	15, 30

Evaluation of Alcohol: Its Effectiveness as a Skin Degerming Agent

41

As a skin disinfectant, 70% alcohol is undoubtedly the most widely used agent. The ubiquitous prepackaged alcohol swabs used by nurses and technicians are evidence that these items are indispensable. The question that often arises is: How really effective is alcohol in routine use? When the skin is swabbed prior to penetration, are all, or mostly all, of the surface bacteria killed? To determine alcohol effectiveness, as it might be used in routine skin disinfection, we are going to perform a very simple experiment here that utilizes four thumbprints and a plate of enriched agar. Class results will be pooled to arrive at a statistical analysis.

Figure 41.1 illustrates the various steps in this test. Note that the Petri plate is divided into four parts. On the left side of the plate an unwashed left thumb is first pressed down on the agar in the lower quadrant

of the plate. Next the left thumb is pressed down on the upper left quadrant. With the left thumb we are trying to establish the percentage of bacteria that are removed by simple contact with the agar.

On the right side of the plate an unwashed right thumb is pressed down on the lower right quadrant of the plate. The next step is to either dip the right thumb into alcohol or to scrub it with an alcohol swab and dry it. Half of the class will use the dipping method and the other half will use alcohol swabs. Your instructor will indicate what your assignment will be. The last step is to press the dried right thumb on the upper right quadrant of the plate.

After inoculating the plate it is incubated at 37° C for 24–48 hours. Colony counts will establish the effectiveness of the alcohol.

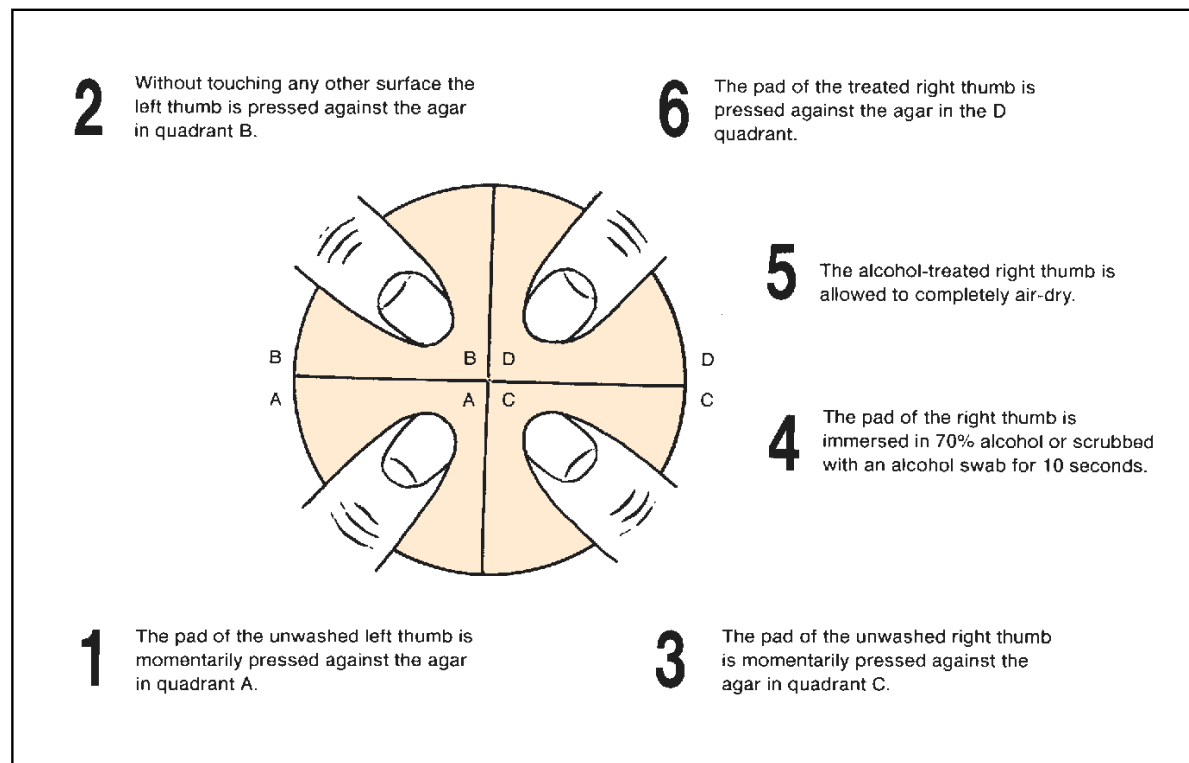


Figure 41.1 Procedure for testing the effectiveness of alcohol on the skin

Exercise 41 • Evaluation of Alcohol: Its Effectiveness as a Skin Degerming Agent

Materials:

1 Petri plate of veal infusion agar
small beaker
70% ethanol
alcohol swab

1. Perform this experiment with unwashed hands.
2. With a china marking pencil, mark the bottom of the Petri plate with two perpendicular lines that divide it into four quadrants. Label the left quadrants **A** and **B** and the right quadrants **C** and **D** as shown in figure 41.1. (*Keep in mind that when you turn the plates over to label them, the A and B quadrants will be on the right and C and D will be on the left.*)
3. Press the pad of your left thumb against the agar surface in the A quadrant.
4. Without touching any other surface, press the left thumb into the B quadrant.
5. Press the pad of your right thumb against the agar surface of the C quadrant.
6. Disinfect the right thumb by one of the two following methods:
 - dip the thumb into a beaker of 70% ethanol for 5 seconds, or
 - scrub the entire pad surface of the right thumb with an alcohol swab.
7. Allow the alcohol to completely evaporate from the skin.
8. Press the right thumb against the agar in the D quadrant.
9. Incubate the plate at 37° C for 24–48 hours.
10. Follow the instructions on the Laboratory Report for evaluating the plate and answer all of the questions.

Evaluation of Antiseptics: The Filter Paper Disk Method

42

The term *antiseptic* has, unfortunately, been somewhat ill-defined. Originally, the term was applied to any agent that prevents sepsis, or putrefaction. Since sepsis is caused by growing microorganisms, it follows that an antiseptic inhibits microbial multiplication without necessarily killing them. By this definition, we can assume that antiseptics are essentially bacteriostatic agents. Part of the confusion that has resulted in its definition is that the United States Food and Drug Administration rates antiseptics essentially the same as disinfectants. Only when an agent is to be used in contact with the body for a long period of time do they rate its bacteriostatic properties instead of its bactericidal properties.

If we are to compare antiseptics on the basis of their bacteriostatic properties, the filter paper disk method (figure 42.1) is a simple, satisfactory method to use. In

this method a disk of filter paper ($\frac{1}{2}$ " diameter) is impregnated with the chemical agent and placed on a seeded nutrient agar plate. The plate is incubated for 48 hours. If the substance is inhibitory, a clear zone of inhibition will surround the disk. The size of this zone is an expression of the agent's effectiveness and can be compared quantitatively against other substances.

In this exercise we will measure the relative effectiveness of three agents (phenol, formaldehyde, and iodine) against two organisms: *Staphylococcus aureus* (gram-positive) and *Pseudomonas aeruginosa* (gram-negative). Table 42.1 will be used to assign each student one chemical agent to be tested against one organism. Note that space has been provided in the table for different agents to be written in as substitutes for the three agents listed. Your instructor may wish to make substitutions. Proceed as follows:

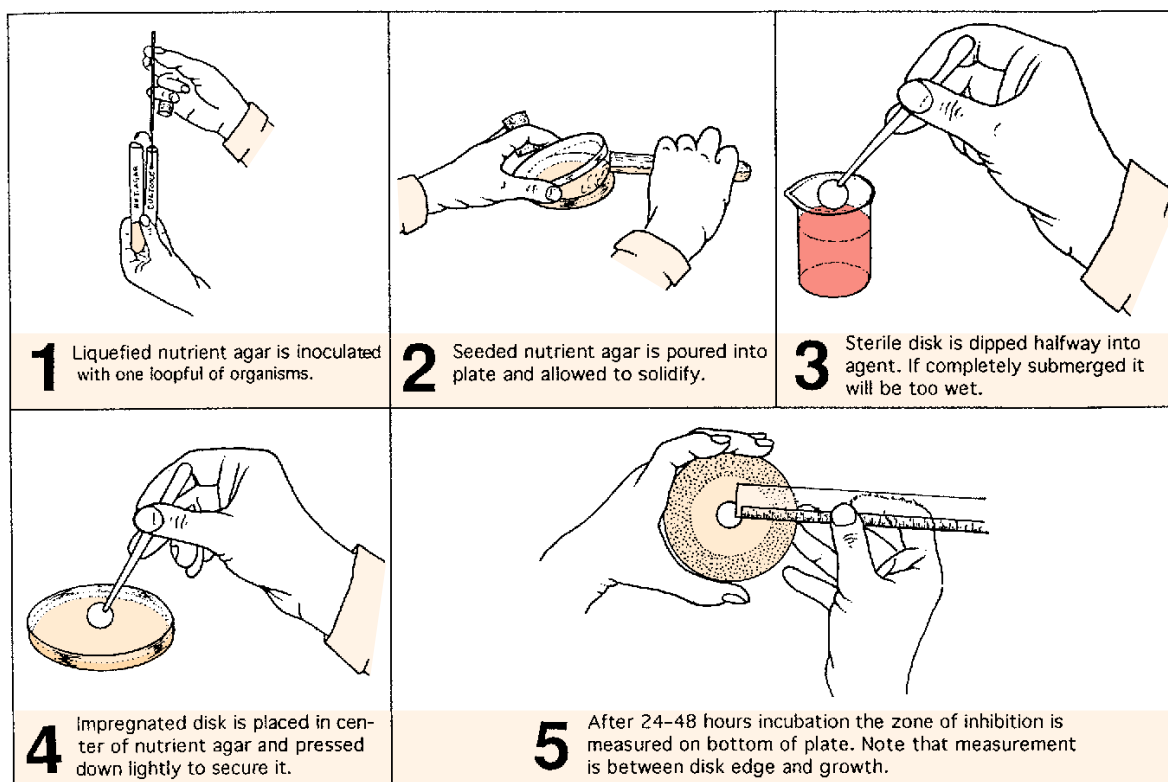


Figure 42.1 Filter paper disk method of evaluating an antiseptic

Exercise 42 • Evaluation of Antiseptics: The Filter Paper Disk Method

FIRST PERIOD (Disk Application)

Materials:

per student:

- 1 nutrient agar pour and 1 Petri plate
 - broth culture of *S. aureus* or *P. aeruginosa* on demonstration table:
 - Petri dish containing sterile disks of filter paper (½" dia)
 - forceps and Bunsen burner
 - chemical agents in small beakers (5% phenol, 5% formaldehyde, 5% aqueous iodine)
- Consult table 42.1 to determine your assignment.
 - Liquefy a nutrient agar pour in a water bath and cool to 50° C.
 - Label the bottom of a Petri plate with the names of the organism and chemical agent.
 - Inoculate the agar pour with one loopful of the organism and pour into the plate.

- After the medium has solidified in the plate, pick up a sterile disk with *lightly flamed* forceps, dip the disk *halfway* into a beaker of the chemical agent, and place the disk in the center of the medium.

To secure the disk to the medium, press lightly on it with the forceps.

- Incubate the plate at 37° C for 48 hours.

SECOND PERIOD (Evaluation)

- Measure the zone of inhibition from the edge of the disk to the edge of the growth (see illustration 5, figure 42.1).
- Exchange plates with other members of the class so that you will have an opportunity to complete the table on the Laboratory Report.

Table 42.1 Student Assignments

Chemical Agent		Student Number	
	Substitution	<i>S. aureus</i>	<i>P. aeruginosa</i>
5% Phenol		1, 7, 13, 19, 25	2, 8, 14, 20, 26
5% Formaldehyde		3, 9, 15, 21, 27	4, 10, 16, 22, 28
5% Iodine		5, 11, 17, 23, 29	6, 12, 18, 24, 30

Antimicrobial Sensitivity Testing: The Kirby-Bauer Method

43

Once the causative organism of a specific disease in a patient has been isolated, it is up to the attending physician to administer a chemotherapeutic agent that will inhibit or kill the pathogen without causing serious harm to the individual. The method must be relatively simple to use, be very reliable, and yield results in as short a time as possible. The Kirby-Bauer method of sensitivity testing is such a method. It is used for testing both antibiotics and drugs. **Antibiotics** are chemotherapeutic agents of low molecular weight produced by microorganisms that inhibit or kill other microorganisms. **Drugs**, on the other hand, are antimicrobial agents that are man-made. Both types of agents will be tested in this laboratory session according to the procedure shown in figure 43.1.

The effectiveness of an antimicrobial in sensitivity testing is based on the size of the zone of inhibition that surrounds a disk that has been impregnated with a specific concentration of the agent. The zone of inhibition, however, varies with the diffusibility of the agent, the size of the inoculum, the type of medium, and many other factors. Only by taking all these variables into consideration can a reliable method be worked out.

The **Kirby-Bauer method** is a standardized system that takes all variables into consideration. It is sanctioned by the U.S. FDA and the Subcommittee on Antimicrobial Susceptibility Testing of the National Committee for Clinical Laboratory Standards. Although time is insufficient here to consider all facets of this test, its basic procedure will be followed.

The recommended medium in this test is Mueller Hinton II agar. Its pH should be between 7.2 and 7.4, and it should be poured to a uniform thickness of 4 mm in the Petri plate. This requires 60 ml in a 150 mm plate and 25 ml in a 100 mm plate. For certain fastidious microorganisms, 5% defibrinated sheep blood is added to the medium.

Inoculation of the surface of the medium is made with a cotton swab from a broth culture. In clinical applications, the broth turbidity has to match a defined standard. Care must also be taken to express excess broth from the swab prior to inoculation.

High-potency disks are used that may be placed on the agar with a mechanical dispenser or sterile forceps. To secure the disks to the medium, it is necessary to press them down onto the agar.

After 16 to 18 hours incubation the plates are examined and the diameters of the zones are measured to the nearest millimeter. To determine the significance of the zone diameters, one must consult a table (Appendix A).

In this exercise we will work with four microorganisms: *Staphylococcus aureus*, *Escherichia coli*, *Proteus vulgaris*, and *Pseudomonas aeruginosa*. Each student will inoculate one plate with one of the four organisms and place the disks on the medium by whichever method is available. Since each student will be doing only a portion of the total experiment, student assignments will be made. Proceed as follows:

FIRST PERIOD (Plate Preparation)

Materials:

- 1 Petri plate of Mueller-Hinton II agar nutrient broth cultures (with swabs) of *S. aureus*, *E. coli*, *P. vulgaris*, and *P. aeruginosa*
- disk dispenser (BBL or Difco)
- cartridges of disks (BBL or Difco)
- forceps and Bunsen burner
- zone interpretation charts (Difco or BBL)

1. Select the organisms you are going to work with from the following table.

Organism	Student Number
<i>S. aureus</i>	1, 5, 9, 13, 17, 21, 25
<i>E. coli</i>	2, 6, 10, 14, 18, 22, 26
<i>P. vulgaris</i>	3, 7, 11, 15, 19, 23, 27
<i>P. aeruginosa</i>	4, 8, 12, 16, 20, 24, 28

2. Label your plate with the name of your organism.
3. Inoculate the surface of the medium with the swab after expressing excess fluid from the swab by pressing and rotating the swab against the inside walls of the tube above the fluid level. Cover the surface of the agar evenly by swabbing in three directions. A final sweep should be made of the agar rim with the swab.
4. Allow **3 to 5 minutes** for the agar surface to dry before applying disks.

Exercise 43 • Antimicrobial Sensitivity Testing: The Kirby-Bauer Method

5. Dispense disks as follows:
 - a. If an automatic dispenser is used, remove the lid from the plate, place the dispenser over the plate, and push down firmly on the plunger. With the sterile tip of forceps, tap each disk lightly to secure it to medium.
 - b. If forceps are used, sterilize them first by flaming before picking up the disks. Keep each disk at least 15 mm from the edge of the plate. Place no more than 13 on a 150 mm plate, nor more than 5 on a 100 mm plate. Apply light pressure to each disk on the agar with the tip of a sterile forceps or inoculating loop to secure it to medium.
6. Invert and incubate the plate for 16 to 18 hours at 37° C.

SECOND PERIOD (Interpretation)

After incubation, measure the zone diameters with a metric ruler to the nearest whole millimeter. The zone of complete inhibition is determined without magnification. Ignore faint growth or tiny colonies that can be

detected by very close scrutiny. Large colonies growing within the clear zone might represent resistant variants or a mixed inoculum and may require reidentification and retesting in clinical situations. Ignore the “swarming” characteristics of *Proteus*, measuring only to the margin of heavy growth.

Record the zone measurements on the table of the Laboratory Report and on the chart on the demonstration table, which has been provided by the instructor.

Use table 43.1 or 43.2 for identifying the various disks. Although BBL and Difco use essentially the same code numbers, there are slight differences in the two charts. Careful comparison of the charts will reveal that each company has certain antibiotics that are not listed by the other company.

To determine which antibiotics your organism is sensitive to (S), or resistant to (R), or intermediate (I), consult Table VII in Appendix A. It is important to note that the significance of a zone of inhibition varies with the type of organism. If you cannot find your antibiotic on the chart, consult a chart that is supplied by BBL or Difco that is on the demonstration table or bulletin board. Table VII is incomplete.

Table 43.1 Code for BBL Disks

AMD-10	Amdinocillin	E-15	Erythromycin
AN-30	Amikacin	GM-120	Gentamicin
AmC-30	Amoxicillin/ Clavulanic Acid	IPM-10	Imipenem
AM-10	Ampicillin	K-30	Kanamycin
SAM-20	Ampicillin/ Sulbactam	LOM-10	Lomefloxacin
AZM-15	Azithromycin	LOR-30	Loracarbef
AZ-75	Azlocillin	DP-5	Methicillin
ATM-30	Aztreonam	MZ-75	Mezlocillin
B-10	Bacitracin	MI-30	Minocycline
CB-100	Carbenicillin	MOX-30	Moxalactam
CEC-30	Cefactor	NF-1	Nafcillin
MA-30	Cefamandole	NA-30	Nalidixic Acid
CZ-30	Cefazolin	N-30	Neomycin
CFM-5	Cefixime	NET-30	Netilmicin
CMZ-30	Cefmetrazole	NOR-10	Norfloxacin
CID-30	Cefonicid	NB-30	Novobiocin
CFP-75	Cefoperazone	OFX-5	Ofloxacin
CTX-30	Cefotaxime	OX-1	Oxacillin
CTT-30	Cefotetan	OA-2	Oxolinic Acid
FOX-30	Cefoxitin	P-10	Penicillin
CPD-10	Cefpodoxime	PIP-100	Piperacillin
CPR-30	Cefprozil	PB-300	Polymyxin B
CAZ-30	Ceftazidime	RA-5	Rifampin
ZOX-30	Ceftizoxime	SPT-100	Spectinomycin
CRO-30	Ceftriaxone	S-300	Streptomycin
CXM-30	Cefuroxime	G-25	Sulfisoxazole
CF-30	Cephalothin	Te-30	Tetracycline
C-30	Chloramphenicol	TIC-75	Ticarcillin
CIN-100	Cinoxacin	TIC-85	Ticarcillin/ Clavulanic Acid
CIP-5	Ciprofloxacin	NN-10	Tobramycin
CLR-15	Clarithromycin	TMP-5	Trimethoprim
CC-2	Clindamycin	SXT	Trimethoprim/ Sulfamethoxazole
CL-10	Colistin	Va-30	Vancomycin

Table 43.2 Code for Difco Disks

AN 30	Amikacin	E 15	Erythromycin
AMC 30	Amoxicillin/ Clavulanic Acid	FLX 5	Fleroxacin
AM 10	Ampicillin	GM 10	Gentamycin
SAM 20	Ampicillin/ Sulbactam	IPM 10	Imipenem
AZM 15	Azithromycin	K 30	Kanamycin
AZ 75	Azlocillin	LOM 10	Lomefloxacin
ATM 30	Aztreonam	LOR 30	Loracarbef
CB 100	Carbenicillin	MZ 75	Mezlocillin
CEC 30	Cefactor	Mi 30	Minocycline
MA 30	Cefamandole	MOX 30	Moxalactam
CZ 30	Cefazolin	NF 1	Nafcillin
FEP 30	Cefepime	NA 30	Nalidixic Acid
CAT 10	Cefetamet	NET 30	Netilmicin
CFM 5	Cefixime	FD 300	Nitrofurantoin
CMZ 30	Cefmetrazole	NOR 10	Norfloxacin
CID 30	Cefonicid	OFX 5	Ofloxacin
CFP 75	Cefoperazone	P-10	Penicillin G
CTX 30	Cefotaxime	PTZ 110	Piperacillin/ Tazobactam
CTT 30	Cefotetan	RA 5	Rifampin
FOX 30	Cefoxitin	S 10	Streptomycin
CPD 10	Cefpodoxime	G 300	Sulfisoxazole
CPR 30	Cefprozil	TEC 30	Telcoplanin
CAZ 30	Ceftazidime	TE 30	Tetracycline
OX 30	Ceftizoxime	TIC 75	Ticarcillin
CRO 30	Ceftriaxone	TIC 85	Ticarcillin/ Clavulanic Acid
CXM 30	Cefuroxime	TN 10	Tobramycin
CF 30	Cephalothin	TMP 5	Trimethoprim
C 30	Chloramphenicol	SxT	Trimethoprim/ Sulfamethoxazole
CIN 100	Cinoxacin	VA 30	Vancomycin
CLR 15	Clarithromycin		
CC 2	Clindamycin		
D 30	Doxycycline		
ENX 10	Enoxacin		

Antimicrobial Sensitivity Testing: The Kirby-Bauer Method • Exercise 43

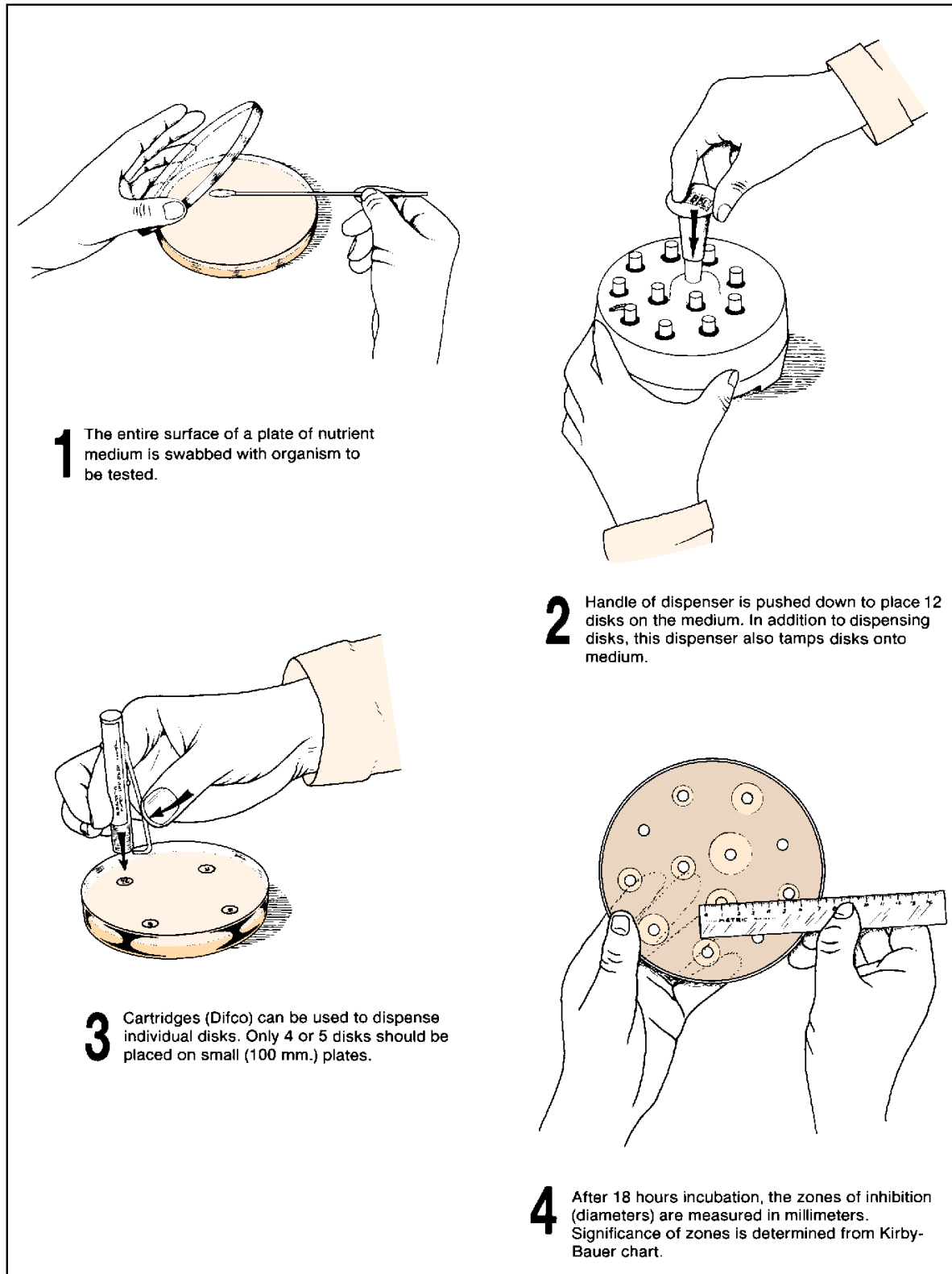


Figure 43.1 Antimicrobial sensitivity testing

44

Effectiveness of Hand Scrubbing

The importance of hand disinfection in preventing the spread of disease is accredited to the observations of Semmelweis at the Lying-In Hospital in Vienna in 1846 and 1847. He noted that the number of cases of puerperal fever was closely related to the practice of sanitary methods. Until he took over his assignment in this hospital, it was customary for medical students to go directly from the autopsy room to a patient's bedside and assist in deliveries without scrubbing and disinfecting their hands. When the medical students were on vacation, only the nurses, who were not permitted in the autopsy room, attended the patients. Semmelweis noted that during this time, deaths due to puerperal fever fell off markedly.

As a result of his observations, he established a policy that no medical students would be allowed to examine obstetric patients or assist in deliveries until they had cleansed their hands with a solution of chloride of lime. This ruling caused the death rate from puerperal infections to drop from 12% to 1.27% in one year.

Today it is routine practice to wash hands prior to the examination of any patient and to do a complete surgical scrub prior to surgery. Scrubbing the hands involves the removal of **transient** (contaminant) and **resident** microorganisms. Depending on the condition of the skin and the numbers of bacteria present, it takes from 7 to 8 minutes of washing with soap and water to remove all transients, and they can be killed with relative ease using suitable antiseptics. Residents, on the other hand, are firmly entrenched and are removed slowly by washing. These organisms, which consist primarily of staphylococci of low pathogenicity, are less susceptible than the transients to the action of antiseptics.

In this exercise, an attempt will be made to evaluate the effectiveness of the length of time in removal of organisms from the hands using a surgical scrub technique. One member of the class will be selected to perform the scrub. Another student will assist by supplying the soap, brushes, and basins, as needed. During the scrub, at 2-minute intervals, the hands will be scrubbed into a basin of sterile water. Bacterial counts will be made of these basins to determine the effectiveness of the previous 2-minute scrub in reduc-

ing the bacterial flora of the hands. Members of the class not involved in the scrub procedure will make the inoculations from the basins for the plate counts.

SCRUB PROCEDURE

The two members of the class who are chosen to perform the surgical scrub will set up their materials near a sink for convenience. As one student performs the scrub, the other will assist in reading the instructions and providing materials as needed. The basic steps, which are illustrated in figure 44.1, are also described in detail below. Before beginning the scrub, both students should read all the steps carefully.

Materials:

- 5 sterile surgical scrub brushes, individually wrapped
- 5 basins (or 2000 ml beakers), containing 1000 ml each of sterile water. These basins should be covered to prevent contamination
- 1 dispenser of green soap
- 1 tube of hand lotion

Step 1 To get some idea of the number of transient organisms on the hands, the scrubber will scrub all surfaces of each hand with a sterile surgical scrub brush for 30 seconds into Basin A. No green soap will be used for this step. The successful performance of this step will depend on

- spending the same amount of time on each hand (30 seconds),
- maintaining the same amount of activity on each hand, and
- scrubbing under the fingernails, as well as working over their surfaces.

After completion of this 60-second scrub, notify Group A that their basin is ready for the inoculations.

Step 2 Using the *same* brush as above, begin scrubbing with green soap for 2 minutes, using cool tap water to moisten and rinse the hands. One minute is devoted to each hand.

The assistant will make one application of green soap to each hand as it is being scrubbed.

Effectiveness of Hand Scrubbing • Exercise 44

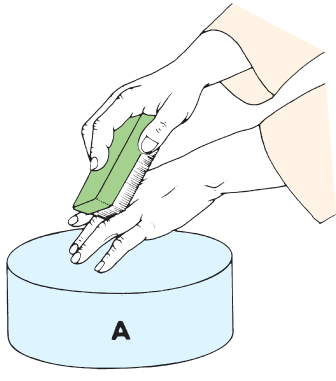
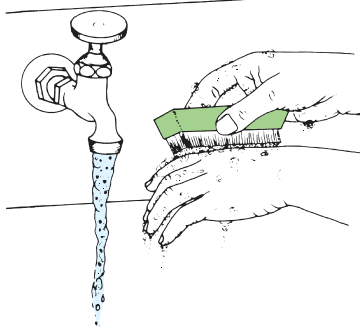
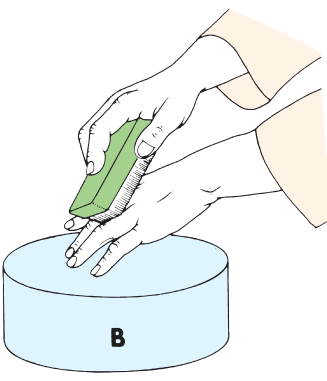
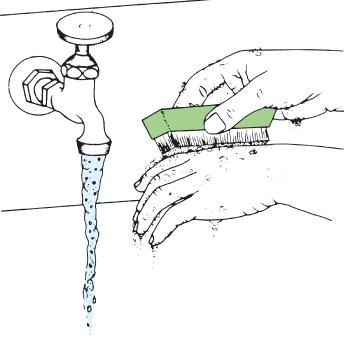
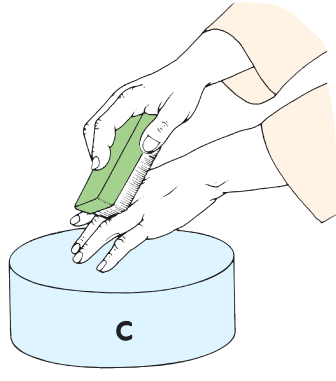
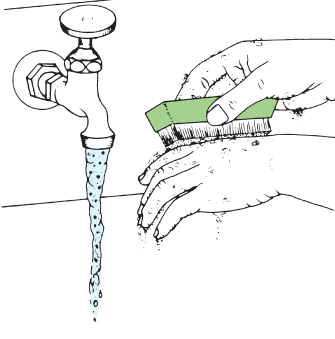
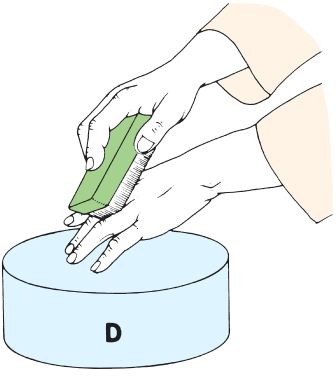
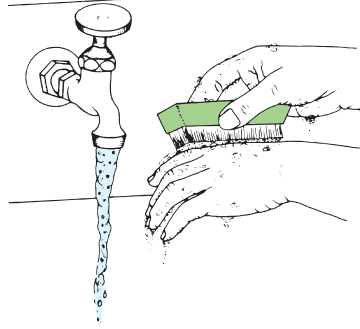
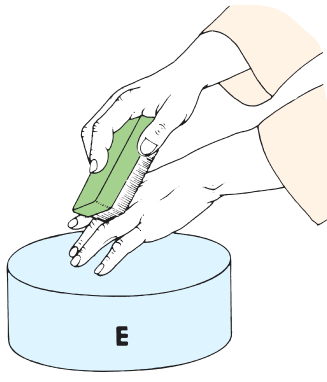
 A		 B
1 Sixty-second hand scrub into Basin A. No soap.	2 Two-minute soap scrub with running water.	3 Sixty-second hand scrub into Basin B. No soap.
	 C	
4 Same as 2.	5 Sixty-second hand scrub into Basin C. No soap.	6 Same as 2.
 D		 E
7 Sixty-second hand scrub into Basin D. No soap.	8 Same as 2.	9 Sixty-second hand scrub into Basin E. No soap.

Figure 44.1 Hand scrubbing routine